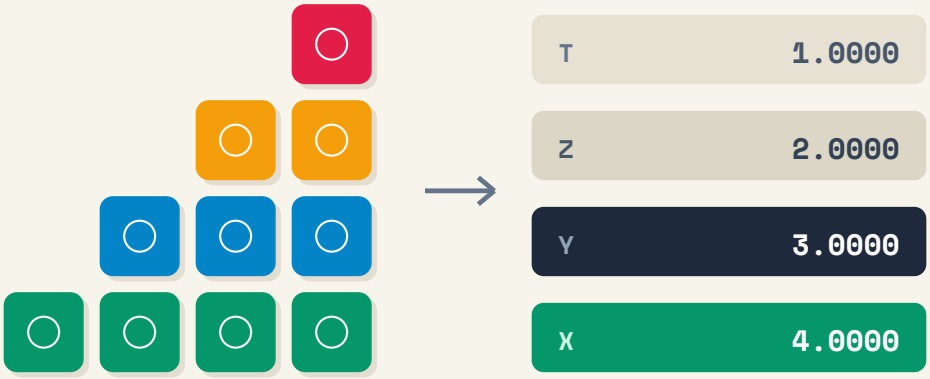
UNIFIED TEACHER'S CONCEPT GUIDE • GRADES K–12

TEACHING WITH THE STACK

Hands-On Mathematics & Concept Facilitation Guide (K–12)



StackCalc

Teaching with the Stack

A Unified K–12 Concept-Based Teacher's Guide paired with Curriculum-Aligned Student Workbooks.

WHY THIS TEACHER'S GUIDE IS ORGANIZED BY CONCEPT (WHILE WORKBOOKS ALIGN BY UNIT)

Following best practice from Singapore Math, CGI, and NRICH, teacher guidance is most effective when organized by mathematical concept domain rather than fragmented across textbook chapters. When a student struggles with common denominators in Grade 7 rational equations, the underlying conceptual gap is fraction equivalence (Domain 6). This unified guide places the physical block models, RPN stack invariants, and misconception diagnostics for every concept in one permanent desk reference—while students work inside dedicated, chapter-aligned workbooks.

THE 4-BOOK CURRICULUM & STUDENT WORKBOOK LINEUP (US LETTER 8.5" × 11")

Book 1 • Teaching with the Stack (This Volume — Unified Master Teacher's Guide by Concept)

Combines Part I (12 K–12 Concept Domains with Core & Extension Lessons, Misconception Diagnostics & Hands-On Remediations), Part II (Middle School Concept & Section Coaching for all 45 Big Ideas Math sections & 900 problems), and Part III (Master Answer Keys & Keycap Reference).

Book 2 • Hands-On Math Lab (General K–12 Student Exploration Workbook)

Student-facing lab workbook for Units 01–12. Each unit provides a 2-page spread: Page A features the companion book story, visual block rules, guided RPN stack walkthrough, and warm-up; Page B provides the 4-problem hands-on investigation worksheet. Includes a back-of-book Student Self-Check Key.

Book 3 • Middle School Math Lab (Big Ideas Math Grades 6–8 Student Practice Workbook)

114-page student workbook aligned section-by-section to 10 Middle School units (45 sections, 135 worked examples with SC-6 keycaps, 900 Practice A & B problems, and a 5-page compact Student Self-Check Answer Key in Appendix E).

Book 4 • Stack Speed & Fluency (Student Speed & Proficiency Trials Workbook)

15 timed UIL-style Calculator Applications sprints (150 problems across Stack Depth Levels 1, 2, and 4), Inside-Out RPN Speed Tactics, Competitor Progress Graph, and Self-Check Answer Key.

ALIGNED TO OUR ILLUSTRATED TRADE BOOKS (AVAILABLE ON AMAZON)

- **Elementary & Middle School (Domains 1–7, SC-6): From Blocks to the Stack: Hands-On Math with the SC-6**
- **High School, AP & Engineering (Domains 8–12, SC-32): Thinking in the Stack: Science & Engineering on the SC-32**
- Every concept domain in Part I opens with the historical story and real-world hook from its companion book chapter.
- **Free Printable Chapter & Unit Packets + Classroom Reproduction Rights: https://stackcalc.io/curriculum/**

Complete Guide Contents & Concept Index

Directory of Front Matter, Cross-Curriculum Concordances, Part I (Domains 1–12), Part II (Middle School Sections), and Part III (Master Keys).

Part / Concept Domain	Grades & Calculator	Companion Book & Student Workbook Link	Pages
Front Matter & Reluctant Teacher RPN Primer	All Educators	4-Book Architecture, Why RPN & 3-Min Primer	pp. i–v
Studio Block Games & Orthographic Check	Initial 3 Kits	Place Value, Fraction & Prime Studio Games	pp. vi–viii
Cross-Curriculum Pacing Concordances	PK–12 / AP	Singapore Dimensions, Bridges & Big Ideas	pp. 1–3
Part I • Domain 01: Cardinality, Counting & Step Accumulation	Pre-K – Kindergarten (SC-6)	Hands-On Lab Unit 01 • From Blocks to the Stack	pp. 4–5
Part I • Domain 02: Part-Part-Whole Number Bonds & Missing Addends	Grades K–1 (SC-6)	Hands-On Lab Unit 02 • From Blocks to the Stack	pp. 6–7
Part I • Domain 03: The Ten-Frame, Subitizing & Make-10 Bridging	Grade 1 (SC-6)	Hands-On Lab Unit 03 • From Blocks to the Stack	pp. 8–9
Part I • Domain 04: Two-Row Number Rack: Doubles & Equal Groups	Grades 1–2 (SC-6)	Hands-On Lab Unit 04 • From Blocks to the Stack	pp. 10–11
Part I • Domain 05: Place Value & The 4-Level Stack Elevator	Grades 3–4 (SC-6)	Hands-On Lab Unit 05 • From Blocks to the Stack	pp. 12–13
Part I • Domain 06: Fraction Studio: 84 mm Towers & Two-Dot Entry	Grades 4–5 (SC-6)	Hands-On Lab Unit 06 • From Blocks to the Stack	pp. 14–15
Part I • Domain 07: Postfix Trees, Order of Operations & Factors	Grades 5–6 (SC-6)	Hands-On Lab Unit 07 • From Blocks to the Stack	pp. 16–17
Part I • Domain 08: Tactile Algebra: Area Tiles, Zero Pairs & Roots	Grades 7–8 (SC-32)	Hands-On Lab Unit 08 • Thinking in the Stack	pp. 18–19
Part I • Domain 09: Coordinate Geometry, Trigonometry & Vectors	Grades 9–11 (SC-32)	Hands-On Lab Unit 09 • Thinking in the Stack	pp. 20–21
Part I • Domain 10: Exponents, Logarithms & Continuous Growth	Grades 10–12 (SC-32)	Hands-On Lab Unit 10 • Thinking in the Stack	pp. 22–23
Part I • Domain 11: Combinatorics, Probability & Series Statistics	Grades 11–12 / AP (SC-32)	Hands-On Lab Unit 11 • Thinking in the Stack	pp. 24–25
Part I • Domain 12: Digital Logic, Base-N & Computer Architecture	Grades 9–12 / AP CS (SC-32)	Hands-On Lab Unit 12 • Thinking in the Stack	pp. 26–27
Part II • Middle School Coaching Overview	Grades 6–8 (SC-6)	How to Coach 45 Sections & 900 Problems	p. 28
Part II • Middle School Units 1–5 Coaching	Grades 6–8 (SC-6)	Big Ideas Units 1–5 Worked Setups & Keys	pp. 29–33
Part II • Middle School Units 6–10 Coaching	Grades 6–8 (SC-6)	Big Ideas Units 6–10 Worked Setups & Keys	pp. 34–38
Part II • Misconception Matrix & Roster	Grades 6–8 (SC-6)	Middle School Remediation & Mastery Tracker	pp. 39–40
Part III • General Lab Workbook Answer Key	Domains 01–12	Complete Core, Extension & 48 Lab Solutions	pp. 41–42
Part III • Speed & Proficiency Trials Key	Trials 1–12, C1–C3	All 150 Timed Sprint Answers & Keystrokes	pp. 43–44
Part III • Manipulative & Keycap Reference	SC-6 & SC-32	Classroom Kit Guide & Master Keycap Atlas	p. 45
Part III • Yearly Pacing & Observation Log	Teacher Log	Reproducible Concept Pacing & Notes Grid	p. 46
Part III • Send-Home Parent & Family Guide	Family Handout	Photocopiable 'Why No [=] Sign?' Welcome Letter	p. 47

Why There Is No [=] Key: You Already Teach RPN

If you grew up with a standard algebraic calculator, here is why stack-based math matches your classroom teaching.

IF YOU GREW UP WITH AN [=] BUTTON, READ THIS FIRST!

Almost every teacher grew up using an algebraic calculator where you type 8 + 5 =. When educators first see Reverse Polish Notation (RPN)—an [ENTER] key and no [=] button—their natural reaction is hesitation: 'Why is the plus sign at the end? I don't have time to learn or teach a strange calculator system!' Here is the reassuring truth: RPN is not a new way to do math. It is the exact order in which you already teach math with physical blocks and on the whiteboard every day.

1. THE 'NOUN, NOUN, VERB' SECRET: PHYSICAL BLOCKS & PAPER ARE ALREADY RPN

How You Teach with Physical Blocks

1. 'Put 8 blocks on the tray.'
2. 'Now bring 5 more blocks.'
3. 'Now push them together (+)!'

You can't push blocks together before the second group is on the desk!

How You Teach Vertical Paper Math

1. Write 8 on the top line (Line Y).
2. Write 5 on the line below (Line X).
3. Draw the line and add (+ -> 13).

Both numbers exist on paper before the operation happens!

How the 4-Line Stack Calculator Works

1. Type [8] [ENTER] -> 8 sits on Line Y.
2. Type [5] -> 5 sits on Line X.
3. Press [+] -> adds Y + X = 13!

Line Y is the Waiting Shelf; Line X is the Workbench. Zero hidden state!

2. HOW THE STACK CURES THE #1 CLASSROOM CALCULATOR HEADACHE (ORDER OF OPERATIONS)

STANDARD [=] CALCULATOR ON (8 + 4) / (9 - 3):

• The Trap: A student types 8 + 4 / 9 - 3 = left-to-right and gets 5.444... (WRONG!) because the calculator silently divided 4 / 9 first.
• Or Parentheses Soup: Students blindly copy ((8+4)/(9-3)) without knowing which operation happens first.
• Hidden Mistakes: A 1-line screen erases each number as soon as + is pressed, so teachers cannot see where a student miskeyed.

4-LINE STACK CALCULATOR ON (8 + 4) / (9 - 3):

• True Inside-Out PEMDAS: With no parentheses keys, the student MUST decide what to solve first!
• Room 1: [8] [ENTER] [4] [+] -> 12 waits visibly on Line Y.
• Room 2: [9] [ENTER] [3] [-] -> 6 sits visibly on Line X.
• Combine: Both 12 and 6 are visible together! Press [/] -> 12 / 6 = 2. It mirrors paper-and-pencil steps!

3. ANSWERING THE THREE BIGGEST TEACHER WORRIES

Worry 1: 'Will this confuse my students when they take state tests with a standard calculator?'

Answer: The exact opposite—it makes them dramatically better! Because the stack has no parentheses keys, students learn to decompose expressions from the inside out using true PEMDAS priority. When handed a basic state-test calculator, they already know which sub-expression to compute first.

Worry 2: 'What if a student gets stuck mid-problem and asks me for help?'

Answer: The 4-line screen (T, Z, Y, X) is a 2-second teacher X-ray. Instead of a mysterious single number on a screen, you can immediately see their first quantity sitting on Line Y and their second quantity on Line X.

Worry 3: 'How long will it take me to feel comfortable teaching with it?'

Answer: About 3 minutes! Turn to Page iv right now and try the 4 hands-on teacher exercises before your first class.

The 3-Minute Teacher Hands-On RPN Primer

Three golden rules, four 45-second try-it exercises, and a word-for-word Day 1 script to read to your class.

THE 3 GOLDEN RULES OF THE STACK (EVERYTHING YOU NEED TO KNOW)

1. [ENTER] = 'Put It on the Waiting Shelf': Press [ENTER] ONLY to separate two numbers you type back-to-back.

2. Action Keys Work Immediately: Press [+], [-], [x], or [/] AFTER both numbers are in place—never press [ENTER] after an operator!

3. Automatic Stack Lift: When you type a new number after an operation, your previous answer automatically lifts to Line Y!

TRY THESE 4 EXERCISES ON YOUR CALCULATOR RIGHT NOW (3 MINUTES TOTAL)

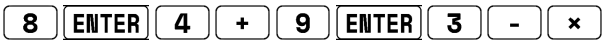
Try-It 1 (30 sec): Basic Addition — 8 + 5 = 13

Type [8] [ENTER] to lift 8 onto Shelf Y. Type [5] on Workbench X. Press [+] -> 8 + 5 = 13!

Press Keys: 

Try-It 2 (45 sec): Two-Group PEMDAS Without Parentheses — (8 + 4) × (9 - 3) = 72

First group: [8] [ENTER] [4] [+] gives 12. Just start typing [9] [ENTER] [3] [-]! 12 lifts to Y while 6 forms on X. Press [x] -> 72!

Press Keys: 

Try-It 3 (45 sec): Two-Dot Fractions on the SC-6 — 1/3 + 1/4 = 7/12

Pressing [.] twice enters a fraction ([.] [1] [.] [3] = 1/3). Press [ENTER], type 1/4, and press [+] -> displays 7/12 as a real fraction!

Press Keys: 

Try-It 4 (45 sec): Fixing Reversed Numbers Without Starting Over — 20 ÷ 4 = 5

Typed 4 then 20 when you meant 20 / 4? Never clear and start over! Press the Swap key to flip Y and X (Y: 20, X: 4), then press [/] -> 5!

Press Keys: 

YOUR 60-SECOND DAY 1 SCRIPT (READ ALOUD TO INTRODUCE THE STACK TO STUDENTS)

"Class, look at your calculator—notice there is no equals sign and no parentheses! That's because this calculator works just like your two hands on your desk. The bottom line, Line X, is your Workbench where you type a number. The line right above it, Line Y, is your Waiting Shelf. When we want to add 8 + 5, we first type 8 and press [ENTER] to push 8 up onto the Waiting Shelf. Then we type 5 on our Workbench, and press [+] to combine them together! Put the two numbers on the shelves first, then press the action button!"

The 4-Step Concrete-to-Stack Teaching Loop

How physical manipulatives and the 4-level RPN stack work together to build deep conceptual understanding.

1. THE 4-STEP CONCRETE–PICTORIAL–ABSTRACT + STACK VERIFICATION LOOP

STEP 1: BOOK STORY & PHYSICAL MANIPULATIVES (10 MIN)

Introduce the concept using the historical or real-world story from the companion book. Students build the problem physically using their classroom blocks (Ten-Frames, 84 mm Fraction Towers, Algebra Area Tiles, or Trig Volvelle) before touching a pencil or calculator.

STEP 2: PICTORIAL SKETCH & WRITTEN PREDICTION (10 MIN)

In their Student Workbook (Hands-On Math Lab or Middle School Math Lab), students sketch their physical block model, write their mathematical strategy, and record their predicted answer before pressing any calculator keys.

STEP 3: RPN STACK EXECUTION & LIVE REGISTER OBSERVATION (10 MIN)

Students enter operands onto the 4-level stack (T, Z, Y, X) on the SC-6 or SC-32 calculator. Because every intermediate result remains visible in Register Y while the next term is built in Register X, the calculator mirrors their physical blocks step by step.

STEP 4: MISCONCEPTION DIAGNOSIS & EXTENSION CHALLENGE (15 MIN)

If a student's calculator result differs from their written prediction, use the Target Misconceptions & Hands-On Remediations table on the second page of each Concept Domain to guide immediate self-correction with the blocks.

2. TURNKEY CLASSROOM & HOMESCHOOL IMPLEMENTATION MODELS

Model A: Core Curriculum Supplement (Singapore Dimensions, Bridges, or Big Ideas Math)

Look up your current textbook lesson in the Cross-Curriculum Pacing Concordances (pp. 1–3). Open this Teacher's Guide to the matching Concept Domain (Part I) or Middle School Unit (Part II), and assign the matching Student Workbook page.

Model B: Weekly Hands-On STEM Math Lab (36-Week Sequence)

Spend 3 weeks per Concept Domain (Domains 1–12): Week 1 = Core Concept Lesson + Guided Walkthrough; Week 2 = Follow-Up Extension + Hands-On Lab Worksheet (Book 2); Week 3 = 5-Minute Speed & Fluency Trial (Book 4).

Model C: Small-Group Intervention & Math Center Rotation (20 Minutes)

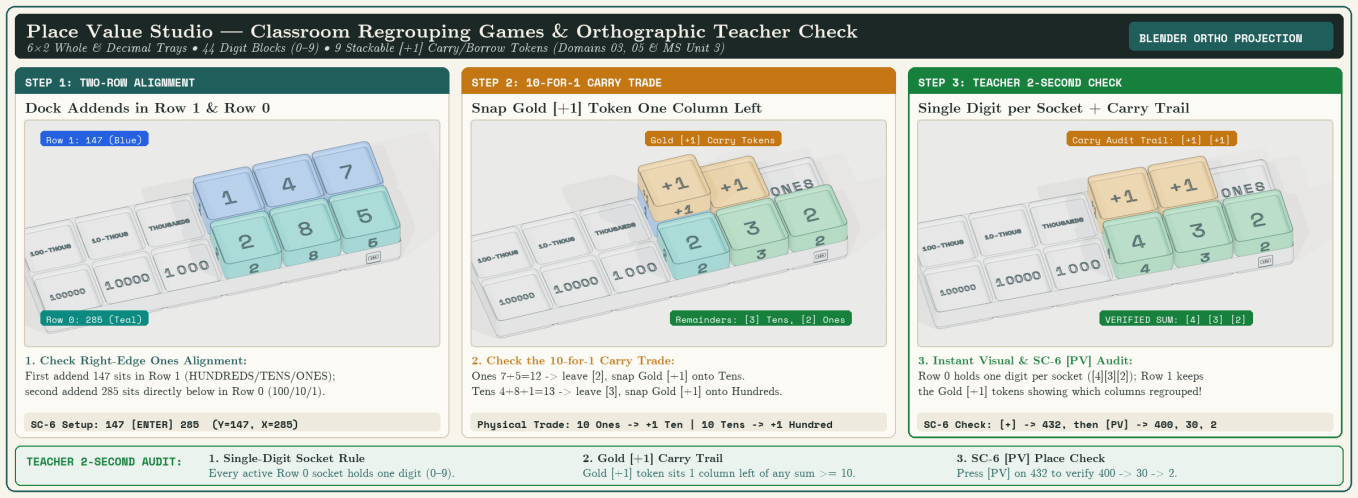
Use the Target Misconceptions & Remediations boxes in Parts I and II to run targeted 20-minute hands-on block intervention stations for students who struggle with abstract symbol manipulation.

Model D: Homeschool & Family Reading + Lab Loop

Read one chapter of From Blocks to the Stack (Grades K–6) or Thinking in the Stack (Grades 7–12) together, build the physical block setup from Part I of this guide, and complete the matching unit in the Hands-On Math Lab student workbook.

Place Value Studio: Games & Orthographic Check

Classroom games and 2-second Blender orthographic visual checks for the 6×2 Whole/Decimal Trays, Digit Blocks (0–9), and [+1] Carry Tokens.



CLASSROOM GAMES & HANDS-ON ACTIVITIES (TIED TO CONCEPT DOMAINS & WORKBOOKS)

Game 1: 'Race to 1,000 Carry Relay' (Domains 03 & 05 • Grades 2–5 • 10-Min Partner Relay)

Partners start at 000 on the 6×2 Whole Number Tray. Take turns rolling two dice (or spinning 0–9) to form a 2-digit or 3-digit addend in Row 0 (100/10/1) and add it to the running total in Row 1. Whenever a column sum reaches 10+, the player calls 'Regroup!', leaves the ones digit in Row 0, and snaps a Gold [+1] Carry Token one column left. First pair to click [1] into THOUSANDS (1000) and verify on the SC-6 wins!

Check Keys (e.g., 147 + 285 = 432): `1` `4` `7` `ENTER` `2` `8` `5` `+` `123`

Game 2: 'The Decimal Point Shifter (×10 & ÷10 Slide)' (Domain 05 & MS Unit 03 • Grades 4–6 • 10 Min)

Dock the Whole Number Tray (100,000 to 1) and Decimal Tray (0.1 to 0.000001) side-by-side at the raised tactile decimal point (•). Call out a starting decimal (e.g., 4.75) and a power-of-10 action (× 10, × 100, or ÷ 10). Students physically slide every digit block one or two columns left (for ×10/×100) or right (for ÷10) across the stationary decimal point—seeing why digits shift columns rather than the point moving!

Check Keys (4.75 × 10 = 47.5): `4` `•` `7` `5` `ENTER` `1` `0` `×` `123`

Game 3: 'The Borrowing Bank Vault (Zero-Placeholder Heist)' (Domain 05 • Grades 3–6 • 12 Min)

Set a 'Vault Target' containing zeros in Row 1 (e.g., 502) and a subtraction code in Row 0 (e.g., 168). Because Ones (2 - 8) and Tens (0 - 6) cannot subtract directly, the 'Banker' partner must un-group 1 Hundred ([5] -> [4]), trade across the zero Tens ([0] -> [9]), and snap the Gold [+1] regrouping token onto the Ones column (12 - 8 = 4) to unlock 334.

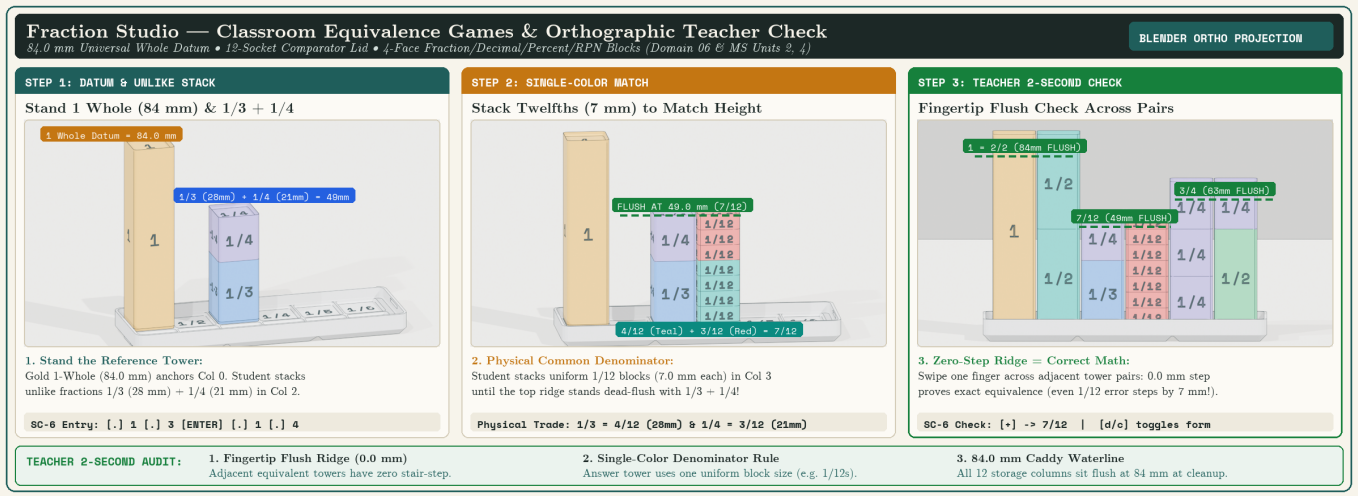
Check Keys (502 − 168 = 334): `5` `0` `2` `ENTER` `1` `6` `8` `-` `123`

HOW TO CHECK STUDENTS ARE DOING IT RIGHT (2-SECOND WALK-BY AUDIT)

- Check 1 — Single-Digit Socket Rule: Glance at Row 0—every active column must hold exactly ONE digit block (0–9). If a student stacked two digit blocks without trading, pause and ask: 'Did this column reach 10?'
- Check 2 — Gold [+1] Carry Trail: Every Gold [+1] token in Row 1 must sit one column to the LEFT of the column that crossed 10.
- Check 3 — SC-6 [PV] Place Decomposition: Have the student press [PV] on their SC-6 answer (432 -> 400 -> 30 -> 2) while pointing to each tray column from left to right.

Fraction Studio: Games & Orthographic Check

Classroom games and 2-second Blender orthographic flush checks for the 84.0 mm Proportional Fraction Towers & 12-Socket Comparator Lid.



CLASSROOM GAMES & HANDS-ON ACTIVITIES (TIED TO CONCEPT DOMAINS & WORKBOOKS)

Game 1: 'The 84 mm Flush Tower Challenge (Make 1 Whole)' (Domain 06 • Grades 3–5 • 10 Min)

Stand the Gold 1-Whole block (84.0 mm) in Col 0 of the 12-Socket Comparator Lid. Call out a constraint—e.g., 'Build 1 Whole in Col 1 using exactly 4 blocks from at least 3 different denominator colors!' (e.g., 1/2 + 1/4 + 1/6 + 1/12 = 42 + 21 + 14 + 7 = 84.0 mm). Students swipe a finger across the top ridge to prove 0.0 mm step, then verify with two-dot fraction entry on the SC-6.

Check Keys (1/2 + 1/4 + 1/6 + 1/12 = 1): [.] [1] [.] [2] [ENTER] [.] [1] [.] [4] [+] [.] [1] [.] [6] [+] [.] [1] [.] [1] [2] [+]

Game 2: 'Two-Dot Common Denominator Duel (1/3 + 1/4 = 7/12)' (Domain 06 & MS Unit 02 • Grades 4–6)

Partner A stacks two unlike fractions in Col 2 (e.g., 1/3 [28 mm] + 1/4 [21 mm] = 49.0 mm). Partner B must find a SINGLE-COLOR block family (twelfths, 7.0 mm each) and stack enough identical blocks in Col 3 (seven 1/12 blocks = 49.0 mm) to stand 100% flush with Partner A's tower. Both partners key the sum on the SC-6 and press [d/c] to inspect proper/improper forms.

Check Keys (1/3 + 1/4 = 7/12): [.] [1] [.] [3] [ENTER] [.] [1] [.] [4] [+] [f] [d/c]

Game 3: 'Mystery Tower Blind Fingertip Audit & 4-Face Spin' (Domain 06 & MS Unit 04 • Grades 3–6)

Student A builds a target fraction tower (e.g., three 1/4 blocks = 3/4 = 63.0 mm) and challenges Student B to build an equal-height tower using NO fourths (e.g., 1/2 + 1/4, or four 1/6 + one 1/12 = 63.0 mm). Student A closes their eyes and runs one fingertip across both tops to judge if they are flush, then both rotate a block 90° at a time to read Fraction (1/4), Decimal (0.25), Percent (25%), and RPN Key (.1.4).

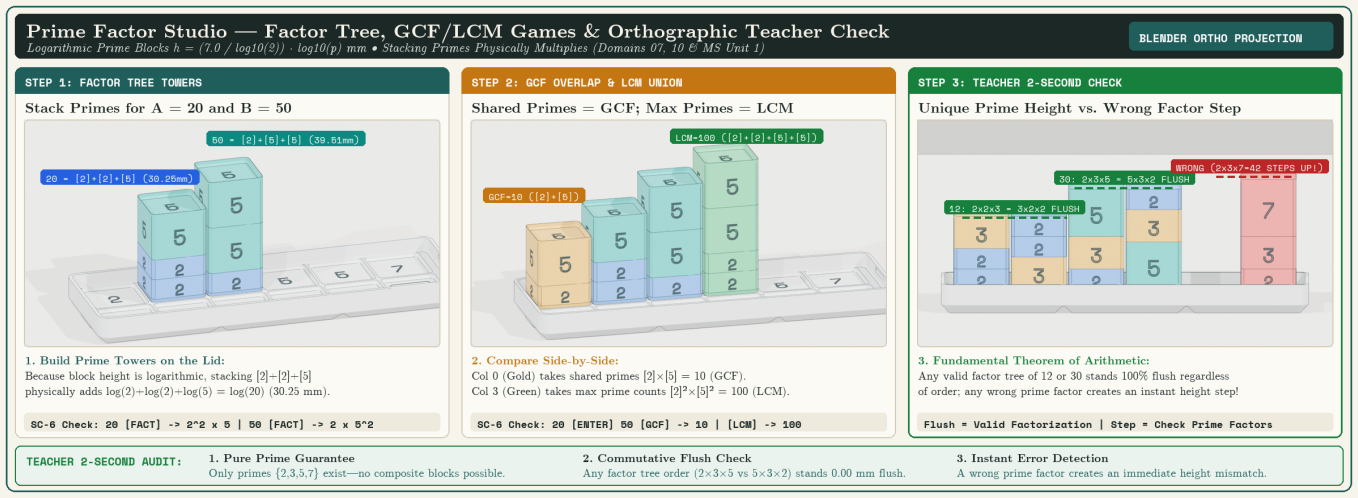
Check Keys (4/6 + 1/12 = 3/4): [.] [4] [.] [6] [ENTER] [.] [1] [.] [1] [2] [+]

HOW TO CHECK STUDENTS ARE DOING IT RIGHT (2-SECOND WALK-BY AUDIT)

- Check 1 — The 0.0 mm Fingertip Flush Test: Even the smallest fraction error (1/12) creates a bold 7.0 mm physical stair-step! If two adjacent towers are dead-flush across the top ridge, the student's equivalence or common denominator is 100% mathematically correct.
- Check 2 — Single-Color Answer Tower: When adding unlike fractions (1/3 + 1/4), check that the student's answer tower in the adjacent socket uses ONE uniform block color (seven 1/12 blocks) rather than mixed denominators.
- Check 3 — 3-Second Cleanup Waterline: When packed into the 12-column caddy, every column sits dead-flush at 84.0 mm—any missing block leaves an obvious gap.

Prime Factor Studio: Games & Orthographic Check

Classroom games and 2-second Blender orthographic checks for the Logarithmic Prime Factor Blocks (h = k·log10(p)) & 12-Socket Comparator Lid.



CLASSROOM GAMES & HANDS-ON ACTIVITIES (TIED TO CONCEPT DOMAINS & WORKBOOKS)

Game 1: 'The Commutative Skyscraper Race (2 × 3 × 5 = 30)' (Domains 07 & 10 • Grades 3–6 • 10 Min)

Call out a composite number (e.g., 12, 30, or 42). Partner A factors it using one factor tree (e.g., 30 = 6 × 5 -> [2] + [3] + [5]) and stacks the prime blocks in Col 2. Partner B uses a different factor tree (30 = 10 × 3 -> [5] + [2] + [3]) in Col 3. Because block height is proportional to log10(p), both towers stand 100% flush at 34.35 mm—proving commutativity and unique factorization physically!

Check Keys (Factor 30 -> 2 × 3 × 5): `3` `0` `ENTER` `g` `人`

Game 2: 'Impossible Impostor (Fundamental Theorem Challenge)' (Domain 10 & MS Unit 01 • Grades 5–8)

Student A builds a prime tower (e.g., 12 = [2] + [2] + [3] = 25.10 mm) and challenges Student B to match its exact height using DIFFERENT prime blocks (e.g., trying [5] + [2] = 10 = 23.25 mm or [2] + [7] = 14 = 26.65 mm). Students quickly discover that NO other combination of primes will ever stand flush—experiencing the Fundamental Theorem of Arithmetic in their hands!

Check Keys (Compare 12 vs 14): `1` `2` `g` `人` `1` `4` `g` `人`

Game 3: 'Side-by-Side GCF Overlap & LCM Union Duel' (Domain 10 & MS Unit 01 • Grades 5–8 • 12 Min)

Given A = 20 and B = 50, students stack Tower A ([2]+[2]+[5]) in Col 1 and Tower B ([2]+[5]+[5]) in Col 2. To build GCF(20,50) = 10 in Col 0, they copy only the SHARED prime blocks present in both towers ([2]+[5]). To build LCM(20,50) = 100 in Col 3, they stack the tallest count of each prime ([2]+[2]+[5]+[5]), then verify [FACT], [GCF], and [LCM] on the SC-6!

Check Keys (GCF & LCM of 20 and 50): `2` `0` `ENTER` `5` `0` `GCF` `2` `0` `ENTER` `5` `0` `f` `LCM`

HOW TO CHECK STUDENTS ARE DOING IT RIGHT (2-SECOND WALK-BY AUDIT)

- Check 1 — Pure Prime Guarantee: Because the kit contains ONLY prime blocks ({2, 3, 5, 7}), students can never stop a factor tree early at a composite number like 4, 6, 8, or 9.
- Check 2 — Logarithmic Flush Check: Any two valid factor trees for the same number stand dead-flush (0.000 mm step), while any wrong prime factor creates an immediate height step (see red tower in Step 3 above).
- Check 3 — GCF + LCM Height Symmetry: On the comparator lid, GCF is always the shortest tower (left), LCM is the tallest tower (right), and h(GCF) + h(LCM) = h(A) + h(B).

CROSS-CURRICULUM PACING & ALIGNMENT INDEX

Singapore Dimensions Math Pacing Concordance

Locate your Dimensions Math PK–8 book, chapter, and lesson to find the matching Concept Domain in Part I and Student Lab Unit in Book 2.

Grade	Textbook Lesson	Mathematical Topic	Manipulative Kit	Teacher Guide	Student Lab
Pre-K	Dimensions PK Book A, Ch 2, Lesson 2.1	**Counting to 5 with Concrete Items**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Pre-K	Dimensions PK Book B, Ch 4, Lesson 4.2	**Comparing Quantities: More or Less**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Kindergarten	Dimensions KA, Ch 1, Lesson 1.2	**One-to-One Counting & Frames**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Kindergarten	Dimensions KA, Ch 2, Lesson 2.3	**Zero and Empty Sets**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Kindergarten	Dimensions KA, Ch 4, Lesson 4.1	**Number Bonds of 5**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Kindergarten	Dimensions KB, Ch 7, Lesson 7.2	**Addition within 10 on the Stack**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Kindergarten	Dimensions KB, Ch 8, Lesson 8.1	**Subtraction as Missing Part**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Grade 1	Dimensions 1A, Ch 1, Lesson 1.3	**Counting & Comparing Numbers to 10**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Grade 1	Dimensions 1A, Ch 2, Lesson 2.1	**Making Number Bonds to 6 and 8**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Grade 1	Dimensions 1A, Ch 2, Lesson 2.2	**Making Number Bonds to 10**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Grade 1	Dimensions 1A, Ch 2, Lesson 2.4	**Decomposing Numbers & Missing Parts**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Grade 1	Dimensions 1A, Ch 3, Lesson 3.1	**Addition as Part-Part-Whole**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Grade 1	Dimensions 1A, Ch 4, Lesson 4.2	**Subtraction as Comparing Sets**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Grade 1	Dimensions 1A, Ch 7, Lesson 7.1	**Ten-Frames & Place Value (11-20)**	Ten-Frame Tray & Counters	Dom 03 (pp.8–9)	Lab Unit 03
Grade 1	Dimensions 1A, Ch 8, Lesson 8.1	**Making 10 to Add**	Ten-Frame Tray & Counters	Dom 03 (pp.8–9)	Lab Unit 03
Grade 1	Dimensions 1A, Ch 8, Lesson 8.3	**Bridging Through 10**	Ten-Frame Tray & Counters	Dom 03 (pp.8–9)	Lab Unit 03
Grade 1	Dimensions 1B, Ch 12, Lesson 12.1	**Multiplication as Equal Groups**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 1	Dimensions 1B, Ch 14, Lesson 14.1	**Halves & Fourths (Equal Shares)**	84 mm Fraction Towers	Dom 06 (pp.14–15)	Lab Unit 06
Grade 2	Dimensions 2A, Ch 2, Lesson 2.2	**Mental Addition with Number Bonds**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Grade 2	Dimensions 2A, Ch 4, Lesson 4.1	**Doubles & Multiplying by 2**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 2	Dimensions 2B, Ch 6, Lesson 6.3	**Near-Doubles Strategies (Doubles + 1)**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 2	Dimensions 2B, Ch 7, Lesson 7.1	**Unit Fractions (Halves, Thirds, Fourths)**	84 mm Fraction Towers	Dom 06 (pp.14–15)	Lab Unit 06
Grade 3	Dimensions 3A, Ch 2, Lesson 2.4	**Multi-Digit Addition on Stack**	Base-10 Place Value Blocks	Dom 05 (pp.12–13)	Lab Unit 05
Grade 3	Dimensions 3A, Ch 3, Lesson 3.5	**Four Operations & Word Problems**	Base-10 Place Value Blocks	Dom 05 (pp.12–13)	Lab Unit 05

How to Use the Singapore Math Concordance

Turn to the listed Domain pages in Part I of this Teacher's Guide for the lesson plan, stack trace, and misconception diagnostics. Assign the matching Unit in Book 2 (Hands-On Math Lab) or Book 3 (Middle School Math Lab) for student practice.

CROSS-CURRICULUM PACING & ALIGNMENT INDEX

Bridges in Mathematics Pacing Concordance

Locate your Bridges K–5 unit, module, and session to find the matching Concept Domain in Part I and Student Lab Unit in Book 2.

Grade	Textbook Lesson	Mathematical Topic	Manipulative Kit	Teacher Guide	Student Lab
Kindergarten	Unit 1, Mod 1, Session 1	**Welcome to Math: Counting Objects**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Kindergarten	Unit 1, Mod 1, Session 2	**Counting & Five-Frames**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Kindergarten	Unit 1, Mod 2, Session 1	**Ten-Frames & Counting On**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Kindergarten	Unit 1, Mod 2, Session 3	**Zero & The Empty Ten-Frame**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Kindergarten	Unit 1, Mod 3, Session 2	**The Step Counter: Adding One More**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Kindergarten	Unit 2, Mod 1, Session 2	**Comparing Sets: Which Has More?**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Kindergarten	Unit 2, Mod 2, Session 3	**Number Bonds to 5**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Kindergarten	Unit 4, Mod 2, Session 1	**Part-Part-Whole Addition**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Grade 1	Unit 1, Mod 2, Session 3	**Ten-Frames & Subitizing**	Ten-Frame Tray & Counters	Dom 03 (pp.8–9)	Lab Unit 03
Grade 1	Unit 2, Mod 1, Session 2	**Combinations to Ten**	Ten-Frame Tray & Counters	Dom 03 (pp.8–9)	Lab Unit 03
Grade 1	Unit 2, Mod 1, Session 4	**Finding the Missing Addend**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Grade 1	Unit 2, Mod 3, Session 3	**Making 10 Addition Strategies**	Ten-Frame Tray & Counters	Dom 03 (pp.8–9)	Lab Unit 03
Grade 1	Unit 3, Mod 1, Session 1	**Introducing the Number Rack**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 1	Unit 3, Mod 1, Session 2	**Double Flips on the Number Rack**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 1	Unit 3, Mod 1, Session 3	**Doubles Plus or Minus One**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 1	Unit 3, Mod 2, Session 1	**5-Structure on the Rack**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 1	Unit 6, Mod 1, Session 2	**Two-Row Rack Comparisons**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 1	Unit 7, Mod 1, Session 3	**Place Value: Tens & Ones**	Ten-Frame Tray & Counters	Dom 03 (pp.8–9)	Lab Unit 03
Grade 2	Unit 1, Mod 1, Session 2	**Number Rack Fact Fluency**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 2	Unit 1, Mod 1, Session 4	**Near-Doubles Fact Strategies**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 2	Unit 1, Mod 2, Session 1	**Combinations to 20**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 2	Unit 3, Mod 2, Session 4	**Two-Digit Subtraction on Stack**	Base-10 Place Value Blocks	Dom 05 (pp.12–13)	Lab Unit 05
Grade 3	Unit 1, Mod 2, Session 3	**Multi-Step Addition & Subtraction**	Base-10 Place Value Blocks	Dom 05 (pp.12–13)	Lab Unit 05
Grade 3	Unit 2, Mod 1, Session 2	**Multiplication Arrays & Groups**	Base-10 Place Value Blocks	Dom 05 (pp.12–13)	Lab Unit 05

How to Use the Bridges in Mathematics Concordance

Turn to the listed Domain pages in Part I of this Teacher's Guide for the lesson plan, stack trace, and misconception diagnostics. Assign the matching Unit in Book 2 (Hands-On Math Lab) or Book 3 (Middle School Math Lab) for student practice.

CROSS-CURRICULUM PACING & ALIGNMENT INDEX

Big Ideas Math & AP Standards Pacing Concordance

Locate your grade, chapter, and section to find the matching Concept Domain in Part I (plus Part II for Grades 6–8).

Grade	Textbook Lesson	Mathematical Topic	Manipulative Kit	Teacher Guide	Student Lab
Kindergarten	Chapter 1, Section 1.2	**Count and Write 1 to 5**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Kindergarten	Chapter 3, Section 3.1	**Model and Count 6 to 10**	Ten-Frame Tray & Counters	Dom 01 (pp.4–5)	Lab Unit 01
Kindergarten	Chapter 5, Section 5.1	**Number Bonds to 10 on the Frame**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Kindergarten	Chapter 7, Section 7.2	**Represent Addition as Putting Together**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Grade 1	Chapter 1, Section 1.4	**Part-Part-Whole Addition**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Grade 1	Chapter 2, Section 2.3	**Think Addition to Subtract**	Math Dominoes & 12-Frame Tray	Dom 02 (pp.6–7)	Lab Unit 02
Grade 1	Chapter 3, Section 3.1	**Doubles Strategies with ENTER**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 1	Chapter 3, Section 3.2	**Doubles Plus 1 & Doubles Minus 1**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 1	Chapter 3, Section 3.4	**Make a 10 to Add on Ten-Frame**	Ten-Frame Tray & Counters	Dom 03 (pp.8–9)	Lab Unit 03
Grade 1	Chapter 4, Section 4.2	**Use 10 to Subtract**	Ten-Frame Tray & Counters	Dom 03 (pp.8–9)	Lab Unit 03
Grade 2	Chapter 1, Section 1.2	**Mental Math: Doubles & Near-Doubles**	Two-Row Number Rack	Dom 04 (pp.10–11)	Lab Unit 04
Grade 2	Chapter 1, Section 1.4	**Make a Ten Addition Strategies**	Ten-Frame Tray & Counters	Dom 03 (pp.8–9)	Lab Unit 03
Grade 2	Chapter 3, Section 3.3	**Two-Digit Addition with Lift**	Base-10 Place Value Blocks	Dom 05 (pp.12–13)	Lab Unit 05
Grade 3	Chapter 1, Section 1.3	**Multiplication as Repeated Addition**	Base-10 Place Value Blocks	Dom 05 (pp.12–13)	Lab Unit 05
Grade 3	Chapter 8, Section 8.2	**Equal Parts & Unit Fractions**	84 mm Fraction Towers	Dom 06 (pp.14–15)	Lab Unit 06
Grade 3	Chapter 8, Section 8.5	**Equivalent Fractions on Blocks**	84 mm Fraction Towers	Dom 06 (pp.14–15)	Lab Unit 06
Grade 4	Chapter 3, Section 3.4	**Solve Multi-Step Word Problems**	4-Register Stack Board	Dom 05 (pp.12–13)	Lab Unit 05
Grade 4	Chapter 3, Section 3.5	**Order of Operations Without Parens**	4-Register Stack Board	Dom 05 (pp.12–13)	Lab Unit 05
Grade 4	Chapter 8, Section 8.3	**Add & Subtract Common Fractions**	84 mm Fraction Towers	Dom 06 (pp.14–15)	Lab Unit 06
Grade 4	Chapter 8, Section 8.6	**Mixed Numbers: Two-Dot Entry**	84 mm Fraction Towers	Dom 06 (pp.14–15)	Lab Unit 06
Grade 5	Chapter 5, Section 5.2	**Add Fractions with Unlike Denominators**	84 mm Fraction Towers	Dom 06 (pp.14–15)	Lab Unit 06
Grade 5	Chapter 6, Section 6.2	**Multiply Fractions & Height Scaling**	84 mm Fraction Towers	Dom 06 (pp.14–15)	Lab Unit 06
Grade 5	Chapter 6, Section 6.5	**Divide Fractions with Reciprocals**	84 mm Fraction Towers	Dom 06 (pp.14–15)	Lab Unit 06
Grade 5	Chapter 12, Section 12.1	**Graphing Points on Coordinate Plane**	Algebra Area Tiles	Dom 08 (pp.18–19)	Lab Unit 08

How to Use the Big Ideas Math Concordance

Turn to the listed Domain pages in Part I of this Teacher's Guide for the lesson plan, stack trace, and misconception diagnostics. Assign the matching Unit in Book 2 (Hands-On Math Lab) or Book 3 (Middle School Math Lab) for student practice.

1-to-1 Correspondence: The Concrete Step Counter

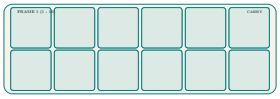
Core Concept Lesson • Bridges in Mathematics (Numbers to 5 & 10) • Standard: CCSS.MATH.K.CC.B.4

COMPANION BOOK STORY: How Ten Fingers & an Empty Shell Created Place Value & The 1-to-10 Staircase

Long before paper or keyboards existed, ancient shepherds counted flocks with clay pebbles ('calculi' in Latin—the root of 'calculator'!) and Maya astronomers drew a shell for zero so an empty tray had its own symbol. Dropping 1 counter at a time into the Ten-Frame while pressing [1] [+] builds the exact 1-to-10 Staircase from Chapter 3 of From Blocks to the Stack.

60-SEC BELL-RINGER HOOK: Close your eyes and clap slowly 4 times. Before opening your eyes, how many pegs are on the Ten-Frame and what number waits on Line X?

1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION



Kit: Ten-Frame Tray & Counting Pegs

Objective: Establish one-to-one correspondence between physical pegs on the frame and calculator screen increments.

Hands-On Setup: Place 1 peg into the first well of the Ten-Frame. Have students observe the single object.

Predict First (Ask Before Keys): When we place 1 more peg into the next well, how many pegs will be on our frame? (Prediction: 2)

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"Place 1 peg in the tray first, then press [1] [+] to match the peg you just touched."

2B. AVOID SAYING (RULE WITHOUT REASON):

"Just press 1 + 1 + 1 as fast as you can until the screen says 5."

3. CANONICAL SC-6 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: [1] [ENTER] [1] [+] [1] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[1]	1	First peg placed: 1 enters register X
[ENTER]	1	1 copied to register Y: holds the current count
[1]	1	Second peg placed: 1 enters register X
[+]	2	Press [+]: 1 + 1 = 2! Screen verifies 2 pegs on board
[1]	1	Third peg placed in frame: 1 enters X
[+]	3	Press [+]: 2 + 1 = 3! Frame and screen match perfectly

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

Each physical counter corresponds to a single increment. The calculator's stack preserves the running count in Y while new items enter X.

Core Guided Investigation: Place 4 pegs on your Ten-Frame. Add 1 more peg. Write the keystrokes to show 4 + 1 on the calculator.

Expected Answer: 5 pegs.

Verification Keys: [4] [ENTER] [1] [+]

More, Less & Zero: Finding Differences

Follow-Up Extension (Singapore Math • CCSS.MATH.K.CC.C.6) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Place 5 pegs in the top row (Row 1) and 2 pegs in the bottom row (Row 2). Prediction: Look at the two rows. Which row has more? How many more pegs are in Row 1 than Row 2?

Extension Key Sequence: [5] [ENTER] [2] [-]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[5]	5	Row 1 quantity (5) enters register X
[ENTER]	5	5 lifted to register Y (first set)
[2]	2	Row 2 quantity (2) enters register X
[-]	3	Minus subtracts X from Y: Row 1 has 3 more pegs!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Reciting number names faster than placing physical counters (losing 1-to-1 correspondence).

Hands-On Block & Stack Fix: Have the student drop 1 peg into the Ten-Frame well first, say the number aloud, and only then press [1] [+] so the physical peg and Register X stay locked in step.

Misconception 2: Confusing the action of adding 1 more with the total quantity on the tray (cardinality).

Hands-On Block & Stack Fix: Ask the student to sweep their hand around all pegs in the Ten-Frame and compare that total to Register X on the screen.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 01 ANSWERS & KEYS)

Lab P1: Place 3 pegs on your frame. Add 2 more. Write the keystrokes to find the total. **Ans: 5**

Check Keys: [3] [ENTER] [2] [+]

Lab P2: A frame has 5 pegs. If you remove 2 pegs, how many remain? **Ans: 3**

Check Keys: [5] [ENTER] [2] [-]

Lab P3: Clear the calculator by pressing [C]. What number represents an empty ten-frame? **Ans: 0 (Zero)**

Check Keys: [C]

Lab P4: Step count by ones starting at 1: [1] [ENTER] [1] [+]. Press [1] [+] two more times. What is X? **Ans: 4**

Check Keys: [1] [+] [1] [+]

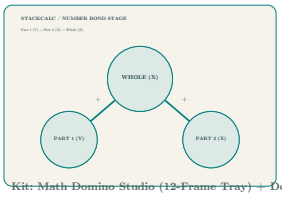
Composing Numbers: Part + Part = Whole

Core Concept Lesson • Singapore Math (Number Bonds & Part-Part-Whole) • Standard: CCSS.MATH.1.OA.C.6

COMPANION BOOK STORY: Spotting Number Partners at a Glance: Subitizing Domino Pips & Friends of Ten

Your brain has a built-in superpower called subitizing (from the Latin 'subitus', meaning 'sudden'): you can recognize 1 to 5 domino pips instantly without counting one by one! In From Blocks to the Stack, the left half of a domino sits on Shelf Y (Part 1), the right half sits on Shelf X (Part 2), and pressing [+] snaps both halves together into the Whole.

60-SEC BELL-RINGER HOOK: Cover the right half of a 7-dot domino with your thumb so only 4 dots show. How does the stack uncover the hidden dots?



Kit: Math Domino Studio (12-Frame Tray) + Domino Tiles

1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION

Objective: Physically compose parts into a whole and connect to RPN binary addition.

Hands-On Setup: Place the [4|3] domino in the 12-frame tray. Count 4 pips on the left (Part 1) and 3 pips on the right (Part 2).

Predict First (Ask Before Keys): Combine both parts on the tray: 4 + 3 = 7 total pips.

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"4 dots sit on Waiting Shelf Y and 3 dots sit on Workbench X—press [+] to slide both parts into the Whole!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Move the 4 across the equals sign and change its sign to minus."

3. CANONICAL SC-6 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: 4 ENTER 3 +

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
4	4	Part 1 (4) enters register X
ENTER	4	4 copied up to register Y; represents Part 1
3	3	Part 2 (3) enters register X
+	7	Plus combines Part 1 (Y) + Part 2 (X) = Whole (7) in X!

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

The RPN stack has dedicated registers for parts: register Y holds Part 1, register X holds Part 2, and [+] merges them into the Whole.

Core Guided Investigation: Pick the [5|4] domino tile. Place it on your 12-frame tray to find the whole. Write the keystrokes.

Expected Answer: 9 pips.

Verification Keys: 5 ENTER 4 +

Decomposing the Whole: Finding the Missing Part

Follow-Up Extension (Bridges in Mathematics • CCSS.MATH.1.OA.B.4) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Place 10 counters in the 12-frame tray. Cover or slide 6 counters to represent known Part 1. Prediction: How many counters remain uncovered in the tray? (10 - 6 = 4).

Extension Key Sequence: [1] [0] [ENTER] [6] [-]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	Enter Whole (10)
[0]	10	10 in register X
[ENTER]	10	Whole (10) lifted to register Y
[6]	6	Known Part 1 (6) enters register X
[-]	4	Minus subtracts known Part from Whole: Missing Part is 4!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Pressing [+] immediately after the first number without pressing [ENTER] to separate Part 1 and Part 2.

Hands-On Block & Stack Fix: Point to the center divider line on the Math Domino: pressing [ENTER] is the physical divider that pushes Part 1 up to Register Y before Part 2 is typed into Register X.

Misconception 2: Adding the whole and the known part when solving for a missing addend (e.g., computing 10 + 6 instead of 10 − 6).

Hands-On Block & Stack Fix: Place 10 counters in the tray and cover 6; show that the Whole (10) goes into Register Y, the known part (6) enters X, and [–] reveals the hidden part.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 02 ANSWERS & KEYS)

Lab P1: Make 6: Part 1 is 2. What is Part 2? Write the keystrokes. **Ans: 4**

Check Keys: [6] [ENTER] [2] [-]

Lab P2: Part 1 is 7, Part 2 is 2. What is the Whole? **Ans: 9**

Check Keys: [7] [ENTER] [2] [+]

Lab P3: Three parts make a whole: 2, 3, and 4. Add them on the stack. **Ans: 9**

Check Keys: [2] [ENTER] [3] [+] [4] [+]

Lab P4: The Whole is 10. Part 1 is 7. What is Part 2? **Ans: 3**

Check Keys: [1] [0] [ENTER] [7] [-]

Complements to 10: Making Ten

Core Concept Lesson • Big Ideas Math (Addition Strategies to 10) • Standard: CCSS.MATH.1.OA.C.6

COMPANION BOOK STORY: Making a Ten First (8 + 5 = 10 + 3 = 13) & The Double-Height '10' Carry Block Trade

Why is 10 + 3 so much easier for our brains than 8 + 5? Because base-10 place value groups everything into full tens! In Chapter 3 of From Blocks to the Stack, students slide 2 dots from the right tray into the 2 empty pockets of the left tray to complete a full 10, then physically swap all 10 loose dots for one double-height '10' Carry Block.

60-SEC BELL-RINGER HOOK: Flash a Ten-Frame holding 8 pegs for 1 second, then hide it. How did your eyes know it was 8 without counting 1-by-1?



1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION

Objective: Quickly determine complements to 10 using physical 2x5 array subitizing and RPN subtraction.

Hands-On Setup: Place 7 unnumbered counters into the Ten-Frame: fill the top row of 5, then place 2 in the bottom row.

Predict First (Ask Before Keys): Look at the empty wells without counting one-by-one. 3 empty spaces mean 7 + 3 = 10.

Studio Games: p. vi • Two 10-Frame Trays (2x5 each, 10 slots) + 3x1 Main Storage Bin (12 slots) + 20 Unnumbered Dots + 2 Double-Height '10' Carry Blocks

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"How many empty wells do you see in the Ten-Frame? Let's borrow 2 pegs from 5 to fill the 10-tray first!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Memorize 8 + 5 = 13 from your flash cards."

3. CANONICAL SC-6 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: `1` `0` `ENTER` `7` `-`

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
`1`	1	Enter 1
`0`	10	Full frame capacity (10) in X
`ENTER`	10	10 lifted to Y
`7`	7	Occupied counters (7) in X
`-`	3	10 - 7 = 3: Exactly 3 empty wells to make 10!

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

The ten-frame physically structures 10 as two groups of 5. Subtraction verifies the visual complement.

Core Guided Investigation: A ten-frame has 6 counters. How many more counters are needed to make 10?

Expected Answer: 4 counters.

Verification Keys: `1` `0` `ENTER` `6` `-`

Bridging Through 10: The Make-Ten & Carry Strategy

Follow-Up Extension (Singapore Math • CCSS.MATH.1.OA.C.6) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Place 8 unnumbered counters in Ten-Frame 1. Prepare 5 counters beside the board. Prediction: Take 2 from the 5 to fill Ten-Frame 1 to 10. Trade the full frame for one '10' block in the Carry Bay! 3 loose counters remain. 10 + 3 = 13.

Extension Key Sequence: [8] [ENTER] [2] [+] [3] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[8]	8	8 counters on frame
[ENTER]	8	8 lifted to Y
[2]	2	2 counters borrowed to make 10
[+]	10	8 + 2 = 10 (Full frame traded for '10' carry block!)
[3]	3	Remaining 3 counters
[+]	13	10 + 3 = 13 total!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Counting on by ones across 10 (e.g., 8 + 5 as 9, 10, 11, 12, 13) instead of decomposing 5 into 2 + 3 to complete the Ten-Frame.

Hands-On Block & Stack Fix: Have the student physically slide 2 counters from the second group into the 2 empty wells of the first Ten-Frame, then key [8] [ENTER] [2] [+] [3] [+].

Misconception 2: Reading teen numbers as two separate single digits (seeing 14 as 1 and 4 rather than 1 full Ten-Frame plus 4 ones).

Hands-On Block & Stack Fix: Snap a full 10-tray beside 4 loose counters and key [10] [ENTER] [4] [+] so students see 14 formed from 10 in Register Y and 4 in Register X.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 03 ANSWERS & KEYS)

Lab P1: What is the complement to 10 for 4? Write the keystrokes. **Ans: 6**

Check Keys: [1] [0] [ENTER] [4] [-]

Lab P2: Solve 7 + 5 using Make-Ten: first make 10, then add 2. **Ans: 12**

Check Keys: [7] [ENTER] [3] [+] [2] [+]

Lab P3: A ten-frame has 9 counters. How many to make 10? **Ans: 1**

Check Keys: [1] [0] [ENTER] [9] [-]

Lab P4: Solve 8 + 6 using Make-Ten on the calculator. **Ans: 14**

Check Keys: [8] [ENTER] [2] [+] [4] [+]

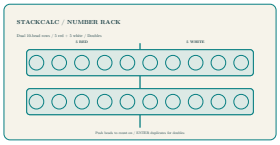
Doubles Strategy: The ENTER Duplicate Key

Core Concept Lesson • Bridges in Mathematics (Number Racks & Fact Strategies) • Standard: CCSS.MATH.1.OA.C.6

COMPANION BOOK STORY: Two Hands, One Workbench: Why 5+5 Bead Rows Mirror the Stack & Instant [ENTER] [+] Doubles

Designed in the Netherlands by math educator Adrian Treffers, the Rekenrek ('calculating rack') splits each row into 5 red and 5 white beads—mirroring the 5 fingers on your left and right hands. In Chapters 1 & 2 of From Blocks to the Stack, pressing [ENTER] clones Shelf X onto Shelf Y just like matching the bottom bead row to the top row for instant doubles!

60-SEC BELL-RINGER HOOK: Slide 6 beads on the top row and 6 on the bottom row in one motion. Which single key clones a number on the calculator?



Kit: Two-Row Number Rack (Rekenrek)

1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION

Objective: Model doubles physically using dual bead rows and computationally using ENTER duplication.

Hands-On Setup: Push 6 beads on the top row (5 red, 1 white). Push 6 beads on the bottom row.

Predict First (Ask Before Keys): Observe the matching rows. 6 + 6 = (5 + 5) + (1 + 1) = 10 + 2 = 12.

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"Press [6] [ENTER]—look how [ENTER] clones your top row of 6 onto Shelf Y so both rows are ready to add!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Press 6, then plus, then 6, then equals."

3. CANONICAL SC-6 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: [6] [ENTER] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[6]	6	Top row quantity (6) enters register X
[ENTER]	6	ENTER duplicates 6 into register Y! Both rows now on stack
[+]	12	Plus combines top and bottom rows: 6 + 6 = 12!

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

ENTER creates an instant identical copy of register X in register Y. This perfectly represents a physical double.

Core Guided Investigation: Push 7 beads on each row of the Number Rack. Use [ENTER] [+] to double it on the calculator.

Expected Answer: 14 beads.

Verification Keys: [7] [ENTER] [+]

Near-Doubles: Doubles Plus One

Follow-Up Extension (Bridges in Mathematics • CCSS.MATH.2.OA.B.2) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Push 7 beads on top row (5 red, 2 white). Push 8 beads on bottom row (5 red, 3 white). Prediction: 7 and 8 is double 7 plus 1 more: 14 + 1 = 15.

Extension Key Sequence: [7] [ENTER] [+] [1] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[7]	7	Enter base number 7
[ENTER]	7	Duplicate 7
[+]	14	Double: 7 + 7 = 14
[1]	1	Near-double difference (1)
[+]	15	14 + 1 = 15!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Retyping the number to double it (e.g., [7] [ENTER] [7] [+]) instead of recognizing that [ENTER] duplicates Register X into Register Y.

Hands-On Block & Stack Fix: Slide 7 beads on both rows of the Number Rack simultaneously, then press [7] [ENTER] and point to the screen: both Y and X already hold 7, so pressing [+] doubles it immediately.

Misconception 2: Confusing '3 groups of 4' with '3 + 4' when transitioning from repeated addition to multiplication.

Hands-On Block & Stack Fix: Build 3 rows of 4 beads on the rack; have the student key [4] [ENTER] [4] [+] [4] [+] = 12 first, then compare directly with [4] [ENTER] [3] [×] = 12.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 04 ANSWERS & KEYS)

Lab P1: Calculate double 8 using [ENTER] [+]. **Ans: 16**

Check Keys: [8] [ENTER] [+]

Lab P2: Solve near-double 8 + 9: double 8 then add 1. **Ans: 17**

Check Keys: [8] [ENTER] [+] [1] [+]

Lab P3: Calculate double 9 using [ENTER] [+]. **Ans: 18**

Check Keys: [9] [ENTER] [+]

Lab P4: Solve 5 + 6 using doubles plus one. **Ans: 11**

Check Keys: [5] [ENTER] [+] [1] [+]

Multi-Step Word Problems Without Parentheses

Core Concept Lesson • Singapore Math (Multi-Step Word Problems) • Standard: CCSS.MATH.4.OA.A.3

COMPANION BOOK STORY: Inside the 4-Floor Number Elevator (T, Z, Y, X) & Young Gauss's 10-Second Shortcut

In 1786, 9-year-old Carl Friedrich Gauss stunned his teacher by adding 1 through 100 in seconds—folding the list into 50 pairs of 101! In Chapters 2 & 4 of From Blocks to the Stack, students discover how the 4-Floor Number Elevator (Shelves X, Y, Z, T) automatically lifts finished answers up to Shelf Y when you type the next number, solving multi-step stories with zero parentheses.

60-SEC BELL-RINGER HOOK: Beat the Smartphone Challenge: Ask a volunteer to type 8 + 4 × 5 left-to-right on a basic phone calculator (it says 60!). Why is the true answer 28?



1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION

Objective: Evaluate chained arithmetic expressions without parentheses by managing stack register flow.

Hands-On Setup: Use Place Value unit blocks to represent quantities alongside the 4-line stack display.

Predict First (Ask Before Keys): Problem: (8 - 3) × 2. Predict: 8 - 3 = 5, then 5 × 2 = 10.

Studio Games: p. vi • Place Value Blocks & 4-Register Stack

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"Solve the inner bundle first and let it wait on Shelf Y; when you type the next number, watch Automatic Stack Lift push it up!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Type the expression from left to right and hope the calculator knows PEMDAS."

3. CANONICAL SC-6 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: [8] [ENTER] [3] [-] [2] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[8]	8	8 in X
[ENTER]	8	8 copied to Y
[3]	3	3 entered into X
[-]	5	8 - 3 = 5 in X
[2]	2	Automatic lift! 5 lifts to Y; 2 enters X
[×]	10	5 × 2 = 10 in X!

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

Unlike traditional algebraic calculators that require parenthesis keys, RPN evaluates inner expressions first, lifting results naturally.

Core Guided Investigation: A baker has 15 cookies, sells 6, and packs the rest into 3 bags. How many per bag? Keystrokes for (15 - 6) / 3.

Expected Answer: 3 cookies per bag.

Verification Keys: [1] [5] [ENTER] [6] [-] [3] [÷]

Automatic Stack Lift & Register Retention

Follow-Up Extension (Big Ideas Math • CCSS.MATH.4.OA.A.3) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Observe the 4-line stack screen as values automatically lift into register Y: (12 / 4) + 7. Prediction: 12 / 4 = 3. Entering 7 will lift 3 to Y. 3 + 7 = 10.

Extension Key Sequence: [1] [2] [ENTER] [4] [÷] [7] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	First digit of 12 enters X
[2]	12	12 in register X
[ENTER]	12	12 copied to Y
[4]	4	4 enters X
[÷]	3	12 / 4 = 3 in X
[7]	7	7 entered: 3 lifts to Y automatically!
[+]	10	3 + 7 = 10!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Pressing [ENTER] after an operator (+, −, ×, ÷) before typing the next number in a chained calculation, wasting a stack register.

Hands-On Block & Stack Fix: Demonstrate Automatic Stack Lift: after [18] [ENTER] [6] [−] produces 12, typing [3] automatically lifts 12 to Register Y—no [ENTER] needed before [3] [×]!

Misconception 2: Reversing operand order in subtraction or division (computing X − Y instead of Y − X).

Hands-On Block & Stack Fix: Remind students that Register Y is always the starting quantity on top and Register X is the operator's action; if entered backward, press [x swap y] before [−] or [÷].

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 05 ANSWERS & KEYS)

Lab P1: Calculate (14 - 5) × 3 on the stack. **Ans: 27**

Check Keys: [1] [4] [ENTER] [5] [-] [3] [×]

Lab P2: Calculate 30 / (2 + 3): add denominator first! **Ans: 6**

Check Keys: [2] [ENTER] [3] [+] [3] [0] [↑↓] [÷]

Lab P3: Calculate (9 + 7) / (8 - 4) using 4 registers. **Ans: 4**

Check Keys: [9] [ENTER] [7] [+] [8] [ENTER] [4] [-] [÷]

Lab P4: A farmer collects 20 eggs, breaks 2, and divides among 6 cartons. **Ans: 3 eggs**

Check Keys: [2] [0] [ENTER] [2] [-] [6] [÷]

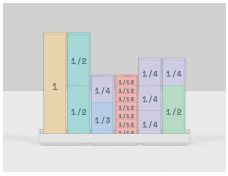
Mixed Numbers & Height Equivalence: 3/4 + 2 1/2

Core Concept Lesson • Singapore Math (Fraction Addition & Mixed Numbers) • Standard: CCSS.MATH.5.NF.A.1

COMPANION BOOK STORY: The Pharaoh's Bakery, Pythagoras's Guitar Strings & Why Your Fraction Towers Stand 84.0 mm Tall

Four thousand years ago, Egyptian scribes had to divide 3 loaves of bread fairly among 4 workers—and instead of cutting every loaf into quarters, they gave each worker 1/2 + 1/4! In Chapter 5 of From Blocks to the Stack, students discover why the Whole block is engineered to 84.0 mm (LCM of 1,2,3,4,6,8,12 -> 84 mm), letting 42 mm (1/2) + 21 mm (1/4) stack flush with three 21 mm (3/4) blocks.

60-SEC BELL-RINGER HOOK: Hold up the 1-Whole tower and a millimeter ruler: Why did engineers make the Whole block exactly 84 mm tall instead of 100 mm?



1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION

Objective: Represent proper and mixed fractions physically with block towers and computationally with two-dot notation.

Hands-On Setup: Place three 1/4 blocks (3 × 21 = 63 mm). Beside them, stack two Whole blocks and one 1/2 block (84 + 84 + 42 = 210 mm).

Predict First (Ask Before Keys): Combine heights: 63 mm + 210 mm = 273 mm. In wholes: 3 1/4 (273 / 84 = 3.25).

Studio Games: p. vii • 84 mm Fraction Towers & Twin Comparator Well

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"Stand your 1/2 (42 mm) and 1/4 (21 mm) towers in the comparator well—do they match three 1/4 towers (63 mm) in height?"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Just cross-multiply the denominators and don't ask why."

3. CANONICAL SC-6 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: [•] [3] [•] [4] [ENTER] [2] [•] [1] [•] [2] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[•]	0.	Leading dot indicates 0 wholes
[3]	0.3	Numerator = 3
[•]	0.3.	Second dot sets fraction entry mode
[4]	3/4	Denominator = 4: displays 3/4 (or 0.75)
[ENTER]	3/4	3/4 lifted to register Y
[2]	2	Whole = 2
[•]	2.	Dot
[1]	2.1	Numerator = 1
[•]	2.1.	Dot
[2]	2 1/2	Denominator = 2: displays 2 1/2
[+]	3 1/4	Plus combines: 3 1/4 (decimal 3.25)!

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

Two dots eliminate ambiguity between decimals and fractions. Physical block heights provide an infallible self-check against the 84 mm comparator pillar.

Core Guided Investigation: Build 1/2 and 1/4 with blocks. Verify their combined height against three 1/4 blocks, then check on the calculator.

Expected Answer: 3/4 (height 63 mm).

Verification Keys: [•] [1] [•] [2] [ENTER] [•] [1] [•] [4] [+]

Fraction Multiplication & Reciprocals

Follow-Up Extension (Bridges in Mathematics • CCSS.MATH.5.NF.B.4) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Set up 3/4 of a tower (63 mm) and divide into 3 equal parts (21 mm each), taking 2 parts: (3/4) × (2/3). Prediction: 3/4 × 2/3 = 6/12 = 1/2. The resulting block height matches one 1/2 block (42 mm).

Extension Key Sequence: [•] [3] [•] [4] [ENTER] [•] [2] [•] [3] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[•]	0.	Start proper fraction 3/4
[3]	0.3	Numerator 3
[•]	0.3.	Fraction separator
[4]	3/4	3/4 in register X
[ENTER]	3/4	3/4 lifted to register Y
[•]	0.	Start proper fraction 2/3
[2]	0.2	Numerator 2
[•]	0.2.	Fraction separator
[3]	2/3	2/3 in register X
[×]	1/2	Product 6/12 = 1/2 (matches 42 mm block)!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Adding numerators and denominators directly (e.g., 1/2 + 1/4 = 2/6) because 6 looks larger.

Hands-On Block & Stack Fix: Place the 1/2 tower (42 mm) + 1/4 tower (21 mm) beside two 1/6 towers (28 mm) in the comparator well; the physical height difference proves 1/2 + 1/4 = 3/4 (63 mm).

Misconception 2: Entering a proper fraction like 3/4 as [3] [.] [4] (which is decimal 3.4) instead of using Two-Dot fraction notation [.][3][.][4].

Hands-On Block & Stack Fix: Teach the rule: 'First dot = whole number zero, second dot = fraction bar.' Typing [.][3][.][4] displays exact 3/4 on the stack.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 06 ANSWERS & KEYS)

Lab P1: Add 1/3 + 1/6 using fraction blocks and verify on the calculator. **Ans: 1/2 (42 mm)**

Check Keys: [•] [1] [•] [3] [ENTER] [•] [1] [•] [6] [+]

Lab P2: Calculate 1 1/2 - 3/4. What block height corresponds to this? **Ans: 3/4 (63 mm)**

Check Keys: [1] [•] [1] [•] [2] [ENTER] [•] [3] [•] [4] [-]

Lab P3: Compare 3/8 (31.5 mm) to 1/2 (42 mm). What is the decimal equivalent of 3/8? **Ans: 0.375**

Check Keys: [•] [3] [•] [8] [FDISP]

Lab P4: Multiply 2/3 × 3/4. Check against your fraction blocks. **Ans: 1/2 (42 mm)**

Check Keys: [•] [2] [•] [3] [ENTER] [•] [3] [•] [4] [×]

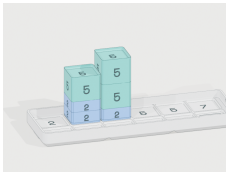
Chained Operations: (a + b) × (c - d) on the Stack

Core Concept Lesson • Big Ideas Math (Numerical & Algebraic Expressions) • Standard: CCSS.MATH.6.EE.A.2

COMPANION BOOK STORY: Why 24 Is the Champion Packing Number, The Typesetter's Bargain & Dijkstra's 1961 Railway Siding

In the 1500s, typesetters invented parentheses only to squash 2D fraction bars onto flat lines of lead type—then in 1924 Jan Łukasiewicz proved that placing operators after their numbers eliminates every parenthesis, and in 1961 Edsger Dijkstra modeled the 4-register stack as a railway shunting yard! Alongside the 24-Tile Array Studio (where 24 packs into 1×24, 2×12, 3×8, and 4×6 rectangles), students evaluate nested expressions from the inside out.

60-SEC BELL-RINGER HOOK: Why do 24 tiles pack into four different rectangles (1×24, 2×12, 3×8, 4×6), while 23 tiles only make one skinny line?



Studio Games: p. viii • Counting & Array Tiles + 4-Register Stack

1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION

Objective: Map an algebraic expression to a postfix sequence and execute on the 4-register stack without parentheses.

Hands-On Setup: Arrange Counting & Array tiles to model the two sub-expressions: (6 + 1) = 7 rows by (9 - 5) = 4 columns.

Predict First (Ask Before Keys): Left sub-expression: 6 + 1 = 7. Right sub-expression: 9 - 5 = 4. Combined 7 × 4 array: 7 × 4 = 28.

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"Build the left group (6 + 1) on Shelf Y and the right group (9 − 5) on Shelf X, then press [×] to form the 7 × 4 rectangle!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Count open and close parentheses before typing anything."

3. CANONICAL SC-6 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: 6 ENTER 1 + 9 ENTER 5 - ×

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
6	6	Value a (6) placed into Register X
ENTER	6	ENTER lifts 6 into Register Y
1	1	Value b (1) placed into Register X
+	7	Left operator (+): Combines Y(6) + X(1) = 7 in X
9	9	Value c (9) entered into X; 7 lifted to Y
ENTER	9	ENTER lifts 9 into Z and 7 into T
5	5	Value d (5) placed into Register X
-	4	Right operator (-): Combines Z/Y(9) - X(5) = 4 in X; 7 drops to Y
×	28	Root operator (×): Combines Y(7) × X(4) = 28 in X!

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

Parentheses exist in written algebra only to compensate for linear notation. On the 4-line stack, precedence is intrinsic.

Core Guided Investigation: Evaluate (8 + 2) × (14 - 9) by solving the inner quantities first, then checking on the stack.

Expected Answer: 50.

Verification Keys: 8 ENTER 2 + 1 4 ENTER 9 - ×

Deep Stack Evaluation & Register Limits

Follow-Up Extension (Singapore Math • CCSS.MATH.6.EE.A.2) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Track the 4 stack registers (T, Z, Y, X) for (12 - 4) / (2 × (1 + 1)). Prediction: Numerator: 12 - 4 = 8. Denominator: 2 × 2 = 4. Result: 8 / 4 = 2.

Extension Key Sequence: [1] [2] [ENTER] [4] [-] [2] [ENTER] [1] [ENTER] [1] [+] [×] [÷]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	First digit of 12
[2]	12	12 in register X
[ENTER]	12	12 lifted to Y
[4]	4	4 in register X
[-]	8	Numerator 12 - 4 = 8 held in X
[2]	2	Factor 2 enters X; 8 lifts to Y
[ENTER]	2	2 lifted to Y; 8 lifts to Z
[1]	1	Inner 1 enters X
[ENTER]	1	1 lifted to Y; 2 in Z; 8 in T
[1]	1	Second 1 enters X
[+]	2	Innermost sum: 1 + 1 = 2 in X
[×]	4	Denominator combined: 2 × 2 = 4 in X; 8 in Y
[÷]	2	8 / 4 = 2!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Evaluating expressions strictly left-to-right (computing 10 + 5 × 2 as 15 × 2 = 30) instead of respecting operator structure.

Hands-On Block & Stack Fix: Build the expression tree with Array Tiles: push 10 to Register Y ([10] [ENTER]), evaluate the 5 × 2 array in X ([5] [ENTER] [2] [×] -> 10), then press [+] -> 20.

Misconception 2: Losing track of intermediate results when evaluating two parenthesized branches like (8 + 2) × (9 − 4).

Hands-On Block & Stack Fix: Show how Register Y holds the first branch (8 + 2 = 10) automatically while the second branch (9 − 4 = 5) is computed in X, ready for [×] = 50.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 07 ANSWERS & KEYS)

Lab P1: Evaluate (4 + 3) × (10 - 6) and check on the stack. **Ans: 28**

Check Keys: [4] [ENTER] [3] [+] [1] [0] [ENTER] [6] [-] [×]

Lab P2: Solve (20 - 4) / (2 × 4) using inner-first evaluation. **Ans: 2**

Check Keys: [2] [0] [ENTER] [4] [-] [2] [ENTER] [4] [×] [÷]

Lab P3: Evaluate (18 - 6) / (1 + 2) on the 4-level stack. **Ans: 4**

Check Keys: [1] [8] [ENTER] [6] [-] [1] [ENTER] [2] [+] [÷]

Lab P4: Evaluate 5 + (3 × 4) by computing the inner product first. **Ans: 17**

Check Keys: [3] [ENTER] [4] [×] [5] [+]

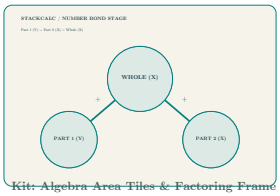
Area Modeling: Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$

Core Concept Lesson • Big Ideas Math (Polynomial Equations & Factoring) • Standard: CCSS.MATH.HSA.APR.A.1

COMPANION BOOK STORY: From Al-Khwarizmi's Physical Squares to the 1535 Duel in Milan & Kahan's [POLY] Solver

In 820 AD in Baghdad, Al-Khwarizmi solved $x^2 + 10x = 39$ by literally arranging physical square and rectangular tiles to 'complete the square'—and in 1535 in Milan, Niccolò Tartaglia defeated Antonio Fiore in a 30-problem public algebra duel overnight! Students build physical $(x + a)(x + b)$ tile rectangles and verify roots with the SC-32 [POLY] solver.

60-SEC BELL-RINGER HOOK: Drop one [+x] bar and one [−x] bar into the tray. How can adding two physical blocks together leave you with zero?



Kit: Algebra Area Tiles & Factoring Frame

1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION

Objective: Model polynomial multiplication and factoring physically using tile arrays and verify with the SC-32 [POLY] solver.

Hands-On Setup: Place 1 x^2 tile, 5 x-tiles, and 6 unit-1 squares on the factoring stage.

Predict First (Ask Before Keys): Arrange into a solid rectangle: width is $(x + 2)$, length is $(x + 3)$. Total Area = $x^2 + 5x + 6$.

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"Arrange the x^2 square, five x strips, and six unit tiles into a complete rectangle—the side lengths ARE the factors $(x + 2)(x + 3)$!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Just FOIL the letters and memorize the quadratic formula."

3. CANONICAL SC-32 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: [1] [ENTER] [5] [ENTER] [6] [↗] [POLY]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[1]	1	Coefficient a = 1 entered into X
[ENTER]	1	1 lifted to Y
[5]	5	Coefficient b = 5 entered into X
[ENTER]	5	b in Y, a in Z
[6]	6	Coefficient c = 6 entered into X
[↗] [POLY]	-2	Press [g:] [POLY]: Root 1 = -2 in Y (factor: x + 2), Root 2 = -3 in X (factor: x + 3)!

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

The physical dimensions of the rectangle (x+2) and (x+3) correspond to roots x = -2 and x = -3 solved by the [POLY] quadratic pipeline.

Core Guided Investigation: Arrange tiles for $x^2 + 7x + 12$ into a rectangle to find its factors, then verify with [POLY].

Expected Answer: $(x + 3)(x + 4)$, roots -3 and -4.

Verification Keys: [1] [ENTER] [7] [ENTER] [1] [2] [↗] [POLY]

Zero Pairs: Factoring Trinomials with Negative Terms

Follow-Up Extension (Singapore Math • CCSS.MATH.HSA.REI.B.4) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Set up 1 x^2 tile and 6 negative unit tiles for $x^2 - x - 6$. Prediction: Add zero pair (+2x and -3x): forms a rectangle of (x - 3) by (x + 2).

Extension Key Sequence: [1] [ENTER] [1] [+/−] [ENTER] [6] [+/−] [g] [POLY]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	Coefficient a = 1
[ENTER]	1	1 lifted to Y
[1]	1	Enter magnitude of b
[+/−]	-1	Coefficient b = -1 in X
[ENTER]	-1	b in Y, a in Z
[6]	6	Enter magnitude of c
[+/−]	-6	Coefficient c = -6 in X
[g] [POLY]	3	Press [g:] [POLY]: Root 1 = 3 in Y (factor: x - 3), Root 2 = -2 in X (factor: x + 2)!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Expanding $(x + 2)(x + 3)$ as $x^2 + 6$ by forgetting the two linear cross-terms (2x and 3x).

Hands-On Block & Stack Fix: Build the rectangle on the Algebra Factoring Stage with 1 [x^2] tile, 5 [x] strips, and 6 [1] unit tiles so the four area quadrants (x^2, 3x, 2x, 6) are physically visible.

Misconception 2: Using the subtraction operator [−] instead of the change-sign key [+/−] when entering negative polynomial coefficients into [POLY].

Hands-On Block & Stack Fix: Point out that [−] immediately subtracts X from Y, whereas [+/−] negates the number currently in Register X without collapsing the stack.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 08 ANSWERS & KEYS)

Lab P1: Factor $x^2 + 6x + 8$ into a rectangle of algebra tiles, then verify with [POLY]. **Ans: (x + 2)(x + 4); roots -2, -4**

Check Keys: [1] [ENTER] [6] [ENTER] [8] [g] [POLY]

Lab P2: Factor $x^2 - 4$ (difference of two squares) using zero-pair x-tiles. **Ans: (x - 2)(x + 2); roots 2, -2**

Check Keys: [1] [ENTER] [0] [ENTER] [4] [+/−] [g] [POLY]

Lab P3: Factor $x^2 + 5x + 4$ using polynomial tiles and verify. **Ans: (x + 1)(x + 4); roots -1, -4**

Check Keys: [1] [ENTER] [5] [ENTER] [4] [g] [POLY]

Lab P4: Factor $x^2 - 5x + 6$ using zero pairs. **Ans: (x - 2)(x - 3); roots 2, 3**

Check Keys: [1] [ENTER] [5] [+/−] [ENTER] [6] [g] [POLY]

Vector Magnitude & Direction: Cartesian to Polar

Core Concept Lesson • Big Ideas Math (Right Triangle Trigonometry & Vectors) • Standard: CCSS.MATH.HSN.VM.A.1

COMPANION BOOK STORY: A Stick in Alexandria: How Eratosthenes Measured the Earth with a Shadow & Crosswind Vectors

In 240 BC, Eratosthenes measured a 7.2° shadow angle in Alexandria at noon on the summer solstice while a well in Syene 800 km away had zero shadow—calculating Earth's circumference within 1% using pure geometry! With the 3-disc Unit Circle Volvelle and [->ϑ,r] polar conversion, students resolve right triangles and aircraft crosswind vectors in a single keystroke.

60-SEC BELL-RINGER HOOK: Rotate the Unit Circle Volvelle to 30°: why is sin(30°) exactly 0.5000—the exact half-height of the circle radius?

1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION



Kit: Unit Circle Trigonometry Volvelle

Objective: Decompose 2D vectors into horizontal and vertical components and convert to polar coordinates (r, ϑ).

Hands-On Setup: Align the Unit Circle Volvelle cursor to horizontal run 4 and vertical rise 3 (forming a right triangle with hypotenuse 5).

Predict First (Ask Before Keys): Run = 4, Rise = 3. Predict magnitude: r = sqrt(4^2 + 3^2) = 5. Predict angle: arctan(3/4) ˜ 36.87°.

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"Put your vertical Rise (y) on Shelf Y and horizontal Run (x) on Workbench X, then press [->ϑ,r] to see hypotenuse r and angle ϑ!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Type arctan(y/x) and guess which quadrant you're in."

3. CANONICAL SC-32 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: [4] [ENTER] [3] [↶] [▸θ,r] [x⇄y]

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
[4]	4	Enter horizontal component x = 4
[ENTER]	4	4 lifted to register Y
[3]	3	Enter vertical component y = 3 in X
[↶] [▸θ,r]	36.87	Rectangular to Polar! Screen displays angle ϑ = 36.87°
[x⇄y]	5.00	Swap registers: Y displays vector magnitude r = 5.00!

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

The calculator's built-in polar conversion executes both the Pythagorean magnitude and arctangent angle in a single microcode step.

Core Guided Investigation: A surveyor walks 60 m East and 80 m North. Find the straight-line displacement magnitude and heading.

Expected Answer: Displacement: 100 m, Heading: 53.13°.

Verification Keys: [6] [0] [ENTER] [8] [0] [↶] [▸θ,r] [x⇄y]

2D Vector Addition: Resultant Forces

Follow-Up Extension (Big Ideas Math • CCSS.MATH.HSN.VM.A.3) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Use the Unit Circle Volvelle to track component vectors F1 = <3, 2> and F2 = <1, 5>. Prediction: Add components: x_net = 3 + 1 = 4; y_net = 2 + 5 = 7. Magnitude r = sqrt($4^2 + 7^2$) = sqrt(65) ~ 8.06.

Extension Key Sequence: [3] [ENTER] [1] [+] [2] [ENTER] [5] [+] [↶] [▸θ,r] [x⇄y]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[3]	3	Horizontal component x1 = 3
[ENTER]	3	Lifted to Y
[1]	1	Horizontal component x2 = 1
[+]	4	Net horizontal force x_net = 4 in X
[2]	2	Vertical component y1 = 2 (4 lifts to Y)
[ENTER]	2	2 lifted to Y; 4 in Z
[5]	5	Vertical component y2 = 5 in X
[+]	7	Net vertical force y_net = 7 in X; x_net = 4 in Y
[↶] [▸θ,r]	60.26	Resultant angle ϑ = 60.26°
[x⇄y]	8.06	Resultant force magnitude r = 8.06!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Evaluating trigonometric functions in Radian mode when angles are given in Degrees (or vice versa).

Hands-On Block & Stack Fix: Have students check the DEG/RAD indicator on the SC-32 status bar and verify on the Unit Circle Volvelle that sin(30°) = 0.5000 before starting vector problems.

Misconception 2: Swapping x and y components when converting rectangular coordinates to polar via [->ϑ,r].

Hands-On Block & Stack Fix: On the SC-32, place the vertical component (y / Rise) in Register Y and the horizontal component (x / Run) in Register X before pressing [->ϑ,r].

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 09 ANSWERS & KEYS)

Lab P1: Find magnitude and angle for vector <12, 5>. **Ans: ϑ = 22.62°, r = 13**

Check Keys: [1] [2] [ENTER] [5] [↶] [▸θ,r]

Lab P2: Find the hypotenuse of a right triangle with legs 7 and 24. **Ans: 25**

Check Keys: [7] [ENTER] [2] [4] [↶] [▸θ,r] [x⇄y]

Lab P3: Calculate sin(30°) on the calculator. **Ans: 0.5**

Check Keys: [3] [0] [SIN]

Lab P4: A vector has magnitude 10 at angle 45°. What are its x and y components? **Ans: x = 7.07, y = 7.07**

Check Keys: [1] [0] [ENTER] [4] [5] [↷] [▸y,x]

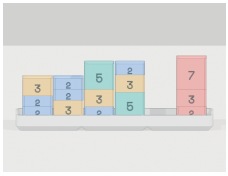
Continuous Compounding & Euler's Constant e

Core Concept Lesson • Big Ideas Math (Exponential & Logarithmic Functions) • Standard: CCSS.MATH.HSF.LE.A.4

COMPANION BOOK STORY: The Scottish Baron Who Doubled the Lifespan of Astronomers (John Napier), Euler's e & The Rule of 72

In 1614, John Napier at Merchiston Castle realized that logarithms turn tedious multiplication into simple height addition—the exact engineering principle of our Prime Factor blocks (where h = 15 × log2(p) mm)! Paired with Jakob Bernoulli's continuous compounding limit e and Fibonacci's Liber Abaci (1202), students model exponential doubling and half-life decay.

60-SEC BELL-RINGER HOOK: Stack three 2-prime towers (15 mm each) beside one 8-composite tower (45 mm). How did adding heights multiply 2 × 2 × 2 = 8?



1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION

Objective: Model continuous exponential accumulation and compound interest using the natural exponential function [e^x].

Hands-On Setup: Stack Fraction Blocks representing exponential doubling: 12 mm, 24 mm, 48 mm, 96 mm.

Predict First (Ask Before Keys): Invest $1,000 at 7% continuous interest for 10 years: A = 1000 × e^(0.07 × 10) = 1000 × e^0.7. Will it double? (e^0.7 ~ 2.01).

Studio Games: p. viii • Proportional Doubling Towers & Logarithmic Blocks

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"Compute the inner growth exponent r × t first, press [e^x] to get the growth factor, and then multiply by principal P!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Multiply the base by the exponent or round intermediate decimals to two places."

3. CANONICAL SC-32 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: 0 • 0 7 ENTER 1 0 × e^x 1 0 0 0 ×

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
0 • 0 7	0.07	Rate r = 0.07 entered in X
ENTER	0.07	Rate lifted to Y
1 0	10	Time t = 10 entered in X
×	0.7	Exponent r × t = 0.70
e^x	2.0138	e^0.70 = 2.01375 (Growth factor: more than doubled!)
1 0 0 0	1000	Principal P = 1000 (growth factor lifts to Y)
×	2013.75	Total accumulated amount A = $2,013.75!

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

The RPN stack computes nested transcendentals from innermost exponent out to outer multiplier without parenthesis keys.

Core Guided Investigation: A bacteria culture starts at 500 and triples every hour. Use [y^x] to calculate population after 4 hours.

Expected Answer: 40,500 bacteria.

Verification Keys: 3 ENTER 4 y^x 5 0 0 ×

Solving for Time: Doubling Time & Natural Log [LN]

Follow-Up Extension (Big Ideas Math • CCSS.MATH.HSF.BF.B.5) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Configure stack registers on the 4-line LCD: target multiplier in Y, growth rate in X. Prediction: At 5% growth (r = 0.05), doubling time is t = ln(2) / 0.05 ~ 0.69315 / 0.05 ~ 13.86 years.

Extension Key Sequence: [2] [LN] [0] [•] [0] [5] [÷]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[2]	2	Doubling target factor 2
[LN]	0.6931	Natural log ln(2) = 0.693147... in X
[0] [•] [0] [5]	0.05	Rate r = 0.05 (ln(2) lifts to Y)
[÷]	13.86	t = ln(2) / r = 13.86 years!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Multiplying the base by the exponent (computing 2^5 as 2 × 5 = 10) or reversing Base and Exponent on [y^x].

Hands-On Block & Stack Fix: Stack five 2-prime logarithmic towers to show that adding 5 tower heights multiplies 2×2×2×2×2 = 32, and verify that Base y goes in Register Y and Exponent x goes in X before [y^x].

Misconception 2: Applying the power [y^x] after multiplying by principal P in $P(1 + r)^t$ (computing $(P \cdot (1+r))^t$ instead of $P \cdot ((1+r)^t)$).

Hands-On Block & Stack Fix: Use the Inside-Out RPN rule: compute $(1 + r)^t$ first ([1.05] [ENTER] [2] [y^x]), then multiply by P ([1000] [×]).

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 10 ANSWERS & KEYS)

Lab P1: Calculate e^2 on the calculator to 4 decimal places. **Ans: 7.3891**

Check Keys: [2] [e^x]

Lab P2: Calculate doubling time for an investment growing at 8% continuous interest. **Ans: 8.66 years**

Check Keys: [2] [LN] [0] [•] [0] [8] [÷]

Lab P3: Compute 10^3.5 using [f:] [10^x]. **Ans: 3162.28**

Check Keys: [3] [•] [5] [↶] [10^x]

Lab P4: Evaluate log10(500) using [f:] [LOG]. **Ans: 2.6990**

Check Keys: [5] [0] [0] [↶] [LOG]

Combinations vs. Permutations: Subsets & Order

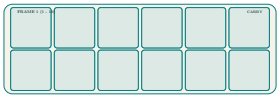
Core Concept Lesson • AP Statistics (Probability, Random Variables & Combinatorics) • Standard: CCSS.MATH.HSS.CP.B.9

COMPANION BOOK STORY: Sir Francis Galton's 1873 Bean Machine, The Asteroid Ceres & The Guinness Brewery Chemist

In 1873, Sir Francis Galton built a board studded with pins (the Quincunx) to show how independent random bounces settle into a bell-shaped curve—while in 1908 at the Guinness Brewery in Dublin, chemist William Sealy Gosset ('Student') invented small-sample statistics with n - 1 degrees of freedom! Students roll fair 16 mm polyhedral dice and accumulate trials with [Σ+].

60-SEC BELL-RINGER HOOK: Roll two 6-sided dice 12 times: why does the sum 7 appear almost three times as often as 2 or 12?

1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION



Kit: Probability Dice & 10-Sector Spinner

Objective: Calculate permutations and combinations by evaluating factorials across the 4-register stack.

Hands-On Setup: Use the 4 precision dice and 10-sector spinner to model multi-stage discrete outcomes.

Predict First (Ask Before Keys): How many ways to choose a committee of 3 from 8 candidates? 8C3 = 8! / (3! × 5!) = 40320 / (6 × 120) = 56.

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"Press [CLΣ] first, then feed each roll in with [Σ+]—watch Register X count n while the stack tracks the running mean!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Re-type all 12 numbers from scratch if you want both the mean and standard deviation."

3. CANONICAL SC-32 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: 8 ↶ x! 3 ↶ x! ÷ 5 ↶ x! ÷

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
8	8	Total candidates n = 8
↶ x!	40320	8! = 40,320 in register X
3	3	Subset r = 3 (40320 lifts to Y)
↶ x!	6	3! = 6 in X
÷	6720	40320 / 6 = 6720 in X
5	5	Remaining n - r = 5 (6720 lifts to Y)
↶ x!	120	5! = 120 in X
÷	56	6720 / 120 = 56 combinations!

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

The RPN stack handles multi-step factorial simplification effortlessly, preventing arithmetic overflow.

Core Guided Investigation: A lottery draws 4 winning numbers from a pool of 10. How many combinations? Calculate 10C4.

Expected Answer: 210 combinations.

Verification Keys: 1 0 ↶ x! 4 ↶ x! ÷ 6 ↶ x! ÷

Statistical Accumulation: Mean & Variance with [Σ+]

Follow-Up Extension (AP Statistics • CCSS.MATH.HSS.ID.A.2) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Roll precision dice to generate discrete data points for [Σ+] accumulation: 4, 6, 7, 8, 10. Prediction: Sum = 4 + 6 + 7 + 8 + 10 = 35. Sample size n = 5. Mean x-bar = 35 / 5 = 7.0.

Extension Key Sequence: 4 Σ+ 6 Σ+ 7 Σ+ 8 Σ+ 1 0 Σ+

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
4 Σ+	1	First score entered: sample size n = 1
6 Σ+	2	Second score: n = 2
7 Σ+	3	Third score: n = 3
8 Σ+	4	Fourth score: n = 4
1 0 Σ+	5	Fifth score: n = 5 total data points

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Forgetting to clear the statistical registers ([CLΣ]) before entering a new dataset with [Σ+].

Hands-On Block & Stack Fix: Notice that pressing [Σ+] displays the sample count n in Register X; if the first data point shows n > 1, press [CLΣ] first so prior data does not corrupt the mean.

Misconception 2: Dividing by r! and then multiplying by (n − r)! when computing C(n, r) = n! / (r!(n − r)!).

Hands-On Block & Stack Fix: Compute n! into Y, compute r! and (n − r)! and multiply them together in X ([×]), then press [÷] once to divide n! by the combined denominator.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 11 ANSWERS & KEYS)

Lab P1: Calculate 6! (6 factorial) on the calculator. **Ans: 720**

Check Keys: 6 ↶ x!

Lab P2: How many ways can 5 runners finish 1st, 2nd, and 3rd? (5P3 = 5! / 2!). **Ans: 60**

Check Keys: 5 ↶ x! 2 ↶ x! ÷

Lab P3: Calculate 7C2 (combinations of 2 from 7). **Ans: 21**

Check Keys: 7 ↶ x! 2 ↶ x! ÷ 5 ↶ x! ÷

Lab P4: Accumulate 5 and 15 with [Σ+]. What is the mean? **Ans: Mean = 10**

Check Keys: 5 Σ+ 1 5 Σ+ ↷ x̄, ȳ

Number Representation: Decimal to Binary and Hexadecimal

Core Concept Lesson • AP Computer Science Principles (Data Representation & Binary Number Systems) • Standard: CCSS.MATH.PRACTICE.MP5

COMPANION BOOK STORY: Leibniz's Binary Medallion, Claude Shannon's 1937 Relay Basement & NASA's 36-Bit IBM 7090 Word

In 1697, Gottfried Wilhelm Leibniz designed a medallion showing that every number can be built from 1 and 0 alone; in 1937, 21-year-old MIT student Claude Shannon proved electrical switches execute Boolean algebra; and in 1961, NASA's IBM 7090 computed orbital trajectories in 36-bit binary words! Using an 8-slot Ten-Frame register and [BASE], students decode bytes, hex nibbles, and bitwise masks.

60-SEC BELL-RINGER HOOK: How can you count from 0 to 31 using only the 5 fingers of one hand? (Hint: each finger is a binary power of 2!)

1. CONCRETE MANIPULATIVE SETUP & STUDENT PREDICTION



Kit: 8-Bit Ten-Frame Register & Place Value Blocks

Objective: Represent integers physically as bit registers and computationally convert between bases with [f:] [BASE].

Hands-On Setup: Use the 8 wells of the Ten-Frame as an 8-bit byte. Place pegs in wells for active '1' bits, leave empty for '0'.

Predict First (Ask Before Keys): Place pegs at positions 0, 1, and 3: binary $00001011_2 = 8 + 2 + 1 = 11_{10}$. Predict hex value: $0B_{16}$.

2A. TEACHER TALK — SAY & ASK WITH BLOCKS:

"Fill pegs for 1s and empty wells for 0s on the Ten-Frame, then toggle [BIN] and [HEX] on the SC-32 to see the exact same bit pattern!"

2B. AVOID SAYING (RULE WITHOUT REASON):

"Convert binary to decimal only by long-division remainders on scratch paper."

3. CANONICAL SC-32 KEYSTROKE SEQUENCE & 4-REGISTER STACK TRACE

Complete Key Sequence: 1 1 ↶ BASE

Key Pressed	Screen (X)	Internal 4-Level Stack Transformation (T, Z, Y, X)
1 1	`11`	Enter decimal integer 11 in DEC mode
↶ BASE	`HEX 0B`	Press [f:] [BASE]: screen displays Hexadecimal 0B!
↶ BASE	`BIN 1011`	Press [f:] [BASE]: screen displays Binary 1011!
↶ BASE	`OCT 13`	Press [f:] [BASE]: screen displays Octal 13!
↶ BASE	`DEC 11`	Press [f:] [BASE]: returns to Decimal 11

4. CONCEPTUAL INSIGHT & RPN PEDAGOGICAL ADVANTAGE

Computer memory stores identical bit patterns; base conversions alter only the human visual representation.

Core Guided Investigation: Enter decimal 255. Use [f:] [BASE] to convert to HEX and BIN. What is the binary pattern?

Expected Answer: HEX: FF, BIN: 11111111 (8 ones).

Verification Keys: 2 5 5 ↶ BASE

Bitwise Logic: AND Masking & Network Subnets

Follow-Up Extension (AP Computer Science Principles • CCSS.MATH.PRACTICE.MP5) + Misconception Remediations & Lab Workbook Key

1. FOLLOW-UP EXTENSION LESSON & HYPOTHESIS

Setup: Row 1 holds byte A (1100 = 12). Row 2 holds mask B (1010 = 10). Prediction: Bitwise AND: bit 3 (1 & 1 = 1), bit 2 (1 & 0 = 0), bit 1 (0 & 1 = 0), bit 0 (0 & 0 = 0). Result: 1000_2 = 8_10.

Extension Key Sequence: [1] [2] [ENTER] [1] [0] [AND]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1] [2]	12	Enter operand A = 12 (binary 1100)
[ENTER]	12	12 lifted to register Y
[1] [0]	10	Enter operand B = 10 (binary 1010) in X
[AND]	8	Bitwise AND evaluates 1100 & 1010 = 1000 (8 decimal)!

2. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Confusing bitwise [AND] / [OR] with decimal addition or multiplication (e.g., thinking 12 AND 10 is 22).

Hands-On Block & Stack Fix: Model 12 (1100_2) and 10 (1010_2) on two rows of the Ten-Frame using pegs for 1s and empty wells for 0s; only columns where BOTH rows have a peg remain 1 in 12 AND 10 = 8 (1000_2).

Misconception 2: Reading hexadecimal [10] as ten instead of decimal 16 ($1 \times 16^1 + 0 \times 16^0$).

Hands-On Block & Stack Fix: Build 1 sixteen-rod and 0 ones on the Place Value tray, then switch from [HEX] [10] to [DEC] on the SC-32 to see 16.

3. HANDS-ON MATH LAB STUDENT WORKBOOK COACHING (BOOK 2 • UNIT 12 ANSWERS & KEYS)

Lab P1: Convert decimal 42 to binary using [f:] [BASE]. **Ans: 101010 (32 + 8 + 2)**

Check Keys: [4] [2] [↰] [BASE]

Lab P2: Convert hexadecimal 2A to decimal. **Ans: 42**

Check Keys: [In] [HEX] [mode:] [2] [A] [->] [switch] [to] [DEC]

Lab P3: Evaluate bitwise AND: 15 (1111) AND 7 (0111). **Ans: 7 (0111)**

Check Keys: [1] [5] [ENTER] [7] [AND]

Lab P4: Evaluate bitwise XOR: 12 (1100) XOR 10 (1010). **Ans: 6 (0110)**

Check Keys: [1] [2] [ENTER] [1] [0] [XOR]

Middle School Section Coaching & 900 Keys

Complete teacher coaching notes, misconception alerts, and worked Practice A & B setups for Book 3 (Middle School Math Lab).

HOW PART II CONNECTS CONCEPT DOMAINS TO DAILY MIDDLE SCHOOL SECTIONS

While Part I organizes K–12 instruction by macro Concept Domain, middle school teachers using Book 3 (Middle School Math Lab — Aligned to Big Ideas Math Grades 6–8) also need rapid section-by-section coaching notes and complete worked solutions for all 45 sections and 900 dual-method problems (450 Practice A + 450 Practice B). Pages 29–40 provide the complete Middle School Teacher & Coach Guide right inside this master volume.

Middle School Unit (Book 3)	Sections & Core Concepts	Related Part I Domains	Coach Page
Unit 01: Number Sense & Operations	Sec 1.1–1.5 (Powers, Order of Ops, GCF/LCM)	Domains 05 & 07	p. 29
Unit 02: Fractions & Mixed Operations	Sec 2.1–2.5 (Multiplying & Dividing Fractions)	Domain 06	p. 30
Unit 03: Decimals & Rational Arithmetic	Sec 3.1–3.4 (Place Value & Decimal Division)	Domains 05 & 06	p. 31
Unit 04: Ratios, Rates & Proportions	Sec 4.1–4.5 (Unit Rates, Scaling & Proportions)	Domains 04 & 06	p. 32
Unit 05: Percents & Financial Literacy	Sec 5.1–5.5 (Discounts, Markup & Interest)	Domains 06 & 10	p. 33
Unit 06: Algebraic Expressions & Props	Sec 6.1–6.5 (Trees, Like Terms & Factoring)	Domains 07 & 08	p. 34
Unit 07: Equations & Inequalities	Sec 7.1–7.4 (1-Step, 2-Step & Balance Check)	Domains 07 & 08	p. 35
Unit 08: Signed Integers & Rationals	Sec 8.1–8.4 (Zero Pairs & Signed Operations)	Domains 03, 05 & 08	p. 36
Unit 09: Geometry: Area, Surface & Vol	Sec 9.1–9.4 (Polygons, Circles & Prisms)	Domains 08 & 09	p. 37
Unit 10: Statistics, Data & Probability	Sec 10.1–10.4 (Mean, Median, IQR, MAD & Dice)	Domain 11	p. 38
Middle School Misconception Matrix	10 High-Frequency Errors & Block Remediations	All Domains	p. 39
Section Mastery & Token Roster	45-Section Pacing & Classroom Check-Off Grid	All Domains	p. 40

Dual-Method Verification Rule for Middle School Classrooms

Every problem in Book 3 requires (1) written mathematical setup and mental/paper estimation first, followed by (2) RPN stack verification on the SC-6 calculator. Pages 29–38 list both the written setup and exact SC-6 keystrokes for all 900 problems.

Unit 1: Numerical Expressions & Factors — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 1 • Hands-On Models: Counting & Array Studio (Counting & Array Tiles) + Prime Factor Studio (Prime Factor Towers)

Target Misconception: Students interpret remainders as meaningless decimal tails (e.g., .875 of a person), read 2^3 as 2×3=6, lose track of PEMDAS parentheses, and confuse GCF (overlap) with LCM (union).

Hands-On Block Remediation: Have students build physical tile rows for division remainders, stack 2D/3D arrays for exponents, use the 4-line elevator (Y holds Room 1 while X solves Room 2) instead of rewriting PEMDAS lines, and stand logarithmic Prime Factor Towers side-by-side on the 12-socket lid.



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 1.1 (pp.4-5): WHOLE NUMBER OPERATIONS & DIVISION WITH REMAINDERS

Methods: W1: Standard Long Division | W2: Partial Quotients | W3: Equal-Groups Array Model

PRACTICE A (p.4): KEY IDEAS (#1-#10)

ExA: 5 R 3

A1: 5 R 1
Setup: 31 ÷ 6 (Q & R)
SC-6: 31 [ENTER] 6 [g][÷R]

A2: 9 R 2
Setup: 47 ÷ 5 (Q & R)
SC-6: 47 [ENTER] 5 [g][÷R]

A3: 8 R 2
Setup: 58 ÷ 7 (Q & R)
SC-6: 58 [ENTER] 7 [g][÷R]

A4: 9 R 4
Setup: 85 ÷ 9 (Q & R)
SC-6: 85 [ENTER] 9 [g][÷R]

A5: 12 R 2
Setup: 146 ÷ 12 (Q & R)
SC-6: 146 [ENTER] 12 [g][÷R]

A6: 7 full lines, 5 cm left
Setup: 250 ÷ 35 (Q & R)
SC-6: 250 [ENTER] 35 [g][÷R]

A7: 117
Setup: (14 × 8) + 5
SC-6: 14 [ENTER] 8 [×] 5 [+]

A8: 9 R 3 (Rem < 8)
Setup: 75 ÷ 8 (Q & R)
SC-6: 75 [ENTER] 8 [g][÷R]

A9: 19 rovers, 2 tires left
Setup: 78 ÷ 4 (Q & R)
SC-6: 78 [ENTER] 4 [g][÷R]

A10: 8 tents (7 full + 1 for the 29th explorer)
Setup: 29 ÷ 4 (Q & R)
SC-6: 29 [ENTER] 4 [g][÷R]

PRACTICE B (p.5): REAL LIFE (#1-#10)

ExB: 17 vans (16 full vans + 1 van with 7 students)

B1: 8 full packs, 5 cells left
Setup: 53 ÷ 6 (Q & R)
SC-6: 53 [ENTER] 6 [g][÷R]

B2: 11 figures, 6 tickets left
Setup: 94 ÷ 8 (Q & R)
SC-6: 94 [ENTER] 8 [g][÷R]

B3: 20 canoes (19 full + 1 with 2 campers)
Setup: 78 ÷ 4 (Q & R)
SC-6: 78 [ENTER] 4 [g][÷R]

B4: 9 full racks, 2 on last
Setup: 110 ÷ 12 (Q & R)
SC-6: 110 [ENTER] 12 [g][÷R]

B5: 28 knights, 2 m left
Setup: 450 ÷ 16 (Q & R)
SC-6: 450 [ENTER] 16 [g][÷R]

B6: 18 full, 19 caches total
Setup: 165 ÷ 9 (Q & R)
SC-6: 165 [ENTER] 9 [g][÷R]

B7: 187 marbles
Setup: (12 × 15) + 7
SC-6: 12 [ENTER] 15 [×] 7 [+]

B8: 8 full trains, 8 on final
Setup: 200 ÷ 24 (Q & R)
SC-6: 200 [ENTER] 24 [g][÷R]

B9: 9 pizzas (5 slices left)
Setup: 67 ÷ 8 (Q & R)
SC-6: 67 [ENTER] 8 [g][÷R]

B10: 11 full crates, 17 apples left
Setup: 325 ÷ 28 (Q & R)
SC-6: 325 [ENTER] 28 [g][÷R]

SEC 1.2 (pp.6-7): POWERS AND EXPONENTS

Methods: W1: Multiply Step-by-Step | W2: Group Factors in Pairs | W3: Square & Cube Array Model

PRACTICE A (p.6): KEY IDEAS (#1-#10)

ExA: 4³ = 16, 4⁴ = 64

A1: 5 × 5 = 25
Setup: 5^2
SC-6: 5 [x²]

A2: 49 (vs. 14)
Setup: 7^2
SC-6: 7 [x²]

A3: 3 × 3 × 3 = 27
Setup: 3^3
SC-6: 3 [x³]

A4: 125
Setup: 5^3
SC-6: 5 [x³]

A5: 32
Setup: 2^5
SC-6: 2 [ENTER] 5 [g][y^x]

A6: 1,296
Setup: 6^4
SC-6: 6 [ENTER] 4 [g][y^x]

A7: 10,000 (4 zeros) and 1
Setup: 10^4
SC-6: 10 [ENTER] 4 [g][y^x]

A8: 16 (2×2×2×2 = 16)
Setup: 2^4
SC-6: 2 [ENTER] 4 [g][y^x]

A9: 96 kN (3 × 32)
Setup: 2^5 × 3
SC-6: 2 [ENTER] 5 [g][y^x] 3 [×]

A10: 8 m³ per room; 64 m³ total
Setup: $2^3 \times 8$
SC-6: 2 [x³] 8 [×]

PRACTICE B (p.7): REAL LIFE (#1-#10)

ExB: 81 shares

B1: 8² = 64 pixels
Setup: 8^2
SC-6: 8 [x²]

B2: 6³ = 216 blocks
Setup: 6^3
SC-6: 6 [x³]

B3: 2^6 = 64 bacteria
Setup: 2^6
SC-6: 2 [ENTER] 6 [g][y^x]

B4: 4 × 4 × 4 × 4 = 256 players
Setup: 4^4
SC-6: 4 [ENTER] 4 [g][y^x]

B5: 144 + 25 = 169 ft²
Setup: $12^2 + 5^2$
SC-6: 12 [x²] 5 [x²] [+]

B6: 5^4 = 625 starlings
Setup: 5^4
SC-6: 5 [ENTER] 4 [g][y^x]

B7: 14 meters
Setup: √(196)
SC-6: 196 [f][√x]

B8: 2^5 = 32 > 5² = 25 (by 7)
Setup: $2^5 - 5^2$
SC-6: 2 [ENTER] 5 [g][y^x] 5 [x²] [-]

B9: 1,024 bytes
Setup: 2^10
SC-6: 2 [ENTER] 10 [g][y^x]

B10: 27 + 343 = 370 in³
Setup: $3^3 + 7^3$
SC-6: 3 [x³] 7 [x³] [+]

SEC 1.3 (pp.8-9): ORDER OF OPERATIONS

Methods: W1: PEMDAS Step-by-Step Funnel | W2: Distributive Property / Fraction-Bar Grouping

PRACTICE A (p.8): KEY IDEAS (#1-#10)

ExA: 72

A1: 11
Setup: 5 + (2 × 3)
SC-6: 5 [ENTER] 2 [ENTER] 3 [×] [+]

A2: 4
Setup: (14 + 6) ÷ (7 - 2)
SC-6: 14 [ENTER] 6 [+] 7 [ENTER] 2 [-] [÷]

A3: 6
Setup: 30 - (4 × 6)
SC-6: 30 [ENTER] 4 [ENTER] 6 [×] [-]

A4: 84
Setup: (9 + 3) × (15 - 8)
SC-6: 9 [ENTER] 3 [+] 15 [ENTER] 8 [-] [×]

A5: 1
Setup: 50 - (3 + 4)²
SC-6: 50 [ENTER] 3 [ENTER] 4 [+] [x²] [-]

A6: 20
Setup: 120 - (4 × (3 + 2)²)
SC-6: 120 [ENTER] 4 [ENTER] 3 [ENTER] 2 [+] [x²] [×] [-]

A7: 2 (24 ÷ 12)
Setup: (48 - (4 × 6)) ÷ (3 + (2 × 4.5))
SC-6: 48 [ENTER] 4 [ENTER] 6 [×] [-] 3 [ENTER] 2 [ENTER] 4.5 [×] [+]..

A8: 24
Setup: (3² + (15 ÷ 3)) - (2 × 4)
SC-6: 3 [x²] 15 [ENTER] 3 [÷] [+] 2 [ENTER] 4 [×] [-]

A9: $92 (60 + 32)
Setup: (5 × 12) + (4 × 8)
SC-6: 5 [ENTER] 12 [×] 4 [ENTER] 8 [×] [+]

A10: $62 (90 - 28)
Setup: (45 × 2) - 28
SC-6: 45 [ENTER] 2 [×] 28 [-]

PRACTICE B (p.9): REAL LIFE (#1-#10)

ExB: 25

B1: 4(9 + 5) = $56
Setup: (9 + 5) × 4
SC-6: 9 [ENTER] 5 [+] 4 [×]

B2: 45 + 4(7) = $73
Setup: 45 + (4 × 7)
SC-6: 45 [ENTER] 4 [ENTER] 7 [×] [+]

B3: 101 points
Setup: (100 - (3 × 8)) + 25
SC-6: 100 [ENTER] 3 [ENTER] 8 [×] [-] 25 [+]

B4: 10³ - 6³ = 64 m³
Setup: $10^3 - 6^3$
SC-6: 10 [x³] 6 [x³] [-]

B5: $73
Setup: (2 × 14) + (5 × 9)
SC-6: 2 [ENTER] 14 [×] 5 [ENTER] 9 [×] [+]

B6: 70 points
Setup: ((3² + 5) × 6) - 14
SC-6: 3 [x²] 5 [+] 6 [×] 14 [-]

B7: 10 bracelets per box
Setup: (72 - 12) ÷ 6
SC-6: 72 [ENTER] 12 [-] 6 [÷]

B8: 42 (÷ and × go left to right)
Setup: ((48 ÷ 6) × 4) + 10
SC-6: 48 [ENTER] 6 [÷] 4 [×] 10 [+]

B9: 12 muffins per basket
Setup: ((5 × 12) + (3 × 8)) ÷ 7
SC-6: 5 [ENTER] 12 [×] 3 [ENTER] 8 [×] [+] 7 [÷]

B10: 74
Setup: ((4³ - 14) ÷ (2 + 3)) + 8²
SC-6: 4 [x²] 14 [-] 2 [ENTER] 3 [+] [÷] 8 [x²] [+]

SEC 1.4 (pp.10-11): PRIME FACTORIZATION

Methods: W1: Factor Tree (Pair A) | W2: Factor Tree (Pair B) | W3: Division Ladder

PRACTICE A (p.10): KEY IDEAS (#1-#10)

ExA: 60 = 2² × 3 × 5

A1: 2² × 3
Setup: Factors(12)
SC-6: 12 [g][FACT]

A2: 2³ × 3²
Setup: Factors(36)
SC-6: 36 [g][FACT]

A3: 29 is prime; 91 is composite (7 × 13)
Setup: Factors(91)
SC-6: 29 [PRIME], 91 [g][FACT]

A4: 3² × 5
Setup: Factors(45)
SC-6: 45 [g][FACT]

A5: 2³ × 3²
Setup: Factors(72)
SC-6: 72 [g][FACT]

A6: 2² × 3 × 7
Setup: Factors(84)
SC-6: 84 [g][FACT]

A7: 2 × 3 × 5 × 7
Setup: Factors(210)
SC-6: 210 [g][FACT]

A8: 2³ × 3² × 7
Setup: Factors(504)
SC-6: 504 [g][FACT]

A9: 7 × 11 × 13
Setup: Factors(1001)
SC-6: 1001 [g][FACT]

A10: 2^4 × 3² × 5²
Setup: Factors(3600)
SC-6: 3600 [g][FACT]

PRACTICE B (p.11): REAL LIFE (#1-#10)

ExB: 2² × 3 × 5 (e.g., 8 × 15 or 10 × 12)

B1: 1×48, 2×24, 3×16, 4×12, 6×8; 2^4 × 3
Setup: Factors(48)
SC-6: 48 [g][FACT]

B2: 2 × 3³
Setup: Factors(54)
SC-6: 54 [g][FACT]

B3: 51 is composite (3 × 17)
Setup: Factors(51)
SC-6: 51 [g][FACT]

B4: 2^4 × 5
Setup: Factors(80)
SC-6: 80 [g][FACT]

B5: 126
Setup: (2 × 3²) × 7
SC-6: 2 [ENTER] 3 [x²] [×] 7 [×]

B6: 2² × 5²
Setup: Factors(100)
SC-6: 100 [g][FACT]

B7: 2² × 5 × 7
Setup: Factors(140)
SC-6: 140 [g][FACT]

B8: 2^4 × 3²
Setup: Factors(144)
SC-6: 144 [g][FACT]

B9: Both give 2 × 3² × 5
Setup: Factors(90)
SC-6: 90 [g][FACT]

B10: 2 × 3 × 5 × 7 = 210
Setup: ((2 × 3) × 5) × 7
SC-6: 2 [ENTER] 3 [×] 5 [×] 7 [×]

SEC 1.5 (pp.12-13): GREATEST COMMON FACTOR (GCF) & LEAST COMMON MULTIPLE (LCM)

Methods: W1: List Factors & Multiples | W2: Prime Factorization | W3: Prime Factor Venn Diagram

PRACTICE A (p.12): KEY IDEAS (#1-#10)

ExA: GCF = 10, LCM = 100

A1: GCF = 6, LCM = 36
Setup: GCF(12, 18); LCM(12, 18)
SC-6: 12 [ENTER] 18 [GCF] / [f][LCM]

A2: GCF = 4, LCM = 24
Setup: GCF(8, 12); LCM(8, 12)
SC-6: 8 [ENTER] 12 [GCF] / [f][LCM]

A3: GCF = 5, LCM = 75
Setup: GCF(15, 25); LCM(15, 25)
SC-6: 15 [ENTER] 25 [GCF] / [f][LCM]

A4: GCF = 7, LCM = 70
Setup: GCF(14, 35); LCM(14, 35)
SC-6: 14 [ENTER] 35 [GCF] / [f][LCM]

A5: 24
Setup: GCF(48, 72)
SC-6: 48 [ENTER] 72 [GCF]

A6: 28
Setup: GCF(84, 140)
SC-6: 84 [ENTER] 140 [GCF]

A7: GCF = 15, LCM = 225
Setup: GCF(45, 75); LCM(45, 75)
SC-6: 45 [ENTER] 75 [GCF] / [f][LCM]

A8: 60 minutes
Setup: LCM(15, 20)
SC-6: 15 [ENTER] 20 [f][LCM]

A9: 180 seconds (3 minutes)
Setup: LCM(45, 60)
SC-6: 45 [ENTER] 60 [f][LCM]

A10: 12 bags (3 red, 4 blue per bag)
Setup: GCF(36, 48)
SC-6: 36 [ENTER] 48 [GCF]

PRACTICE B (p.13): REAL LIFE (#1-#10)

ExB: 6 packs (3 pretzels and 4 fruit bars per pack)

B1: 14 kits (2 bandages, 3 wipes each)
Setup: GCF(28, 42)
SC-6: 28 [ENTER] 42 [GCF]

B2: 60 years
Setup: LCM(12, 15)
SC-6: 12 [ENTER] 15 [f][LCM]

B3: 24 (3 packs of hot dogs, 2 packs of buns)
Setup: LCM(8, 12)
SC-6: 8 [ENTER] 12 [f][LCM]

B4: 16 singers per row
Setup: GCF(32, 48)
SC-6: 32 [ENTER] 48 [GCF]

B5: 10:12 AM (72 minutes later)
Setup: LCM(18, 24)
SC-6: 18 [ENTER] 24 [f][LCM]

B6: 12 inches
Setup: GCF(36, 60)
SC-6: 36 [ENTER] 60 [GCF]

B7: GCF = 7, LCM = 42
Setup: LCM(14, 21)
SC-6: 14 [ENTER] 21 [GCF] 14 [ENTER] 21 [f][LCM]

B8: 60 minutes
Setup: LCM(LCM(4, 6), 10)
SC-6: 4 [ENTER] 6 [f][LCM] 10 [f][LCM]

B9: 15 bouquets (3 gold, 5 silver each)
Setup: GCF(45, 75)
SC-6: 45 [ENTER] 75 [GCF]

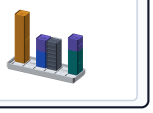
B10: GCF=5, LCM=60; 5×60 = 300 = 15×20
Setup: GCF(15, 20) × LCM(15, 20)
SC-6: 15 [ENTER] 20 [GCF] 15 [ENTER] 20 [f][LCM] [×]

Unit 2: Fractions & Decimals — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 2 • Hands-On Models: Fraction Studio (Fraction Strips) + Place Value Studio (Place Value Blocks)

Target Misconception: Students treat fractions, mixed numbers, and decimals as unrelated rules, expect multiplication to always make numbers bigger, forget which fraction to flip in division, and misalign decimal columns.

Hands-On Block Remediation: Use the 84.0 mm Fraction Studio towers so equivalence and common denominators are physical millimeter height matches. Use [f] [1/x] on Floor 1 (X) so students see only the divisor flip, and use [f] [IP] / [g] [FP] to split whole dollars from decimal cents.



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 2.1 (pp.15-16): EQUIVALENT FRACTIONS, MIXED NUMBERS & LCD ADDITION

Methods: W1: Divide Once by GCF / Use LCD | W2: Step-by-Step Common Factors | W3: Fraction Strips Model

PRACTICE A (p.15): KEY IDEAS (#1-#10)

ExA: 2/3

A1: 2/3
Setup: Simp(8/12)×2
SC-6: [.] 8 [.] 12 [ENTER] [g][SIMP]×2

A2: 8/3 = 2 2/3 trays
Setup: Simp(48/18)
SC-6: [.] 48 [.] 18 [ENTER] [g][SIMP]

A3: 13/5 and 2.6
Setup: Improper(2 3/5)
SC-6: 2 [.] 3 [.] 5 [ENTER] [f][d/c]

A4: 3 5/6 pizzas (3 whole, 5 sixths)
Setup: 23/6
SC-6: [.] 23 [.] 6 [ENTER]

A5: 5/6
Setup: 1/2 + 1/3
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 3 [+]

A6: 1 3/20 gallons (23/20)
Setup: 3/4 + 2/5
SC-6: [.] 3 [.] 4 [ENTER] [.] 2 [.] 5 [+]

A7: 3 5/6 feet
Setup: 1 1/2 + 2 1/3
SC-6: 1 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 3 [+]

A8: 2 7/12 feet
Setup: 5 1/3 - 2 3/4
SC-6: 5 [.] 1 [.] 3 [ENTER] 2 [.] 3 [.] 4 [-]

A9: 7/8 climbed; 1/8 remains
Setup: (1/2 + 1/4) + 1/8
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 4 [+] [.] 1 [.] 8 [+]

A10: 5 1/4 pints
Setup: 3 1/2 + 1 3/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 1 [.] 3 [.] 4 [+]

PRACTICE B (p.16): REAL LIFE (#1-#10)

ExB: 2 11/12 miles (35/12 miles)

B1: 5/8 cup
Setup: 3/8 + 1/4
SC-6: [.] 3 [.] 8 [ENTER] [.] 1 [.] 4 [+]

B2: 3/4 complete; 1/4 remains
Setup: Simp(18/24)
SC-6: [.] 18 [.] 24 [g][SIMP]

B3: 13/8 and 1.625 in
Setup: Improper(1 5/8)
SC-6: 1 [.] 5 [.] 8 [ENTER] [f][d/c] [FDISP]

B4: 11/24 foot
Setup: 5/6 - 3/8
SC-6: [.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]

B5: 3 5/6 pizzas (23/6)
Setup: 2 1/3 + 1 1/2
SC-6: 2 [.] 1 [.] 3 [ENTER] 1 [.] 1 [.] 2 [+]

B6: 2 3/4 feet (11/4)
Setup: 4 1/2 - 1 3/4
SC-6: 4 [.] 1 [.] 2 [ENTER] 1 [.] 3 [.] 4 [-]

B7: 3/5 = 0.6
Setup: Simp(15/25)
SC-6: [.] 15 [.] 25 [g][SIMP] [FDISP]

B8: 1/5 liter (0.2 L)
Setup: (2/5 + 3/10) - 1/2
SC-6: [.] 2 [.] 5 [ENTER] [.] 3 [.] 10 [+] [.] 1 [.] 2 [-]

B9: 4 5/12 yd used; 7/12 yd left
Setup: 2 2/3 + 1 3/4
SC-6: 2 [.] 2 [.] 3 [ENTER] 1 [.] 3 [.] 4 [+]

B10: 4 5/8 miles
Setup: (3/8 + 2 1/2) + 1 3/4
SC-6: [.] 3 [.] 8 [ENTER] 2 [.] 1 [.] 2 [+] 1 [.] 3 [.] 4 [+]

SEC 2.2 (pp.17-18): MULTIPLYING FRACTIONS AND MIXED NUMBERS

Methods: W1: Multiply Straight Across | W2: Cross-Cancel Common Factors First | W3: Area Model / Distributive Split

PRACTICE A (p.17): KEY IDEAS (#1-#10)

ExA: 3/8

A1: 1/3
Setup: 1/2 × 2/3
SC-6: [.] 1 [.] 2 [ENTER] [.] 2 [.] 3 [×]

A2: 1/2
Setup: 2/3 × 3/4
SC-6: [.] 2 [.] 3 [ENTER] [.] 3 [.] 4 [×]

A3: 1/3
Setup: 3/5 × 5/9
SC-6: [.] 3 [.] 5 [ENTER] [.] 5 [.] 9 [×]

A4: 6 cups
Setup: 16 × 3/8
SC-6: 16 [ENTER] [.] 3 [.] 8 [×]

A5: 4
Setup: 1 3/4 × 2 2/7
SC-6: 1 [.] 3 [.] 4 [ENTER] 2 [.] 2 [.] 7 [×]

A6: 3
Setup: 2 1/4 × 1 1/3
SC-6: 2 [.] 1 [.] 4 [ENTER] 1 [.] 1 [.] 3 [×]

A7: 9 cups
Setup: 2 1/4 × 4
SC-6: 2 [.] 1 [.] 4 [ENTER] 4 [×]

A8: 3 1/3 (Less than 5 because 2/3 < 1)
Setup: 5 × 2/3
SC-6: 5 [ENTER] [.] 2 [.] 3 [×]

A9: 14 7/8 miles (119/8)
Setup: 3 1/2 × 4 1/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 4 [.] 1 [.] 4 [×]

A10: 1/4 of the input speed
Setup: (1/2 × 2/3) × 3/4
SC-6: [.] 1 [.] 2 [ENTER] [.] 2 [.] 3 [×] [.] 3 [.] 4 [×]

PRACTICE B (p.18): REAL LIFE (#1-#10)

ExB: 4 cups

B1: 2/5 of the garden
Setup: 2/3 × 3/5
SC-6: [.] 2 [.] 3 [ENTER] [.] 3 [.] 5 [×]

B2: 2/5 of a full battery
Setup: 4/5 × 1/2
SC-6: [.] 4 [.] 5 [ENTER] [.] 1 [.] 2 [×]

B3: 4 3/8 cups (35/8)
Setup: 1 3/4 × 2 1/2
SC-6: 1 [.] 3 [.] 4 [ENTER] 2 [.] 1 [.] 2 [×]

B4: 3 in²
Setup: 2 1/4 × 1 1/3
SC-6: 2 [.] 1 [.] 4 [ENTER] 1 [.] 1 [.] 3 [×]

B5: 4 miles
Setup: 3 1/5 × 1 1/4
SC-6: 3 [.] 1 [.] 5 [ENTER] 1 [.] 1 [.] 4 [×]

B6: 2/3 yd²
Setup: 3/4 × 8/9
SC-6: [.] 3 [.] 4 [ENTER] [.] 8 [.] 9 [×]

B7: 10 gallons
Setup: (24 × 5/8) × 2/3
SC-6: 24 [ENTER] [.] 5 [.] 8 [×] [.] 2 [.] 3 [×]

B8: 1/8 in²
Setup: 1/2⁸
SC-6: [.] 1 [.] 2 [x^]

B9: 8 ft²
Setup: 3 1/2 × 2 2/7
SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 2 [.] 7 [×]

B10: 1/12 of the day (2 hours)
Setup: 1/3 × 1/4
SC-6: [.] 1 [.] 3 [ENTER] [.] 1 [.] 4 [×]

SEC 2.3 (pp.19-20): DIVIDING FRACTIONS AND MIXED NUMBERS

Methods: W1: Multiply by the Reciprocal | W2: Common-Denominator Division | W3: Fraction Strip / Number-Line Model

PRACTICE A (p.19): KEY IDEAS (#1-#10)

ExA: 6

A1: 2
Setup: 1/2 ÷ 1/4
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 4 [÷]

A2: 4
Setup: 2/3 ÷ 1/6
SC-6: [.] 2 [.] 3 [ENTER] [.] 1 [.] 6 [÷]

A3: 2
Setup: 5/6 × 1/(5/12)
SC-6: [.] 5 [.] 6 [ENTER] [.] 5 [.] 12 [f][1/x] [×]

A4: 8
Setup: 6 ÷ 3/4
SC-6: 6 [ENTER] [.] 3 [.] 4 [÷]

A5: 1 7/8 (15/8)
Setup: 3/4 ÷ 2/5
SC-6: [.] 3 [.] 4 [ENTER] [.] 2 [.] 5 [÷]

A6: 7/8 cup
Setup: 3 1/2 ÷ 4
SC-6: 3 [.] 1 [.] 2 [ENTER] 4 [÷]

A7: 2 2/3 bike frames
Setup: 2/3 ÷ 1/4
SC-6: [.] 2 [.] 3 [ENTER] [.] 1 [.] 4 [÷]

A8: 3
Setup: 5 1/4 ÷ 1 3/4
SC-6: 5 [.] 1 [.] 4 [ENTER] 1 [.] 3 [.] 4 [÷]

A9: 5 full slats (5 7/37 -> 3 1/2 in left)
Setup: 96 ÷ 18 1/2
SC-6: 96 [ENTER] 18 [.] 1 [.] 2 [÷]

A10: 6 glasses
Setup: 3 3/4 ÷ 5/8
SC-6: 3 [.] 3 [.] 4 [ENTER] [.] 5 [.] 8 [÷]

PRACTICE B (p.20): REAL LIFE (#1-#10)

ExB: 6 shelves

B1: 6 slices
Setup: 3/4 ÷ 1/8
SC-6: [.] 3 [.] 4 [ENTER] [.] 1 [.] 8 [÷]

B2: 12 pebbles
Setup: 4 1/2 ÷ 3/8
SC-6: 4 [.] 1 [.] 2 [ENTER] [.] 3 [.] 8 [÷]

B3: 6 runners
Setup: 3 3/4 ÷ 5/8
SC-6: 3 [.] 3 [.] 4 [ENTER] [.] 5 [.] 8 [÷]

B4: 7 segments
Setup: 5 1/4 ÷ 3/4
SC-6: 5 [.] 1 [.] 4 [ENTER] [.] 3 [.] 4 [÷]

B5: 40 layers
Setup: 6 ÷ 3/20
SC-6: 6 [ENTER] [.] 3 [.] 20 [÷]

B6: 6 sections
Setup: 7 1/2 ÷ 1 1/4
SC-6: 7 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 4 [÷]

B7: 2 meters
Setup: 4 1/2 ÷ 2 1/4
SC-6: 4 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [÷]

B8: 8 cups
Setup: 2 2/3 ÷ 1/3
SC-6: 2 [.] 2 [.] 3 [ENTER] [.] 1 [.] 3 [÷]

B9: 5/12 (12/5 × 5/12 = 1)
Setup: 1/(2 2/5)
SC-6: 2 [.] 2 [.] 5 [f][1/x]

B10: 12 pieces
Setup: (10 1/2 - 1 1/2) ÷ 3/4
SC-6: 10 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 2 [-] [.] 3 [.] 4 [÷]

SEC 2.4 (pp.21-22): ADDING, SUBTRACTING, AND MULTIPLYING DECIMALS

Methods: W1: Vertical Place-Value Algorithm | W2: Convert to Fractions/Hundredths | W3: Place-Value Split / Area Model

PRACTICE A (p.21): KEY IDEAS (#1-#10)

ExA: 18.45

A1: 345.67 mm
Setup: (345 + 0.6) + 0.07
SC-6: 345 [ENTER] 0.6 [+] 0.07 [+]

A2: 17.55
Setup: 12.8 + 4.75
SC-6: 12.8 [ENTER] 4.75 [+]

A3: 11.35
Setup: 20 - 8.65
SC-6: 20 [ENTER] 8.65 [-]

A4: 2.7
Setup: 4.5 × 0.6
SC-6: 4.5 [ENTER] 0.6 [×]

A5: 18.74
Setup: (25.4 - 9.86) + 3.2
SC-6: 25.4 [ENTER] 9.86 [-] 3.2 [+]

A6: 8.1 (Estimate: 4 × 2 = 8)
Setup: 3.6 × 2.25
SC-6: 3.6 [ENTER] 2.25 [×]

A7: 0.06 (3/50)
Setup: 0.15 × 0.4
SC-6: 0.15 [ENTER] 0.4 [×] [FDISP]

A8: 35.79 g (35.7864)
Setup: 28.86 × 1.24
SC-6: 28.86 [ENTER] 1.24 [×]

A9: $6.65
Setup: 20 - (8.75 + 4.60)
SC-6: 20 [ENTER] 8.75 [ENTER] 4.60 [+] [-]

A10: 161 meters
Setup: ((30 × 1.20) + (40 × 1.35)) + (50 × 1.42)
SC-6: 30 [ENTER] 1.20 [×] 40 [ENTER] 1.35 [×] [+] 50 [ENTER] 1.42 [×..

PRACTICE B (p.22): REAL LIFE (#1-#10)

ExB: $8.10

B1: $41.24
Setup: 18.75 + 22.49
SC-6: 18.75 [ENTER] 22.49 [+]

B2: $6.35
Setup: 20 - 13.65
SC-6: 20 [ENTER] 13.65 [-]

B3: 0.55 seconds faster
Setup: 13.42 - 12.87
SC-6: 13.42 [ENTER] 12.87 [-]

B4: $7.80
Setup: 2.40 × 3.25
SC-6: 2.40 [ENTER] 3.25 [×]

B5: 1.19 cm²
Setup: 1.4 × 0.85
SC-6: 1.4 [ENTER] 0.85 [×]

B6: $41.40
Setup: 11.5 × 3.60
SC-6: 11.5 [ENTER] 3.60 [×]

B7: 30.35 meters
Setup: (24.8 - 6.35) + 11.9
SC-6: 24.8 [ENTER] 6.35 [-] 11.9 [+]

B8: $35.75
Setup: (2 × 14.50) + 6.75
SC-6: 2 [ENTER] 14.50 [×] 6.75 [+]

B9: 2.5 mm gap
Setup: 170.5 - (4 × 42)
SC-6: 170.5 [ENTER] 4 [ENTER] 42 [×] [-]

B10: $35.00 profit
Setup: (3.20 - 1.80) × 25
SC-6: 3.20 [ENTER] 1.80 [-] 25 [×]

SEC 2.5 (pp.23-24): DIVIDING WHOLE NUMBERS AND DECIMALS

Methods: W1: Power-of-10 Decimal Shift & Long Division | W2: Equivalent Fraction Scaling | W3: Reciprocal Fraction Division

PRACTICE A (p.23): KEY IDEAS (#1-#10)

ExA: 5 R 3 = 5 3/4 = 5.75

A1: 4 R 7, 4 7/8, and 4.875
Setup: 39 ÷ 8
SC-6: 39 [ENTER] 8 [÷] [FDISP]

A2: 2.3
Setup: 18.4 ÷ 8
SC-6: 18.4 [ENTER] 8 [÷]

A3: 45 quarters
Setup: 11.25 ÷ 0.25
SC-6: 11.25 [ENTER] 0.25 [÷]

A4: 8
Setup: 7.2 ÷ 0.9
SC-6: 7.2 [ENTER] 0.9 [÷]

A5: 130
Setup: 15.6 ÷ 0.12
SC-6: 15.6 [ENTER] 0.12 [÷]

A6: 6.3 (6 3/10)
Setup: 9.45 ÷ 1.5
SC-6: 9.45 [ENTER] 1.5 [÷] [FDISP]

A7: $20.80
Setup: 124.80 ÷ 6
SC-6: 124.80 [ENTER] 6 [÷]

A8: 24 (24 × 0.2 = 4.8)
Setup: 4.8 ÷ 0.2
SC-6: 4.8 [ENTER] 0.2 [÷]

A9: 6 laps
Setup: 4.5 ÷ 0.75
SC-6: 4.5 [ENTER] 0.75 [÷]

A10: $1.25 per bottle; 16 bottles
Setup: 7.50 ÷ 6; 20 ÷ 1.25
SC-6: 7.50 [ENTER] 6 [÷] -> 1.25; 20 [ENTER] 1.25 [÷]

PRACTICE B (p.24): REAL LIFE (#1-#10)

ExB: 27 quarters

B1: 4 R 3 = 4 3/8 = 35/8 = 4.375
Setup: 35 ÷ 8
SC-6: 35 [ENTER] 8 [÷] [FDISP]

B2: 16 quarters
Setup: 28 ÷ 1.75
SC-6: 28 [ENTER] 1.75 [÷]

B3: $8.65 each
Setup: 34.60 ÷ 4
SC-6: 34.60 [ENTER] 4 [÷]

B4: 72 layers
Setup: 14.4 ÷ 0.20
SC-6: 14.4 [ENTER] 0.20 [÷]

B5: 15 flowers
Setup: 5.25 ÷ 0.35
SC-6: 5.25 [ENTER] 0.35 [÷]

B6: 6 miles per hour
Setup: 4.5 ÷ 0.75
SC-6: 4.5 [ENTER] 0.75 [÷]

B7: 18 sections
Setup: 22.5 ÷ 1.25
SC-6: 22.5 [ENTER] 1.25 [÷]

B8: 30 cups
Setup: 13.5 ÷ 0.45
SC-6: 13.5 [ENTER] 0.45 [÷]

B9: 7 tickets
Setup: 87.50 ÷ 12.50
SC-6: 87.50 [ENTER] 12.50 [÷]

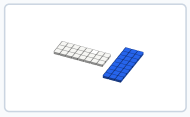
B10: 6
Setup: 10.5 ÷ 1.75
SC-6: 10.5 [ENTER] 1.75 [÷]

Unit 3: Ratios, Rates & Percents — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapters 3, 4 & 5 (Ratios, Rates & Percents) • Hands-On Models: Counting & Array Tiles (Counting & Array Tiles) + Fraction Studio (Fraction Strips)

Target Misconception: Students confuse Part-to-Part (3:2) with Part-to-Whole (3/5), add instead of multiply in ratio tables, divide unit rates upside-down, and memorize 6 disconnected rules for moving decimal points in percents.

Hands-On Block Remediation: Use Blue and Gold Counting Tiles as physical Tape Diagram boxes, step through ratio tables rhythmically with [g] [REP], use [x swap y] when unit-rate division is entered upside-down, and use [f] [%] + [FDISP] to show Fraction/Decimal/Percent are three costumes of one number.



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 3.1 (pp.26-27): RATIOS AND TAPE DIAGRAMS

Methods: W1: Tape Diagram (Equal Boxes) | W2: Equivalent Ratio Fraction | W3: Divide by GCF

PRACTICE A (p.26): KEY IDEAS (#1-#10)

ExA: Part-to-Part = 3:2; Part-to-Whole = 3:5

A1: 2 : 3
Setup: Ratio(12:18)
SC-6: 12 [ENTER] 18 [g][a:b]

A2: 3 : 2
Setup: Ratio(24:16)
SC-6: 24 [ENTER] 16 [g][a:b]

A3: 2 : 5
Setup: Ratio(4:10)
SC-6: 4 [ENTER] 10 [g][a:b]

A4: 18 and 24 (Total = 42)
Setup: 4 × 6
SC-6: 3 [ENTER] 6 [×], 4 [ENTER] 6 [×]

A5: 4 L concentrate, 30 L water
Setup: 34 ÷ 17
SC-6: 34 [ENTER] 17 [÷] -> 2/part

A6: 20 dogs
Setup: (36 ÷ 9) × 5
SC-6: 36 [ENTER] 9 [÷] 5 [×]

A7: 25 seventh graders
Setup: (15 ÷ 3) × 5
SC-6: 15 [ENTER] 3 [÷] 5 [×]

A8: 30 electric guitars (12 acoustic)
Setup: (18 ÷ 3) × 5
SC-6: 18 [ENTER] 3 [÷] 5 [×]

A9: 48 gears and 36 gears
Setup: 84 ÷ 7
SC-6: 84 [ENTER] 7 [÷] -> 12/box

A10: 77 coins total (49 gold, 28 silver)
Setup: (21 ÷ 3) × 11
SC-6: 21 [ENTER] 3 [÷] 11 [×]

PRACTICE B (p.27): REAL LIFE (#1-#10)

ExB: 20 wins (and 12 losses)

B1: Part-to-Part = 3:2; Part-to-Whole = 3:5
Setup: Ratio(15:10)
SC-6: 15 [ENTER] 10 [g][a:b]

B2: 16 strawberries, 12 bananas
Setup: (28 ÷ 7) × 4
SC-6: 28 [ENTER] 7 [÷] 4 [×]

B3: 6 cups yellow (21 cups total)
Setup: (15 ÷ 5) × 2
SC-6: 15 [ENTER] 5 [÷] 2 [×]

B4: 35 secret and 25 open
Setup: (60 ÷ 12) × 7
SC-6: 60 [ENTER] 12 [÷] 7 [×]

B5: 42 gold coins, 18 silver coins
Setup: (24 ÷ 4) × 7
SC-6: 24 [ENTER] 4 [÷] 7 [×]

B6: 3 : 1
Setup: Ratio(48:16)
SC-6: 48 [ENTER] 16 [g][a:b]

B7: 4 : 9 (16 out of 36)
Setup: Ratio(16:36)
SC-6: 16 [ENTER] 36 [g][a:b]

B8: 16 : 9
Setup: Ratio(1920:1080)
SC-6: 1920 [ENTER] 1080 [g][a:b]

B9: 150g almonds, 100g cashews, 250g raisins
Setup: 500 ÷ 10
SC-6: 500 [ENTER] 10 [÷]

B10: 3 : 4
Setup: Ratio(24:32)
SC-6: 24 [ENTER] 32 [g][a:b]

SEC 3.2 (pp.28-29): USING RATIO TABLES AND DOUBLE NUMBER LINES

Methods: W1: Multiplicative Scale Factor | W2: Double Number Line / Repeated Addition | W3: Scale Down Then Up

PRACTICE A (p.28): KEY IDEAS (#1-#10)

ExA: 40 meters

A1: 4:9, 8:18, 12:27, 16:36
Setup: Repeat(+4), Repeat(+9)
SC-6: 4 [g][REP] [+], 9 [g][REP] [+]

A2: 63
Setup: (28 ÷ 4) × 9
SC-6: 28 [ENTER] 4 [÷] 9 [×]

A3: 25
Setup: (60 ÷ 12) × 5
SC-6: 60 [ENTER] 12 [÷] 5 [×]

A4: 60 pages
Setup: (8 ÷ 2) × 15
SC-6: 8 [ENTER] 2 [÷] 15 [×]

A5: 48 ounces
Setup: (18 ÷ 3) × 8
SC-6: 18 [ENTER] 3 [÷] 8 [×]

A6: $15
Setup: (12 ÷ 4) × 5
SC-6: 12 [ENTER] 4 [÷] 5 [×]

A7: 84 beats
Setup: (60 ÷ 15) × 21
SC-6: 60 [ENTER] 15 [÷] 21 [×]

A8: Ratios scale by multiplication (e.g., ×3 gives 9:15), not ad..
Setup: Ratio(7:9)
SC-6: 7 [ENTER] 9 [g][a:b]

A9: 20 cups
Setup: (12 ÷ 3) × 5
SC-6: 12 [ENTER] 3 [÷] 5 [×]

A10: 40 miles
Setup: (14 ÷ 7) × 20
SC-6: 14 [ENTER] 7 [÷] 20 [×]

PRACTICE B (p.29): REAL LIFE (#1-#10)

ExB: $75

B1: 45 grams
Setup: 15 × 3
SC-6: 15 [ENTER] 3 [×]

B2: 56 scoops
Setup: (16 ÷ 2) × 7
SC-6: 16 [ENTER] 2 [÷] 7 [×]

B3: $15 for 10 notebooks
Setup: (12 ÷ 4) × 5
SC-6: 12 [ENTER] 4 [÷] 5 [×]

B4: 112 beats per minute
Setup: 28 × 4
SC-6: 28 [ENTER] 4 [×]

B5: 260 gallons of sap
Setup: 40 × 6.5
SC-6: 40 [ENTER] 6.5 [×]

B6: 25 pounds
Setup: 150 ÷ 6
SC-6: 150 [ENTER] 6 [÷]

B7: 15 cups of flour
Setup: (6 ÷ 2) × 5
SC-6: 6 [ENTER] 2 [÷] 5 [×]

B8: 75 minutes (1 hr 15 min)
Setup: (90 ÷ 18) × 15
SC-6: 90 [ENTER] 18 [÷] 15 [×]

B9: 8 inches
Setup: (60 ÷ 15) × 2
SC-6: 60 [ENTER] 15 [÷] 2 [×]

B10: 30 lemons
Setup: (9 ÷ 3) × 10
SC-6: 9 [ENTER] 3 [÷] 10 [×]

SEC 3.3 (pp.30-31): RATES, UNIT RATES, AND COMPARING RATES

Methods: W1: Divide to Find the Unit Rate (1 Unit) | W2: Common Multiple Comparison | W3: Double Number Line

PRACTICE A (p.30): KEY IDEAS (#1-#10)

ExA: 63 words per minute

A1: 65 mph
Setup: 195 ÷ 3
SC-6: 195 [ENTER] 3 [÷]

A2: $9 per hour
Setup: 36 ÷ 4
SC-6: 36 [ENTER] 4 [÷]

A3: $1.30 per pound
Setup: 7.80 ÷ 6
SC-6: 7.80 [ENTER] 6 [÷]

A4: 15 grams per hour
Setup: 90 ÷ 6
SC-6: 90 [ENTER] 6 [÷]

A5: Supplier B ($0.17/oz vs. $0.18/oz)
Setup: 3.24 ÷ 18; 4.08 ÷ 24
SC-6: 3.24 [ENTER] 18 [÷]; 4.08 [ENTER] 24 [÷]

A6: 10-pack ($0.82/pen vs. $0.85/pen)
Setup: 5.10 ÷ 6; 8.20 ÷ 10
SC-6: 5.10 [ENTER] 6 [÷]; 8.20 [ENTER] 10 [÷]

A7: 300 miles
Setup: (210 ÷ 3.5) × 5
SC-6: 210 [ENTER] 3.5 [÷] 5 [×]

A8: Pump A (9 gal/min vs. 8 gal/min)
Setup: 45 ÷ 5; 64 ÷ 8
SC-6: 45 [ENTER] 5 [÷]; 64 [ENTER] 8 [÷]

A9: 180 mm
Setup: (45 ÷ 15) × 60
SC-6: 45 [ENTER] 15 [÷] 60 [×]

A10: Car B by 1 mpg (29 mpg vs. 28 mpg)
Setup: 168 ÷ 6; 232 ÷ 8
SC-6: 168 [ENTER] 6 [÷]; 232 [ENTER] 8 [÷]

PRACTICE B (p.31): REAL LIFE (#1-#10)

ExB: Pack B ($1.40 per battery vs. $1.45 per battery)

B1: 150 miles per hour
Setup: 450 ÷ 3
SC-6: 450 [ENTER] 3 [÷]

B2: Box B ($0.70/bar vs. $0.75/bar)
Setup: 7.00 ÷ 10
SC-6: 4.50 [ENTER] 6 [÷] 7.00 [ENTER] 10 [÷]

B3: Spool B ($1.95 vs. $2.20 per 100g)
Setup: 19.50 ÷ 10
SC-6: 11 [ENTER] 5 [÷] 19.50 [ENTER] 10 [÷]

B4: 15 gal/min; 375 gallons
Setup: (180 ÷ 12) × 25
SC-6: 180 [ENTER] 12 [÷] 25 [×]

B5: B (57 taps/min vs. A's 55 taps/min)
Setup: 285 ÷ 5
SC-6: 220 [ENTER] 4 [÷] 285 [ENTER] 5 [÷]

B6: Car A (28 mpg vs. 27 mpg)
Setup: 405 ÷ 15
SC-6: 336 [ENTER] 12 [÷] 405 [ENTER] 15 [÷]

B7: $9 per dog -> 11 dogs
Setup: 99 ÷ (54 ÷ 6)
SC-6: 54 [ENTER] 6 [÷] 99 [x swap y] [÷]

B8: 12-pack ($0.42 vs. $0.45 each)
Setup: 5.04 ÷ 12
SC-6: 3.60 [ENTER] 8 [÷] 5.04 [ENTER] 12 [÷]

B9: 4 inches per hour
Setup: 14 ÷ 3.5
SC-6: 14 [ENTER] 3.5 [÷]

B10: 4.5 m/s -> 90 meters
Setup: (27 ÷ 6) × 20
SC-6: 27 [ENTER] 6 [÷] 20 [×]

SEC 3.4 (pp.32-33): PERCENTS, FRACTIONS, AND DECIMALS

Methods: W1: Equivalent Fraction over 100 | W2: Divide Numerator ÷ Denominator | W3: Percent Bar / Benchmark Model

PRACTICE A (p.32): KEY IDEAS (#1-#10)

ExA: 0.35 and 35%

A1: 0.6 and 60%
Setup: 3/5 -> Dec/Frac
SC-6: [.] 3 [.] 5 [ENTER] [FDISP]

A2: 0.36 and 36%
Setup: 9/25 -> Dec/Frac
SC-6: [.] 9 [.] 25 [ENTER] [FDISP]

A3: 0.45 and 9/20
Setup: 45 ÷ 100
SC-6: 45 [ENTER] 100 [÷] [FDISP]

A4: 12.5% and 1/8
Setup: 0.125 -> Dec/Frac
SC-6: 0.125 [ENTER] [FDISP]

A5: 270 seats
Setup: 60% of 450
SC-6: 450 [ENTER] 60 [%]

A6: 3,510 grams
Setup: 65% of 5400
SC-6: 5400 [ENTER] 65 [%]

A7: 84%
Setup: 42 ÷ 50
SC-6: 42 [ENTER] 50 [÷]

A8: 1.5 and 1 1/2
Setup: 150 ÷ 100
SC-6: 150 [ENTER] 100 [÷] [FDISP]

A9: 60%, 0.61, 5/8
Setup: 5/8 -> Dec/Frac
SC-6: [.] 5 [.] 8 [FDISP] -> 0.625

A10: 1,400 mAh
Setup: 35% of 4000
SC-6: 4000 [ENTER] 35 [%]

PRACTICE B (p.33): REAL LIFE (#1-#10)

ExB: 75% (3/4 = 0.75)

B1: 0.85 = 85%
Setup: 17/20 × 100
SC-6: [.] 17 [.] 20 [FDISP] 100 [×]

B2: 0.36 = 36%
Setup: 9/25 × 100
SC-6: [.] 9 [.] 25 [FDISP] 100 [×]

B3: 20 + 40 = 60 points
Setup: 40% of 50 + 80% of 50
SC-6: 50 [ENTER] 40 [%] 50 [ENTER] 80 [%] [+]

B4: 104 students
Setup: 65% of 160
SC-6: 160 [ENTER] 65 [%]

B5: 0.15 and 3/20
Setup: Simp(15/100)
SC-6: [.] 15 [.] 100 [g][SIMP]

B6: 45 students total
Setup: 18 ÷ 0.40
SC-6: 18 [ENTER] 0.40 [÷]

B7: $9.00
Setup: 15% of 60
SC-6: 60 [ENTER] 15 [%]

B8: 7/8 = 0.875 = 87.5%
Setup: 21 ÷ 24
SC-6: 21 [ENTER] 24 [÷] [FDISP]

B9: $480 goal
Setup: 360 ÷ 0.75
SC-6: 360 [ENTER] 0.75 [÷]

B10: 58% < 3/5 (60%) < 0.62 (62%)
Setup: 3/5 -> Dec/Frac
SC-6: [.] 3 [.] 5 [FDISP]

SEC 3.5 (pp.34-35): SOLVING PERCENT PROBLEMS, DISCOUNTS & CONVERSIONS

Methods: W1: Find Discount & Subtract | W2: Multiply by Remaining % | W3: Percent Proportion / Ratio Bar

PRACTICE A (p.34): KEY IDEAS (#1-#10)

ExA: Discount = $15; Sale Price = $45

A1: Discount $17; Sale price $68
Setup: 20% of 85
SC-6: 85 [ENTER] 20 [%] [-]

A2: Tax $10.20; Total $130.20
Setup: 8.5% of 120
SC-6: 120 [ENTER] 8.5 [%] [+]

A3: +400% ((75-15)÷15 = 400%)
Setup: %Chg(15->75)
SC-6: 15 [ENTER] 75 [g][%CHG]

A4: 25% decrease
Setup: %Chg(80->60)
SC-6: 80 [ENTER] 60 [g][%CHG]

A5: 25.4 cm
Setup: 10 (to cm)
SC-6: 10 [f][->cm]

A6: 20 in
Setup: 50.8 (to in)
SC-6: 50.8 [g][->in]

A7: $39.20
Setup: 30% of 56
SC-6: 56 [ENTER] 30 [%] [-]

A8: 72
Setup: 18 ÷ 0.25
SC-6: 18 [ENTER] 0.25 [÷]

A9: $144 ($200 -> $160 -> $144)
Setup: 10% of 20% of 200
SC-6: 200 [ENTER] 20 [%] [-] 10 [%] [-]

A10: 81.28 cm and 2.27 kg
Setup: 32 (to cm); 5 (to kg)
SC-6: 32 [f][->cm]; 5 [f][->kg]

PRACTICE B (p.35): REAL LIFE (#1-#10)

ExB: 30% increase; 81.28 cm

B1: Discount = $15; Sale Price = $60
Setup: 20% of 75
SC-6: 75 [ENTER] 20 [%] [-]

B2: Tip = $9.60; Total = $73.60
Setup: 15% of 64
SC-6: 64 [ENTER] 15 [%] [+]

B3: Tax = $3.60; Total = $51.60
Setup: 7.5% of 48
SC-6: 48 [ENTER] 7.5 [%] [+]

B4: 30% increase
Setup: %Chg(40->52)
SC-6: 40 [ENTER] 52 [g][%CHG]

B5: 30% decrease
Setup: %Chg(90->63)
SC-6: 90 [ENTER] 63 [g][%CHG]

B6: 16.09 km
Setup: 10 × 1.609344
SC-6: 10 [ENTER] 1.609344 [×]

B7: 44.09 lb
Setup: 20 (to lb)
SC-6: 20 [g][->lb]

B8: 3.79 liters; 128 fl oz
Setup: 1 (to L)
SC-6: 1 [f][->L]

B9: $72 (10% is taken off $80, not $100)
Setup: 10% of 20% of 100
SC-6: 100 [ENTER] 20 [%] [-] 10 [%] [-]

B10: 68°F
Setup: 20 (to °F)
SC-6: 20 [g][->°F]

Unit 4: Algebraic Expressions & Properties — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 3 / Chapter 5 (Algebraic Expressions) • Hands-On Models: Algebra Boxes & Factoring Stage (Algebra & Area Tiles) + Counting Tiles (Counting & Array Tiles)

Target Misconception: Students panic when letters appear, erroneously combine $3x + 2$ into $5x$ or $x^2 + x$ into x^8, and don't see why $3(x + 4) = 3x + 12$.

Hands-On Block Remediation: Show that a variable x is literally a physical green x-rod (42×14mm) whereas 1 is a small yellow unit cube (14×14mm) and x^2 is a large blue flat (42×42mm)—different physical shapes cannot be counted as one pile! Build rectangles on the L-frame stage for the Distributive Property.



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 4.1 (pp.37-38): WRITING AND EVALUATING ALGEBRAIC EXPRESSIONS

Methods: W1: Substitute in Parentheses & Follow PEMDAS | W2: Equal-Groups Bar / Algebra Tile Model

PRACTICE A (p.37): KEY IDEAS (#1-#10)

ExA: 27

A1: 33
Setup: $(6 \times 7) - 9$
SC-6: 6 [ENTER] 7 [×] 9 [-]

A2: 54
Setup: $(5 \times 8) + 14$
SC-6: 5 [ENTER] 8 [×] 14 [+]

A3: 13
Setup: $40 - (3 \times 9)$
SC-6: 40 [ENTER] 3 [ENTER] 9 [×] [-]

A4: 47
Setup: $6^2 + 11$
SC-6: 6 [x²] 11 [+]

A5: 27
Setup: $(4^2 \times 2) - 5$
SC-6: 4 [x²] 2 [×] 5 [-]

A6: 55
Setup: $(7 \times 4) + (3 \times 9)$
SC-6: 7 [ENTER] 4 [×] 3 [ENTER] 9 [×] [+]

A7: 8
Setup: $(19 + 13) \div 4$
SC-6: 19 [ENTER] 13 [+] 4 [÷]

A8: 15.1
Setup: $(5 \times 3.5) - 2.4$
SC-6: 5 [ENTER] 3.5 [×] 2.4 [-]

A9: 8 + 4.50h; $26
Setup: $(4.50 \times 4) + 8$
SC-6: 4.50 [ENTER] 4 [×] 8 [+]

A10: 150 cm²
Setup: $5^2 \times 6$
SC-6: 5 [x²] 6 [×]

PRACTICE B (p.38): REAL LIFE (#1-#10)

ExB: 99 points

B1: 5g + 6; $26
Setup: $(5 \times 4) + 6$
SC-6: 5 [ENTER] 4 [×] 6 [+]

B2: 50 - 4r; 14 tickets
Setup: $50 - (4 \times 9)$
SC-6: 50 [ENTER] 4 [ENTER] 9 [×] [-]

B3: $225
Setup: $(5^2 \times 8) + 25$
SC-6: 5 [x²] 8 [×] 25 [+]

B4: 82 points
Setup: $(10 \times 7) + (3 \times 4)$
SC-6: 10 [ENTER] 7 [×] 3 [ENTER] 4 [×] [+]

B5: 144 feet
Setup: $3^2 \times 16$
SC-6: 3 [x²] 16 [×]

B6: 96 in²
Setup: $4^2 \times 6$
SC-6: 4 [x²] 6 [×]

B7: 7p + 4d; $41
Setup: $(7 \times 3) + (4 \times 5)$
SC-6: 7 [ENTER] 3 [×] 4 [ENTER] 5 [×] [+]

B8: 78
Setup: $A = \frac{1}{2}((11 + 15) \times 6)$
SC-6: 11 [ENTER] 15 [+] 6 [g][TRI]

B9: 111 minutes (1 hr 51 min)
Setup: $(12 \times 8) + 15$
SC-6: 12 [ENTER] 8 [×] 15 [+]

B10: 128 - 75 = 53
Setup: $(4^3 \times 2) - (5^2 \times 3)$
SC-6: 4 [x²] 2 [×] 5 [x²] 3 [×] [-]

SEC 4.2 (pp.39-40): SIMPLIFYING EXPRESSIONS BY COMBINING LIKE TERMS

Methods: W1: Reorder & Add Coefficients (Properties) | W2: Sort by Tile Shape / Column Alignment

PRACTICE A (p.39): KEY IDEAS (#1-#10)

ExA: 6x + 8

A1: 9x + 13
Setup: $7 + 2$; $4 + 9$
SC-6: 7 [ENTER] 2 [+]; 4 [ENTER] 9 [+]

A2: 5x + 7
Setup: 8 - 3; 11 - 4
SC-6: 8 [ENTER] 3 [-]; 11 [ENTER] 4 [-]

A3: 8y + 6
Setup: 9 - 1
SC-6: 9 [ENTER] 1 [-]

A4: 8a + 13
Setup: 12 - 4; $5 + 8$
SC-6: 12 [ENTER] 4 [-]; 5 [ENTER] 8 [+]

A5: 6x² + 8x + 7
Setup: 4+2=6; 3+5=8
SC-6: Mental / Table Check

A6: 4n² + 5n + 10
Setup: 6-2=4; 9-4=5
SC-6: Mental / Table Check

A7: 5.3x + 2.7
Setup: $3.5 + 1.8$; 4.2 - 1.5
SC-6: 3.5 [ENTER] 1.8 [+]; 4.2 [ENTER] 1.5 [-]

A8: 53 != 80; 5x + 3 is already in simplest form
Setup: 5(10)+3=53 vs 8(10)=80
SC-6: Mental / Table Check

A9: 6x + 16; 76 meters
Setup: 2+2+1+1=6x; 3+3+5+5=16
SC-6: Mental / Table Check

A10: 9x + 8; 53 cm
Setup: 3+2+4=9x; 2+7-1=8
SC-6: Mental / Table Check

PRACTICE B (p.40): REAL LIFE (#1-#10)

ExB: 5x² + 4x + 6

B1: 9x + 9
Setup: $(3 + 1) + 5$
SC-6: 2 [ENTER] 3 [+] 4 [+] 3 [ENTER] 1 [+] 5 [+]

B2: 8x + 12
Setup: $(4 + 2) \times 2$
SC-6: 3 [ENTER] 1 [+] 2 [×] 4 [ENTER] 2 [+] 2 [×]

B3: 4x + 14
Setup: $9 + 5$
SC-6: 6 [ENTER] 2 [-] 9 [ENTER] 5 [+]

B4: 6x² + 4x + 11
Setup: $3 + 8$
SC-6: 4 [ENTER] 2 [+] 7 [ENTER] 3 [-] 3 [ENTER] 8 [+]

B5: 13c + 31
Setup: $12 + 19$
SC-6: 8 [ENTER] 5 [+] 12 [ENTER] 19 [+]

B6: 7x + 19 (coeff = 7, constant = 19)
Setup: $14 + 5$
SC-6: 9 [ENTER] 2 [-] 14 [ENTER] 5 [+]

B7: (1/2)n + 13
Setup: $4 + 9$
SC-6: [.] 3 [.] 8 [ENTER] [.] 1 [.] 8 [+] 4 [ENTER] 9 [+]

B8: 4y + 3.5
Setup: 5.8 - 2.3
SC-6: 2.4 [ENTER] 1.6 [+] 5.8 [ENTER] 2.3 [-]

B9: 6m + 9 (both equal 39 at m=5)
Setup: $((10 - 4) \times 5) + (15 - 6)$
SC-6: 10 [ENTER] 4 [-] 5 [×] 15 [ENTER] 6 [-] [+]

B10: 3x² + 7
Setup: 10 - 3
SC-6: 5 [ENTER] 2 [-] 10 [ENTER] 3 [-]

SEC 4.3 (pp.41-42): THE DISTRIBUTIVE PROPERTY

Methods: W1: Multiply Each Term Inside a(b + c) = ab + ac | W2: Split-Rectangle Area Model | W3: Repeated Addition of Rows

PRACTICE A (p.41): KEY IDEAS (#1-#10)

ExA: 84

A1: 140 + 21 = 161
Setup: $(7 \times 20) + (7 \times 3)$
SC-6: 7 [ENTER] 20 [×] 7 [ENTER] 3 [×] [+]

A2: 4x + 20
Setup: 4×5
SC-6: 4 [ENTER] 5 [×]

A3: 18x + 24
Setup: 6×3=18; 6×4=24
SC-6: Mental / Table Check

A4: 16y - 40
Setup: 8×2=16; 8×5=40
SC-6: Mental / Table Check

A5: 387
Setup: $(9 \times 40) + (9 \times 3)$
SC-6: 9 [ENTER] 40 [×] 9 [ENTER] 3 [×] [+]

A6: 17x + 6
Setup: 3×4+5=17; 3×2=6
SC-6: Mental / Table Check

A7: 12n + 13
Setup: 4×3=12; 20-7=13
SC-6: Mental / Table Check

A8: 5x + 30 (She forgot to multiply 6 by 5)
Setup: 5×6
SC-6: 5 [ENTER] 6 [×]

A9: 14x + 15 (Both equal 43 when x = 2)
Setup: 5×2+4=14; 5×3=15
SC-6: Mental / Table Check

A10: 6(x + 3.50) = 6x + 21; $60 total
Setup: 6×3.50
SC-6: 6 [ENTER] 3.50 [×] -> 6x+21

PRACTICE B (p.42): REAL LIFE (#1-#10)

ExB: 14x + 15

B1: 8(50 + 3) = 400 + 24 = $424
Setup: $(8 \times 50) + (8 \times 3)$
SC-6: 8 [ENTER] 50 [×] 8 [ENTER] 3 [×] [+]

B2: 6(10 - 0.10) = $60.00 - $0.60 = $59.40
Setup: $(6 \times 10) - (6 \times 0.10)$
SC-6: 6 [ENTER] 10 [×] 6 [ENTER] 0.10 [×] [-]

B3: 7(x + 8) = 7x + 56
Setup: 7×8
SC-6: 7 [ENTER] 8 [×]

B4: 9(2x + 7) = 18x + 63
Setup: 9×7
SC-6: 9 [ENTER] 2 [×] 9 [ENTER] 7 [×]

B5: 4(t - 3) = 4t - 12 ($48 at t = $15)
Setup: $(4 \times 15) - (4 \times 3)$
SC-6: 4 [ENTER] 15 [×] 4 [ENTER] 3 [×] [-]

B6: 14x + 15
Setup: 3×5
SC-6: 3 [ENTER] 4 [×] 2 [+] 3 [ENTER] 5 [×]

B7: 13x + 22
Setup: $(4 \times 3) + (5 \times 2)$
SC-6: 4 [ENTER] 2 [×] 5 [+] 4 [ENTER] 3 [×] 5 [ENTER] 2 [×] [+]

B8: 3x + 5
Setup: $10 \div 2$
SC-6: 6 [ENTER] 2 [÷] 10 [ENTER] 2 [÷]

B9: 2(2x + 5) + 2(x + 3) = 6x + 16
Setup: $(2 \times 5) + (2 \times 3)$
SC-6: 2 [ENTER] 2 [×] 2 [+] 2 [ENTER] 5 [×] 2 [ENTER] 3 [×] [+]

B10: 12(15) = 180 and 120 + 60 = 180
Setup: 12×15
SC-6: 12 [ENTER] 15 [×]

SEC 4.4 (pp.43-44): FACTORING EXPRESSIONS USING THE GCF

Methods: W1: Factor Out the GCF Using Prime Factors | W2: Equal-Rows Area Model / Division Ladder

PRACTICE A (p.43): KEY IDEAS (#1-#10)

ExA: 6(3x + 4)

A1: 4(3 + 5)
Setup: GCF(12, 20)
SC-6: 12 [ENTER] 20 [GCF] -> 4

A2: 4(2x + 3)
Setup: GCF(8, 12)
SC-6: 8 [ENTER] 12 [GCF] -> 4

A3: 5(3y - 5)
Setup: GCF(15, 25)
SC-6: 15 [ENTER] 25 [GCF] -> 5

A4: 12(2x + 3)
Setup: GCF(24, 36)
SC-6: 24 [ENTER] 36 [GCF] -> 12

A5: 16(2n - 3)
Setup: GCF(32, 48)
SC-6: 32 [ENTER] 48 [GCF] -> 16

A6: 14(3a + 1)
Setup: GCF(42, 14)
SC-6: 42 [ENTER] 14 [GCF] -> 14

A7: 12(3m - 5)
Setup: GCF(36, 60)
SC-6: 36 [ENTER] 60 [GCF] -> 12

A8: 12(2x + 3) (4 and 6 still share a factor of 2)
Setup: GCF(4, 6)
SC-6: 4 [ENTER] 6 [GCF] -> 2 left!

A9: Height = 14 in.; Width = (2x + 3) in.
Setup: GCF(28, 42)
SC-6: 28 [ENTER] 42 [GCF] -> 14

A10: (9x + 12) feet
Setup: $36 \div 4$; $48 \div 4$
SC-6: 36 [ENTER] 4 [÷]; 48 [ENTER] 4 [÷]

PRACTICE B (p.44): REAL LIFE (#1-#10)

ExB: 15(3x - 2)

B1: 8(2x + 3) -> 8 bags
Setup: GCF(16, 24)
SC-6: 16 [ENTER] 24 [GCF]

B2: 7(3x + 5) -> length = 3x + 5 m
Setup: GCF(21, 35)
SC-6: 21 [ENTER] 35 [GCF]

B3: 4(5x + 9) -> side = 5x + 9 in
Setup: GCF(20, 36)
SC-6: 20 [ENTER] 36 [GCF]

B4: 3(6y - 9) -> side = 6y - 9 cm
Setup: GCF(18, 27)
SC-6: 18 [ENTER] 27 [GCF]

B5: 8(5x + 8)
Setup: GCF(40, 64)
SC-6: 40 [ENTER] 64 [GCF]

B6: 18(2s + 3) -> 18 player kits
Setup: GCF(36, 54)
SC-6: 36 [ENTER] 54 [GCF]

B7: 6(2x + 3y + 5)
Setup: GCF(GCF(12, 18), 30)
SC-6: 12 [ENTER] 18 [GCF] 30 [GCF]

B8: 7(5d - 2) -> 5d - 2 per row
Setup: GCF(35, 14)
SC-6: 35 [ENTER] 14 [GCF]

B9: 24x + 36 = 12(2x + 3)
Setup: GCF((10 + 14), (18 + 18))
SC-6: 10 [ENTER] 14 [+] 18 [ENTER] 18 [+] [GCF]

B10: 9(3x - 2); at x=2 both equal 36
Setup: GCF(27, 18)
SC-6: 27 [ENTER] 18 [GCF]

Unit 5: Areas of Polygons & Elapsed Time — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 4 (Areas of Polygons) • Hands-On Models: Counting & Array Tiles (Counting & Array Tiles) + Planetary Geared Clock (Geared Modulo-60 Clock)

Target Misconception: Students multiply the slanted side of a parallelogram instead of the vertical height, forget to divide by 2 for triangles, lose track of multi-piece composite areas, and subtract clock times in base-10 instead of base-60.

Hands-On Block Remediation: Slide the corner triangle of a parallelogram to form a straight [f] [RECT] rectangle, use the dedicated [g] [TRI] key ([g][7]) which divides by 2 automatically, accumulate composite chunks with [Σ+], and use the Planetary Clock + [CLK] key for base-60 time.



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 5.1 (pp.46-47): AREAS OF PARALLELOGRAMS AND RECTANGLES

Methods: W1: Area Formula A = bh | W2: Cut-and-Slide Rectangle Decomposition

PRACTICE A (p.46): KEY IDEAS (#1-#10)

ExA: 96 cm²

A1: 135 m²
Setup: A = 15 × 9
SC-6: 15 [ENTER] 9 [f][RECT]

A2: 36 m²
Setup: A = 8 × 4.5
SC-6: 8 [ENTER] 4.5 [f][RECT]

A3: 54.4 m²
Setup: A = 8.5 × 6.4
SC-6: 8.5 [ENTER] 6.4 [f][RECT]

A4: 169 cm²
Setup: 13²
SC-6: 13 [x²]

A5: 26 in.²
Setup: A = 6 1/2 × 4
SC-6: 6 [.] 1 [.] 2 [ENTER] 4 [f][RECT]

A6: 9 cm
Setup: 108 ÷ 12
SC-6: 108 [ENTER] 12 [÷]

A7: 18 ft
Setup: 135 ÷ 7.5
SC-6: 135 [ENTER] 7.5 [÷]

A8: 60 cm² (He multiplied by the slanted side 7.5)
Setup: A = 10 × 6
SC-6: 10 [ENTER] 6 [f][RECT]

A9: 666 ft²
Setup: (9×18.5) × 4
SC-6: 9 [ENTER] 18.5 [f][RECT] 4 [×]

A10: 168 ft²
Setup: (15×12) - (4×3)
SC-6: 15 [ENTER] 12 [f][RECT] 4 [ENTER] 3 [f][RECT] [-]

PRACTICE B (p.47): REAL LIFE (#1-#10)

ExB: 12 inches

B1: 180 ft²
Setup: A = 10 × 18
SC-6: 10 [ENTER] 18 [f][RECT]

B2: 60 ft
Setup: (10 + 20) × 2
SC-6: 10 [ENTER] 20 [+] 2 [×]

B3: 14.4 ft²
Setup: A = 4.5 × 3.2
SC-6: 4.5 [ENTER] 3.2 [f][RECT]

B4: 15 inches
Setup: 135 ÷ 9
SC-6: 135 [ENTER] 9 [÷]

B5: 216 cm²
Setup: (6×4.5) × 8
SC-6: 6 [ENTER] 4.5 [f][RECT] 8 [×]

B6: 228 ft²
Setup: A = 19 × 12
SC-6: 19 [ENTER] 12 [f][RECT]

B7: 2,520 ft²
Setup: (65×40) - (10×8)
SC-6: 65 [ENTER] 40 [f][RECT] 10 [ENTER] 8 [f][RECT] [-]

B8: 22 in²
Setup: 5 1/2 × 4
SC-6: 5 [.] 1 [.] 2 [ENTER] 4 [×]

B9: Both equal 144 cm² (same base and perpendicular height)
Setup: A = 16 × 9
SC-6: 16 [ENTER] 9 [f][RECT]

B10: 28 ft² × $4 = $112.00
Setup: (8×3.5) × 4
SC-6: 8 [ENTER] 3.5 [f][RECT] 4 [×]

SEC 5.2 (pp.48-49): AREAS OF TRIANGLES

Methods: W1: Formula A = (b × h) ÷ 2 | W2: Take Half of Base or Height First (½b × h)

PRACTICE A (p.48): KEY IDEAS (#1-#10)

ExA: 30 cm²

A1: 750 cm²
Setup: A = ½(25 × 60)
SC-6: 25 [ENTER] 60 [g][TRI]

A2: 18 ft²
Setup: A = ½(9 × 4)
SC-6: 9 [ENTER] 4 [g][TRI]

A3: 88 cm²
Setup: A = ½(16 × 11)
SC-6: 16 [ENTER] 11 [g][TRI]

A4: 52.5 m²
Setup: A = ½(12.5 × 8.4)
SC-6: 12.5 [ENTER] 8.4 [g][TRI]

A5: 13 1/2 in.² (13.5)
Setup: A = ½(4 1/2 × 6)
SC-6: 4 [.] 1 [.] 2 [ENTER] 6 [g][TRI]

A6: 9 cm
Setup: (54 × 2) ÷ 12
SC-6: 54 [ENTER] 2 [×] 12 [÷]

A7: 12 m
Setup: (84 × 2) ÷ 14
SC-6: 84 [ENTER] 2 [×] 14 [÷]

A8: 20 cm² (He forgot to divide by 2)
Setup: A = ½(8 × 5)
SC-6: 8 [ENTER] 5 [g][TRI]

A9: 210 ft²
Setup: ½(12×25) + ½(8×15)
SC-6: 12 [ENTER] 25 [g][TRI] 8 [ENTER] 15 [g][TRI] [+]

A10: 1,400 cm²
Setup: A = ½(40 × 70)
SC-6: 40 [ENTER] 70 [g][TRI]

PRACTICE B (p.49): REAL LIFE (#1-#10)

ExB: 8 inches

B1: 150 ft²
Setup: A = ½(12 × 25)
SC-6: 12 [ENTER] 25 [g][TRI]

B2: 14 ft² (1 side); 28 ft² (both)
Setup: A = ½(8 × 3.5)
SC-6: 8 [ENTER] 3.5 [g][TRI]

B3: 141 ft²
Setup: A = ½(30 × 9.4)
SC-6: 30 [ENTER] 9.4 [g][TRI]

B4: 36 inches
Setup: (450 × 2) ÷ 25
SC-6: 450 [ENTER] 2 [×] 25 [÷]

B5: 6 feet
Setup: (18 × 2) ÷ 6
SC-6: 18 [ENTER] 2 [×] 6 [÷]

B6: 10.125 in² (10 1/8 in²)
Setup: A = ½(4.5 × 4.5)
SC-6: 4.5 [ENTER] 4.5 [g][TRI]

B7: 54 ft²
Setup: A = ½(9 × 12)
SC-6: 9 [ENTER] 12 [g][TRI]

B8: 630 cm² (315 cm² per wing)
Setup: ½(18×35) × 2
SC-6: 18 [ENTER] 35 [g][TRI] 2 [×]

B9: 13 in²
Setup: (6 1/2 × 4) ÷ 2
SC-6: 6 [.] 1 [.] 2 [ENTER] 4 [×] 2 [÷]

B10: 64 ft²
Setup: ½(5×6.4) × 4
SC-6: 5 [ENTER] 6.4 [g][TRI] 4 [×]

SEC 5.3 (pp.50-51): AREAS OF TRAPEZOIDS AND COMPOSITE FIGURES

Methods: W1: Decompose & Add Sub-Areas | W2: Trapezoid Formula A = ½h(b1 + b2) | W3: Enclosing Box Minus Cutouts

PRACTICE A (p.50): KEY IDEAS (#1-#10)

ExA: 126 ft²

A1: 72 cm²
Setup: A = ½((10 + 14) × 6)
SC-6: 10 [ENTER] 14 [+] 6 [g][TRI]

A2: 135 ft²
Setup: A = ½((18 + 12) × 9)
SC-6: 18 [ENTER] 12 [+] 9 [g][TRI]

A3: 92 ft²
Setup: Stats(10, 6, 8, 4)
SC-6: 10 [ENTER] 6 [f][RECT] [Σ+] 8 [ENTER] 4 [f][RECT] [Σ+] [f][Σx]

A4: 120 in.²
Setup: Stats(15, 6, 10, 6)
SC-6: 15 [ENTER] 6 [f][RECT] [Σ+] 10 [ENTER] 6 [g][TRI] [Σ+] [f][Σx]

A5: 29.4 m²
Setup: A = ½((8.5 + 5.5) × 4.2)
SC-6: 8.5 [ENTER] 5.5 [+] 4.2 [g][TRI]

A6: 56 in.²
Setup: Stats(10, 8, 6, 4)
SC-6: 10 [ENTER] 8 [f][RECT] [Σ+] 6 [ENTER] 4 [f][RECT] [f][Σ-] [f][Σx]

A7: 10 cm
Setup: (90 × 2) ÷ (11 + 7)
SC-6: 90 [ENTER] 2 [×] 11 [ENTER] 7 [+] [÷]

A8: 88 ft²
Setup: Stats(60, 5, 9, 4)
SC-6: 60 [Σ+] 5 [ENTER] 9 [+] 4 [g][TRI] [Σ+] [f][Σx]

A9: 104 cm² (80 + 24 = 104)
Setup: A = ½((10 + 16) × 8)
SC-6: 10 [ENTER] 16 [+] 8 [g][TRI]

A10: 251 ft²
Setup: Stats(300, 24, 25)
SC-6: 300 [Σ+] 24 [f][Σ-] 25 [f][Σ-] [f][Σx]

PRACTICE B (p.51): REAL LIFE (#1-#10)

ExB: 1,008 square inches

B1: 440 in²
Setup: A = ½((16 + 28) × 20)
SC-6: 16 [ENTER] 28 [+] 20 [g][TRI]

B2: 40 in² per side; 160 in² total
Setup: A = ½((4 + 6) × 8)
SC-6: 4 [ENTER] 6 [+] 8 [g][TRI]

B3: 84 + 50 = 134 ft²
Setup: (14×6) + (10×5)
SC-6: 14 [ENTER] 6 [f][RECT] 10 [ENTER] 5 [f][RECT] [+]

B4: 42 cm² (1 fin); 126 cm² (3 fins)
Setup: A = ½((5 + 9) × 6)
SC-6: 5 [ENTER] 9 [+] 6 [g][TRI]

B5: 90 + 48 = 138 in²
Setup: (15×6) + ½(12×8)
SC-6: 15 [ENTER] 6 [f][RECT] 12 [ENTER] 8 [g][TRI] [+]

B6: 1,152 - 36 = 1,116 in²
Setup: (48×24) - 6²
SC-6: 48 [ENTER] 24 [f][RECT] 6 [x²] [-]

B7: h = 8 ft
Setup: 104 ÷ 13
SC-6: 104 [ENTER] 13 [÷]

B8: 280 + 60 = 340 ft²
Setup: (20×14) + ½(20×6)
SC-6: 20 [ENTER] 14 [f][RECT] 20 [ENTER] 6 [g][TRI] [+]

B9: 416 - 200 = 216 m²
Setup: (26×16) - (20×10)
SC-6: 26 [ENTER] 16 [f][RECT] 20 [ENTER] 10 [f][RECT] [-]

B10: 42 cm²
Setup: ½((4 + 8)×3.5) × 2
SC-6: 4 [ENTER] 8 [+] 3.5 [g][TRI] 2 [×]

SEC 5.4 (pp.52-53): POLYGONS IN THE COORDINATE PLANE & ELAPSED TIME

Methods: W1: Subtract Coordinates / Regroup 60 min = 1 hr | W2: Count Grid Squares / Time-Line Jumps

PRACTICE A (p.52): KEY IDEAS (#1-#10)

ExA: Width = 7, Height = 5, Perimeter = 24, Area = 35

A1: 8 units
Setup: 11 - 3
SC-6: 11 [ENTER] 3 [-]

A2: 54 square units
Setup: A = (10 - 1) × (8 - 2)
SC-6: 10 [ENTER] 1 [-] 8 [ENTER] 2 [-] [f][RECT]

A3: 10:58 AM
Setup: 7.10 + 3.48
SC-6: [CLK] 7.10 [ENTER] 3.48 [+]

A4: 2.5 hours; 4.75 hours
Setup: 2.30 (to HR); 4.45 (to HR)
SC-6: 2.30 [f][->HR]; 4.45 [f][->HR]

A5: 6 hours 50 minutes
Setup: 15.05 - 8.15
SC-6: [CLK] 15.05 [ENTER] 8.15 [-]

A6: 1 hour 45 minutes
Setup: 1.75 (to HMS)
SC-6: 1.75 [g][->HMS]

A7: 2 hours 35 minutes
Setup: 11.25 - 8.50
SC-6: [CLK] 11.25 [ENTER] 8.50 [-]

A8: 24 square units
Setup: A = ½(8 × 6)
SC-6: 8 [ENTER] 6 [g][TRI]

A9: 5.5 hours (5 hr 30 min); $101.75
Setup: 5.5 × 18.50
SC-6: 5.5 [ENTER] 18.50 [×]

A10: Perimeter = 420 yd; Area = 10,800 yd²
Setup: A = 12 × 9
SC-6: 12 [ENTER] 9 [f][RECT]

PRACTICE B (p.53): REAL LIFE (#1-#10)

ExB: 12:30 PM

B1: W=10, H=6, P=32, Area=60
Setup: A = (11 - 1) × (8 - 2)
SC-6: 11 [ENTER] 1 [-] 8 [ENTER] 2 [-] [f][RECT]

B2: Base=8, Height=6, Area=24
Setup: A = ½((10 - 2) × (7 - 1))
SC-6: 10 [ENTER] 2 [-] 7 [ENTER] 1 [-] [g][TRI]

B3: 12:15 PM
Setup: 8.35 + 3.40
SC-6: [CLK] 8.35 [ENTER] 3.40 [+]

B4: 3 hours 35 minutes
Setup: 14.20 - 10.45
SC-6: [CLK] 14.20 [ENTER] 10.45 [-]

B5: 3 hours 50 minutes
Setup: 1.52 + 1.58
SC-6: [CLK] 1.52 [ENTER] 1.58 [+]

B6: 3.5 hr × $14 = $49.00
Setup: 3.5 × 14
SC-6: 3.5 [ENTER] 14 [×]

B7: 2 hours 45 minutes
Setup: 2.75 (to HMS)
SC-6: 2.75 [g][->HMS]

B8: 4 hours 30 minutes
Setup: (1.25 + 2.15) + 0.50
SC-6: [CLK] 1.25 [ENTER] 2.15 [+] 0.50 [+]

B9: (3, 10); Area = 36
Setup: A = (9 - 3) × (10 - 4)
SC-6: 9 [ENTER] 3 [-] 10 [ENTER] 4 [-] [f][RECT]

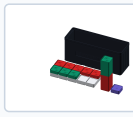
B10: 180 miles
Setup: 2.25 × 80
SC-6: 2.25 [ENTER] 80 [×]

Unit 6: Integers, Rational Numbers & The Coordinate Plane — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 6 • Hands-On Models: Ten-Frame Two-Tone Counters (Ten-Frames & Two-Tone Counters) + Algebra Zero-Pair Tiles (Algebra & Area Tiles)

Target Misconception: Students confuse the subtraction operator [-] with the negative sign [+/-], think -18 > -5 because 18 > 5, confuse Opposite ([+/-]) with Absolute Value ([g] [|x|]), and reverse (x, y) coordinates.

Hands-On Block Remediation: Use two-tone (+1 / -1) Zero-Pair counters and Zero-Pair tiles, plot signed jumps on the SC-6 Number Line ([f] [LINE]), test inequalities with [f] [>] and [g] [<], contrast [+/-] (flip sign) with [g] [|x|] (always positive distance from 0), and use the Elevator Rule (walk across x first, ride up/down y second).



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 6.1 (pp.55-56): POSITIVE AND NEGATIVE INTEGERS & OPPOSITES

Methods: W1: Equal Distance on Opposite Sides of 0 (Number Line) | W2: Additive Inverse & Zero-Pair Counters

PRACTICE A (p.55): KEY IDEAS (#1-#10)

ExA: Opposite of -15 is +15; -15 + 15 = 0

A1: -25; Opposite = +25
Setup: -25; +/-
SC-6: 25 [+/-] -> -25; [+/-] -> 25

A2: +42; and +42
Setup: 42
SC-6: 42 [+/-] [+/-]

A3: 0 m (Sea level)
Setup: -18 + 18
SC-6: 18 [+/-] [ENTER] 18 [+]

A4: 0
Setup: 6 + -6
SC-6: 6 [ENTER] 6 [+/-] [+]

A5: 0 ft
Setup: -120 + 120
SC-6: 120 [+/-] [ENTER] 120 [+]

A6: 0 (0 is 0 units from itself)
Setup: -0
SC-6: 0 [+/-]

A7: +7°F
Setup: -12 + 19
SC-6: 12 [+/-] [ENTER] 19 [+]

A8: 0 (Even par)
Setup: -3 + 3
SC-6: 3 [+/-] [ENTER] 3 [+]

A9: Floor 3
Setup: (14 + -20) + 9
SC-6: 14 [ENTER] 20 [+/-] [+] 9 [+]

A10: +$18,800
Setup: ((14500 - 3200) + 8900) - 1400
SC-6: 14500 [ENTER] 3200 [-] 8900 [+] 1400 [-]

PRACTICE B (p.56): REAL LIFE (#1-#10)

ExB: +$5

B1: 0 ft (opposites / additive inverses)
Setup: -120 + 120
SC-6: 120 [+/-] [ENTER] 120 [+]

B2: -282 ft; opposite is +282 ft
Setup: -282
SC-6: 282 [+/-]

B3: +19 yards
Setup: (12 + -8) + 15
SC-6: 12 [ENTER] 8 [+/-] [+] 15 [+]

B4: -28°C; opposite is +28°C
Setup: -28
SC-6: 28 [+/-]

B5: +$7
Setup: -18 + 25
SC-6: 18 [+/-] [ENTER] 25 [+]

B6: -5 (5 under par)
Setup: (-4 + 2) + -3
SC-6: 4 [+/-] [ENTER] 2 [+] 3 [+/-] [+]

B7: -x = +35; -(-x) = -35
Setup: 35
SC-6: 35 [+/-] [+/-]

B8: +$10
Setup: (25 - 25) + 10
SC-6: 25 [ENTER] 25 [-] 10 [+]

B9: +35 meters
Setup: (-40 + 95) - 20
SC-6: 40 [+/-] [ENTER] 95 [+] 20 [-]

B10: +23
Setup: -14 + -9
SC-6: 14 [+/-] [ENTER] 9 [+/-] [+] [+/-]

SEC 6.2 (pp.57-58): COMPARING AND ORDERING INTEGERS & RATIONAL NUMBERS

Methods: W1: Convert to Decimals & Plot on Number Line | W2: Rewrite with Least Common Denominator (LCD)

PRACTICE A (p.57): KEY IDEAS (#1-#10)

ExA: -5 > -18

A1: -5°C (-12 < -5)
Setup: Compare -5 < -12
SC-6: 5 [+/-] [ENTER] 12 [+/-] [g][<]

A2: -9 < -2
Setup: Compare -2 < -9
SC-6: 2 [+/-] [ENTER] 9 [+/-] [g][<]

A3: -15 < 0
Setup: Compare 0 < -15
SC-6: 0 [ENTER] 15 [+/-] [g][<]

A4: -10, -7, -2, 0, 4
Setup: fLINE
SC-6: Left-to-right on [f][LINE]

A5: -3/4 < -0.6 (-0.6 is greater)
Setup: Compare -0.6 < -0.75
SC-6: 0.6 [+/-] [ENTER] 0.75 [+/-] [g][<]

A6: -2 1/2 < -2.3
Setup: Compare -2.3 < -2.5
SC-6: 2.3 [+/-] [ENTER] 2.5 [+/-] [g][<]

A7: -5/8 > -7/8
Setup: Compare -7/8 > -5/8
SC-6: [.] 7 [.] 8 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [f][>]

A8: -4.1 is closer to 0 (further to the right) than -4.5
Setup: Compare -4.5 < -4.1
SC-6: 4.5 [+/-] [ENTER] 4.1 [+/-] [g][<]

A9: -1 3/4, -1.5, -0.5, 1 1/4
Setup: -1.75, -1.5, -0.5, 1.25
SC-6: Mental / Table Check

A10: Deepest: -15.2 m; Shallowest: -14 1/2 m
Setup: -15.2 < -14.8 < -14.5
SC-6: Mental / Table Check

PRACTICE B (p.58): REAL LIFE (#1-#10)

ExB: -0.75 > -4/5 (Diver B is closer to the surface)

B1: -11°C > -18°C (Tuesday was warmer)
Setup: Compare -18 > -11
SC-6: 18 [+/-] [ENTER] 11 [+/-] [f][>]

B2: Puffin (-45.8 m) is deeper; -45.25 > -45.8
Setup: Compare -45.8 > -45.25
SC-6: 45.8 [+/-] [ENTER] 45.25 [+/-] [f][>]

B3: -2/3 > -5/6 (-4/6 > -5/6)
Setup: Compare -5/6 > -2/3
SC-6: [.] 5 [.] 6 [+/-] [ENTER] [.] 2 [.] 3 [+/-] [f][>]

B4: -9, -6, -1, +2
Setup: Compare -6 < -9
SC-6: 6 [+/-] [ENTER] 9 [+/-] [g][<]

B5: -89°C, -39°C, -6°C, 0°C
Setup: Compare -39 < -89
SC-6: 39 [+/-] [ENTER] 89 [+/-] [g][<]

B6: Sea turtle (-3.1 > -3.25)
Setup: Compare -3.1 < -3.25
SC-6: 3.1 [+/-] [ENTER] 3.25 [+/-] [g][<]

B7: Stock B (-1.08% > -1.8%)
Setup: Compare -1.8 > -1.08
SC-6: 1.8 [+/-] [ENTER] 1.08 [+/-] [f][>]

B8: Examples: -3 1/2 and -3.5
Setup: -3.5
SC-6: 3.5 [+/-]

B9: -2, -7/4 (-1.75), -1.6, 3/2 (1.5)
Setup: -7/4 -> Dec/Frac
SC-6: [.] 7 [.] 4 [+/-] [FDISP]

B10: a is farther from 0 (e.g., -10 < -2)
Setup: |-10|
SC-6: 10 [+/-] [g][|x|]

SEC 6.3 (pp.59-60): ABSOLUTE VALUE AND MAGNITUDE

Methods: W1: Measure Distance from 0 on a Number Line | W2: Evaluate |x| Grouping Bars Before Outside Signs

PRACTICE A (p.59): KEY IDEAS (#1-#10)

ExA: |-14| = 14, |14| = 14, -|-14| = -14

A1: 125
Setup: |-125|
SC-6: 125 [+/-] [g][|x|]

A2: 30
Setup: |-18| + 12
SC-6: 18 [+/-] [g][|x|] [ENTER] 12 [+]

A3: 8.0
Setup: |-4.8| + |-3.2|
SC-6: 4.8 [+/-] [g][|x|] [ENTER] 3.2 [+/-] [g][|x|] [+]

A4: 16
Setup: |-35| - |-19|
SC-6: 35 [+/-] [g][|x|] [ENTER] 19 [+/-] [g][|x|] [-]

A5: |-12| > |9| (12 > 9)
Setup: 12 > 9
SC-6: Mental / Table Check

A6: -24
Setup: |-24|
SC-6: 24 [+/-] [g][|x|] [+/-]

A7: 32
Setup: 50 - |-18|
SC-6: 50 [ENTER] 18 [+/-] [g][|x|] [-]

A8: -|-9|, -6, 0, |-3|, |-8|
Setup: Values: 8, -6, 3, 0, -9
SC-6: Mental / Table Check

A9: Seagull is closer by 7 feet (25 < 32)
Setup: |-32| - 25
SC-6: 32 [+/-] [g][|x|] [ENTER] 25 [-]

A10: Greater debt: -$45 ($45); Greater balance: -$28
Setup: |-45|=45 > |-28|=28
SC-6: Mental / Table Check

PRACTICE B (p.60): REAL LIFE (#1-#10)

ExB: 46 m plunge; and 50 - |-18| = 32

B1: |32| + |-14| = 46 meters
Setup: |32| + |-14|
SC-6: 32 [g][|x|] 14 [+/-] [g][|x|] [+]

B2: -$40 is greater balance; Account A owes more ($65)
Setup: |-65|
SC-6: 65 [+/-] [g][|x|]

B3: |-18| + |14| = 32 meters
Setup: |-18| + |14|
SC-6: 18 [+/-] [g][|x|] 14 [g][|x|] [+]

B4: 35 and -35 (outside minus applies after | |)
Setup: |-35|
SC-6: 35 [+/-] [g][|x|] [+/-]

B5: -1.5 cm (|-1.5| = 1.5 cm is smallest)
Setup: |-1.5|
SC-6: 1.5 [+/-] [g][|x|]

B6: 42 ÷ 6 + 11 = 18
Setup: (|-42| ÷ |-6|) + |-11|
SC-6: 42 [+/-] [g][|x|] 6 [+/-] [g][|x|] [÷] 11 [+/-] [g][|x|] [+]

B7: |25| + |-140| = 165 feet
Setup: |25| + |-140|
SC-6: 25 [g][|x|] 140 [+/-] [g][|x|] [+]

B8: 19 and -19
Setup: |-19|
SC-6: 19 [+/-] [g][|x|]

B9: -|-9| (-9), -6, |5| (5), |-8| (8)
Setup: |-9|
SC-6: 9 [+/-] [g][|x|] [+/-]

B10: 4 5/8
Setup: (3 1/2 + 2 1/4) - 1 1/8
SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [+] 1 [.] 1 [.] 8 [-]

SEC 6.4 (pp.61-62): THE FOUR-QUADRANT COORDINATE PLANE & REFLECTIONS

Methods: W1: Plot & Count Distance to the Mirror Axis | W2: Coordinate Reflection Sign Rules

PRACTICE A (p.61): KEY IDEAS (#1-#10)

ExA: A: Quadrant II, B: Quadrant IV, C: Quadrant III

A1: Quadrant IV
Setup: (+, -) -> Right, Down
SC-6: Mental / Table Check

A2: Quadrant III
Setup: (-, -) -> Left, Down
SC-6: Mental / Table Check

A3: Quadrant II
Setup: (-, +) -> Left, Up
SC-6: Mental / Table Check

A4: (-6, 9)
Setup: -6
SC-6: 6 [+/-] -> -6

A5: (6, -9)
Setup: -9
SC-6: 9 [+/-] -> -9

A6: (-3, 7)
Setup: 7
SC-6: 7 [+/-] [+/-] -> +7

A7: (8, -4)
Setup: Flip x -> (8,4); flip y -> (8,-4)
SC-6: Mental / Table Check

A8: (0, -9) on y-axis; (-7, 0) on x-axis
Setup: 0 sideways vs 0 up/down
SC-6: Mental / Table Check

A9: (-3, 3) in Quadrant II
Setup: 4 - 7; -5 + 8
SC-6: 4 [ENTER] 7 [-]; 5 [+/-] [ENTER] 8 [+]

A10: D(-4, -5) in Quadrant III
Setup: Match x of A (-4) & y of C (-5)
SC-6: Mental / Table Check

PRACTICE B (p.62): REAL LIFE (#1-#10)

ExB: (5, -8) in Quadrant IV

B1: Cave: II, Lighthouse: I, Shipwreck: III, Pier: IV
Setup: 4
SC-6: 8 [+/-] [ENTER] 4

B2: (0, -7) is on y-axis; (9, 0) is on x-axis
Setup: -7
SC-6: 0 [ENTER] 7 [+/-]

B3: (8, -12) in Quadrant IV
Setup: -12
SC-6: 8 [ENTER] 12 [+/-]

B4: (9, 5) in Quadrant I
Setup: 5
SC-6: 9 [+/-] [+/-] [ENTER] 5

B5: Step 1: (-4, 9); Final: (4, 9) in Quad I
Setup: 9
SC-6: 4 [+/-] [+/-] [ENTER] 9 [+/-] [+/-]

B6: (4, -8) in Quadrant IV
Setup: -6 + 10
SC-6: 6 [+/-] [ENTER] 10 [+]

B7: P: Quadrant II; Q: Quadrant III
Setup: 4.5
SC-6: 2.5 [+/-] [ENTER] 4.5

B8: y-axis (x-coordinate changed sign, y stayed -3)
Setup: -7
SC-6: 7 [+/-]

B9: (-5, 5), (-5, -5), and (5, -5)
Setup: -5
SC-6: 5 [+/-]

B10: (x, y) is in Quad II; (-x, -y) is in Quad IV
Setup: -1
SC-6: 1 [+/-]

SEC 6.5 (pp.63-64): DISTANCES IN THE COORDINATE PLANE

Methods: W1: Add/Subtract Absolute Values to the Axis | W2: Absolute Difference of Coordinates |x1 - x2|

PRACTICE A (p.63): KEY IDEAS (#1-#10)

ExA: 15 units; and 8 units

A1: 27 units
Setup: |(-12 - 15)|
SC-6: 12 [+/-] [ENTER] 15 [-] [g][|x|]

A2: 11 units
Setup: |(-4 - -15)|
SC-6: 4 [+/-] [ENTER] 15 [+/-] [-] [g][|x|]

A3: 15 units
Setup: |(9 - -6)|
SC-6: 9 [ENTER] 6 [+/-] [-] [g][|x|]

A4: 9 units
Setup: |(-14 - -5)|
SC-6: 14 [+/-] [ENTER] 5 [+/-] [-] [g][|x|]

A5: 10 units
Setup: |(-3.5 - 6.5)|
SC-6: 3.5 [+/-] [ENTER] 6.5 [-] [g][|x|]

A6: 8 units
Setup: |(-4.5 - 3.5)|
SC-6: 4.5 [+/-] [ENTER] 3.5 [-] [g][|x|]

A7: (4, 3)
Setup: -6 + 10
SC-6: 6 [+/-] [ENTER] 10 [+]

A8: 13 units (They lie on opposite sides of the y-axis: |-5| + |..
Setup: |(-5 - 8)|
SC-6: 5 [+/-] [ENTER] 8 [-] [g][|x|]

A9: Perimeter = 34 units, Area = 72 sq units
Setup: Width 9, Height 8
SC-6: Mental / Table Check

A10: 20 grid units = 10 miles
Setup: 10 + 10 = 20 units × 0.5
SC-6: Mental / Table Check

PRACTICE B (p.64): REAL LIFE (#1-#10)

ExB: Perimeter = 34 units; Area = 72 square units

B1: |-8| + |11| = 19 units
Setup: |(-8 - 11)|
SC-6: 8 [+/-] [ENTER] 11 [-] [g][|x|]

B2: 9 units = 90 meters
Setup: |(-14 - -5)|
SC-6: 14 [+/-] [ENTER] 5 [+/-] [-] [g][|x|]

B3: |15| + |-35| = 50 units
Setup: |(15 - -35)|
SC-6: 15 [ENTER] 35 [+/-] [-] [g][|x|]

B4: W = 10, H = 6, Perimeter = 32 units
Setup: (|(-3 - 7)| + |(4 - -2)|) × 2
SC-6: 3 [+/-] [ENTER] 7 [-] [g][|x|] 4 [ENTER] 2 [+/-] [-] [g][|x|]..

B5: 60 square units
Setup: A = 10 × 6
SC-6: 10 [ENTER] 6 [f][RECT]

B6: 7.75 miles
Setup: 3.5 + 4.25
SC-6: 3.5 [ENTER] 4.25 [+]

B7: 5 3/4 units
Setup: 3 1/2 + 2 1/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [+]

B8: 7 + 10 = 17 units
Setup: (5 + 2) + (4 + 6)
SC-6: 5 [ENTER] 2 [+] 4 [ENTER] 6 [+] [+]

B9: (11, 4) and (-5, 4)
Setup: 3 - 8
SC-6: 3 [ENTER] 8 [+] 3 [ENTER] 8 [-]

B10: Base = 10, Height = 8, Area = 40 units²
Setup: A = ½((4 + 6) × (7 + 1))
SC-6: 4 [ENTER] 6 [+] 7 [ENTER] 1 [+] [g][TRI]

Unit 7: Equations & Inequalities — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 7 • Hands-On Models: Algebra Boxes & Stage (Algebra & Area Tiles) + Counting & Array Tiles (Counting & Array Tiles)

Target Misconception: Students treat '=' as 'write the answer', translate '8 less than n' backwards as 8 - n, resent writing -14 under both sides of x + 14 = 39, and confuse open vs. solid inequality circles.

Hands-On Block Remediation: Model '=' as a Balance Scale using Algebra Boxes & Stage. Solve 1-step equations with inverse operations on both sides, verify your solution with the SC-6 Equality Test [!=] (which flashes NO when Left = Right!), and test inequalities on [f] [LINE] with [f] [>=] and [g] [<=] (which flash YES or NO).



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 7.1 (pp.66-67): WRITING EQUATIONS IN ONE VARIABLE & TESTING SOLUTIONS

Methods: W1: Substitute Candidates to Test Equality | W2: Balance Scale / Mental Unwrapping

PRACTICE A (p.66): KEY IDEAS (#1-#10)

ExA: x = 9 is the solution (41 = 41)

A1: x = 8 (5(8)-7 = 33)
Setup: (5 × 8) - 7
SC-6: 33 [ENTER] 5 [ENTER] 8 [×] 7 [-] [!=]

A2: y = 15 (15÷3 + 4 = 9)
Setup: (15 ÷ 3) + 4
SC-6: 9 [ENTER] 15 [ENTER] 3 [÷] 4 [+] [!=]

A3: x + 14 = 52; x = 38
Setup: x + 14 = 52
SC-6: Mental / Table Check

A4: 6k = 84; k = 14
Setup: 6k = 84
SC-6: Mental / Table Check

A5: w - 9 = 17; w = 26
Setup: w - 9 = 17
SC-6: Mental / Table Check

A6: m / 5 = 13; m = 65
Setup: m / 5 = 13
SC-6: Mental / Table Check

A7: 4p + 6 = 54; p = $12
Setup: (4 × 12) + 6
SC-6: 54 [ENTER] 4 [ENTER] 12 [×] 6 [+] [!=]

A8: x - 12 = 20; x = 32
Setup: x - 12 = 20
SC-6: Mental / Table Check

A9: 10 + 3m = 28; m = 6 movies
Setup: (28 - 10) ÷ 3
SC-6: 28 [ENTER] 10 [-] 3 [÷]

A10: b / 3 = 14.50; b = $43.50
Setup: 14.50 × 3
SC-6: 14.50 [ENTER] 3 [×]

PRACTICE B (p.67): REAL LIFE (#1-#10)

ExB: n - 8 = 25; n = 33 drones

B1: x = 8 (5·8 - 7 = 33)
Setup: (5 × 8) - 7
SC-6: 33 [ENTER] 5 [ENTER] 8 [×] 7 [-] [!=]

B2: 15t + 5 = 65; yes, t = 4 is a solution
Setup: (15 × 4) + 5
SC-6: 65 [ENTER] 15 [ENTER] 4 [×] 5 [+] [!=]

B3: k - 12 = 29; k = 41
Setup: 29 + 12
SC-6: 29 [ENTER] 12 [+]

B4: n ÷ 6 = 14; n = 84
Setup: 14 × 6
SC-6: 14 [ENTER] 6 [×]

B5: 3s = 54; s = 18 cm
Setup: 54 ÷ 3
SC-6: 54 [ENTER] 3 [÷]

B6: m = 15 (100 - 60 = 40)
Setup: 100 - (4 × 15)
SC-6: 40 [ENTER] 100 [ENTER] 4 [ENTER] 15 [×] [-] [!=]

B7: p ÷ 8 = 3; p = 24 slices
Setup: 24 ÷ 8
SC-6: 24 [ENTER] 8 [÷]

B8: 45 + d = 120; d = $75
Setup: 120 - 45
SC-6: 120 [ENTER] 45 [-]

B9: Yes (both sides equal 28)
Setup: (5 × 6) - 2
SC-6: 3 [ENTER] 6 [×] 10 [+] 5 [ENTER] 6 [×] 2 [-] [!=]

B10: b - 4 = 13; b = 17 years old
Setup: 13 + 4
SC-6: 13 [ENTER] 4 [+]

SEC 7.2 (pp.68-69): SOLVING EQUATIONS USING ADDITION OR SUBTRACTION

Methods: W1: Properties of Equality (Inverse Operations) | W2: Balance Scale / Part-Part-Whole Bar Model

PRACTICE A (p.68): KEY IDEAS (#1-#10)

ExA: x = 45

A1: x = 45
Setup: 73 - 28
SC-6: 73 [ENTER] 28 [-]

A2: y = 92
Setup: 58 + 34
SC-6: 58 [ENTER] 34 [+]

A3: n = 45
Setup: 64 - 19
SC-6: 64 [ENTER] 19 [-]

A4: k = 76
Setup: 29 + 47
SC-6: 29 [ENTER] 47 [+]

A5: w = 54.4
Setup: 34.8 + 19.6
SC-6: 34.8 [ENTER] 19.6 [+]

A6: p = 16.65
Setup: 25.4 - 8.75
SC-6: 25.4 [ENTER] 8.75 [-]

A7: m = 11/24
Setup: 5/6 - 3/8
SC-6: [.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]

A8: a = 4 3/4
Setup: 3 1/2 + 1 1/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 4 [+]

A9: m - 18.50 = 26.75; m = $45.25
Setup: 26.75 + 18.50
SC-6: 26.75 [ENTER] 18.50 [+]

A10: h + 42.5 = 110; h = 67.5 ft
Setup: 110 - 42.5
SC-6: 110 [ENTER] 42.5 [-]

PRACTICE B (p.69): REAL LIFE (#1-#10)

ExB: w = 54.4 g; m = 11/24

B1: s + 45 = 210; s = 165 points
Setup: 210 - 45
SC-6: 210 [ENTER] 45 [-]

B2: p - 18 = 67; p = $85
Setup: 67 + 18
SC-6: 67 [ENTER] 18 [+]

B3: h + 1450 = 5200; h = 3,750 feet
Setup: 5200 - 1450
SC-6: 5200 [ENTER] 1450 [-]

B4: f - 64.8 = 185.2; f = 250.0 grams
Setup: 185.2 + 64.8
SC-6: 185.2 [ENTER] 64.8 [+]

B5: t - 0.48 = 12.65; t = 13.13 seconds
Setup: 12.65 + 0.48
SC-6: 12.65 [ENTER] 0.48 [+]

B6: m + 3/8 = 7/8; m = 1/2 cup
Setup: 7/8 - 3/8
SC-6: [.] 7 [.] 8 [ENTER] [.] 3 [.] 8 [-]

B7: 2 3/4 + d = 4 1/2; d = 1 3/4 miles
Setup: 4 1/2 - 2 3/4
SC-6: 4 [.] 1 [.] 2 [ENTER] 2 [.] 3 [.] 4 [-]

B8: b + 38 = 125; b = 87°
Setup: 125 - 38
SC-6: 125 [ENTER] 38 [-]

B9: -40 + d = -115; d = -75 meters
Setup: -115 - -40
SC-6: 115 [+/-] [ENTER] 40 [+/-] [-]

B10: L = 44 cm; 1 length = 22 cm
Setup: (64 - 20) ÷ 2
SC-6: 64 [ENTER] 20 [-] 2 [÷]

SEC 7.3 (pp.70-71): SOLVING EQUATIONS USING MULTIPLICATION OR DIVISION

Methods: W1: Divide/Multiply by Coefficient | W2: Multiply by the Reciprocal | W3: Equal-Groups Tape Diagram

PRACTICE A (p.70): KEY IDEAS (#1-#10)

ExA: x = 17; k = 101.5

A1: x = 17
Setup: 136 ÷ 8
SC-6: 136 [ENTER] 8 [÷]

A2: y = 15
Setup: 180 ÷ 12
SC-6: 180 [ENTER] 12 [÷]

A3: m = 114
Setup: 19 × 6
SC-6: 19 [ENTER] 6 [×]

A4: k = 101.5
Setup: 14.5 × 7
SC-6: 14.5 [ENTER] 7 [×]

A5: w = 14
Setup: 63 ÷ 4.5
SC-6: 63 [ENTER] 4.5 [÷]

A6: x = 24
Setup: 18 ÷ 3/4
SC-6: 18 [ENTER] [.] 3 [.] 4 [÷]

A7: n = 40
Setup: 16 ÷ 2/5
SC-6: 16 [ENTER] [.] 2 [.] 5 [÷]

A8: p = 4/5 (0.8)
Setup: 2/3 ÷ 5/6
SC-6: [.] 2 [.] 3 [ENTER] [.] 5 [.] 6 [÷]

A9: 1.5t = 27; t = 18 minutes
Setup: 27 ÷ 1.5
SC-6: 27 [ENTER] 1.5 [÷]

A10: 3/5 s = 27; s = 45 students
Setup: 27 ÷ 3/5
SC-6: 27 [ENTER] [.] 3 [.] 5 [÷]

PRACTICE B (p.71): REAL LIFE (#1-#10)

ExB: 40 songs

B1: 12s = 168; s = $14 per shirt
Setup: 168 ÷ 12
SC-6: 168 [ENTER] 12 [÷]

B2: p ÷ 9 = 3; p = 27 slices
Setup: 3 × 9
SC-6: 3 [ENTER] 9 [×]

B3: 65t = 325; t = 5 hours
Setup: 325 ÷ 65
SC-6: 325 [ENTER] 65 [÷]

B4: 2.5p = 16.25; p = $6.50 per pound
Setup: 16.25 ÷ 2.5
SC-6: 16.25 [ENTER] 2.5 [÷]

B5: s ÷ 4 = 9; s = $36
Setup: 9 × 4
SC-6: 9 [ENTER] 4 [×]

B6: (3/4)c = 21; c = 28 students
Setup: 21 ÷ 3/4
SC-6: 21 [ENTER] [.] 3 [.] 4 [÷]

B7: (5/6)p = 150; p = $180
Setup: 150 ÷ 5/6
SC-6: 150 [ENTER] [.] 5 [.] 6 [÷]

B8: 13.5w = 108; w = 8 feet
Setup: 108 ÷ 13.5
SC-6: 108 [ENTER] 13.5 [÷]

B9: h ÷ 15 = 3.2; h = 48 feet
Setup: 3.2 × 15
SC-6: 3.2 [ENTER] 15 [×]

B10: 0.40n = 56; n = 140
Setup: 56 ÷ 0.40
SC-6: 56 [ENTER] 0.40 [÷]

SEC 7.4 (pp.72-73): WRITING AND GRAPHING INEQUALITIES

Methods: W1: Inverse Property of Inequality & Number Line Graph | W2: Boundary & Test-Point Substitution

PRACTICE A (p.72): KEY IDEAS (#1-#10)

ExA: x >= 19 (Solid dot at 19, arrow right)

A1: x > 13 (Open circle at 13, arrow right)
Setup: 21 - 8
SC-6: 21 [ENTER] 8 [-]

A2: y <= 24 (Solid dot at 24, arrow left)
Setup: 15 + 9
SC-6: 15 [ENTER] 9 [+]

A3: n >= 13 (Solid dot at 13, arrow right)
Setup: 78 ÷ 6
SC-6: 78 [ENTER] 6 [÷]

A4: m < 36 (Open circle at 36, arrow left)
Setup: 9 × 4
SC-6: 9 [ENTER] 4 [×]

A5: h >= 48
Setup: 'At least' = >=
SC-6: Mental / Table Check

A6: w <= 2,000
Setup: 'At most' = <=
SC-6: Mental / Table Check

A7: Only 6 (At n = 5, 23 > 23 is false)
Setup: Compare 23 > ((4 × 6) + 3)
SC-6: 23 [ENTER] 4 [ENTER] 6 [×] 3 [+] [f][>]

A8: 5 and 6
Setup: (4 × 5) - 3
SC-6: 17 [ENTER] 4 [ENTER] 5 [×] 3 [-] [f][>=]

A9: 8b <= 50; at most 6 books ($2 left over)
Setup: 50 ÷ 8 (Q & R)
SC-6: 50 [ENTER] 8 [g][÷R]

A10: p + 340 >= 500; p >= 160 points
Setup: 500 - 340
SC-6: 500 [ENTER] 340 [-]

PRACTICE B (p.73): REAL LIFE (#1-#10)

ExB: g <= 15 (at most 15 gold bars)

B1: h >= 54 (solid dot at 54, shade right); at least 5 in
Setup: 54 - 49
SC-6: 54 [ENTER] 49 [-]

B2: p + 5 <= 12 -> p <= 7 passengers
Setup: 12 - 5
SC-6: 12 [ENTER] 5 [-]

B3: 22.50 + x >= 35 -> x >= $12.50
Setup: 35 - 22.50
SC-6: 35 [ENTER] 22.50 [-]

B4: 3.50g <= 28 -> g <= 8 games
Setup: 28 ÷ 3.50
SC-6: 28 [ENTER] 3.50 [÷]

B5: 15r > 240 -> r > 16 (at least 17 rows)
Setup: 240 ÷ 15
SC-6: 240 [ENTER] 15 [÷]

B6: b - 25 < 15 -> b < 40%
Setup: 15 + 25
SC-6: 15 [ENTER] 25 [+]

B7: 8 and 10
Setup: (5 × 8) - 10
SC-6: 30 [ENTER] 5 [ENTER] 8 [×] 10 [-] [f][>=]

B8: 2c - 30 >= 70 -> c >= 50 cupcakes
Setup: (70 + 30) ÷ 2
SC-6: 70 [ENTER] 30 [+] 2 [÷]

B9: s <= 20 (solid dot at 20, shade left down to 0)
Setup: Compare 20 <= 20
SC-6: 20 [ENTER] 20 [g][<=]

B10: 45c + 170 <= 800 -> c <= 14 crates
Setup: (800 - 170) ÷ 45
SC-6: 800 [ENTER] 170 [-] 45 [÷]

SEC 7.5 (pp.74-75): WRITING EQUATIONS IN TWO VARIABLES (Y = KX AND Y = X + B)

Methods: W1: Test Constant Ratio y/x vs. Constant Difference y - x | W2: Input-Output Table & Coordinate Graph

PRACTICE A (p.74): KEY IDEAS (#1-#10)

ExA: Table A: y = 7x; Table B: y = x - 4

A1: y = 8x; y = 80
Setup: 16 ÷ 2; 8×10
SC-6: 16 [ENTER] 2 [÷] -> 8; 8×10

A2: y = x + 8; y = 28
Setup: 9 - 1; 20+8
SC-6: 9 [ENTER] 1 [-] -> +8; 20+8

A3: y = 7x; y = 84
Setup: 14 ÷ 2; 7×12
SC-6: 14 [ENTER] 2 [÷] -> 7; 7×12

A4: y = x - 4; y = 26
Setup: 30 - 4
SC-6: 30 [ENTER] 4 [-]

A5: (1, 6), (2, 12), (3, 18), (4, 24)
Setup: Repeat(+6)
SC-6: 6 [g][REP] [+] [+] [+]

A6: Independent: t; Dependent: d
Setup: d depends on t
SC-6: Mental / Table Check

A7: 292.5 miles
Setup: 65 × 4.5
SC-6: 65 [ENTER] 4.5 [×]

A8: y = 4x passes through (0, 0); y = x + 4 crosses at (0, 4)
Setup: Test x = 0 in both
SC-6: Mental / Table Check

A9: 8 hours
Setup: 100 ÷ 12.50
SC-6: 100 [ENTER] 12.50 [÷]

A10: c = 15m + 25; $205
Setup: (15 × 12) + 25
SC-6: 15 [ENTER] 12 [×] 25 [+]

PRACTICE B (p.75): REAL LIFE (#1-#10)

ExB: p = 12.50h; $81.25

B1: y = x + 5 is additive; y = 8m is multiplicative
Setup: 8 × 1
SC-6: 8 [ENTER] 1 [×]

B2: g = 18h (h = indep, g = dep); 81 grams
Setup: 18 × 4.5
SC-6: 18 [ENTER] 4.5 [×]

B3: m = z + 4; m = 12, 16, 19
Setup: 8 + 4
SC-6: 8 [ENTER] 4 [+]

B4: s = 15p; p = 12 sheets
Setup: 180 ÷ 15
SC-6: 180 [ENTER] 15 [÷]

B5: y = x - 6; $18, $29, and $44
Setup: 24 - 6
SC-6: 24 [ENTER] 6 [-]

B6: y = 3.5x; 140 wheel turns
Setup: 3.5 × 40
SC-6: 3.5 [ENTER] 40 [×]

B7: y = 9x; at x = 20, y = 180
Setup: (27 ÷ 3) × 20
SC-6: 27 [ENTER] 3 [÷] 20 [×]

B8: c = 20g + 7; $127
Setup: (20 × 6) + 7
SC-6: 20 [ENTER] 6 [×] 7 [+]

B9: Multiplicative y = kx (when x = 0, y = k·0 = 0)
Setup: 0 × 7
SC-6: 0 [ENTER] 7 [×]

B10: d = 0.2s; 3 miles away
Setup: 0.2 × 15
SC-6: 0.2 [ENTER] 15 [×]

Unit 8: Surface Area & Volume of 3D Solids — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 8 • Hands-On Models: Counting & Array Tiles (Counting & Array Tiles) + 3x1 Storage Boxes

Target Misconception: Students drop one of the 6 faces when finding rectangular prism surface area, tangle prism and pyramid formulas, and confuse 2D Surface Area (cardboard outside) with 3D Volume (cubes inside).

Hands-On Block Remediation: Have students touch the 3 pairs of faces on their physical box while logging each pair into [Σ+] (n=1, 2, 3), use [f] [PRISM] ([f][9]) to multiply 3 stack floors (l, w, h) at once for volume, and unlock [f] [CIRC], [g] [CONE], and [g] [HYP].



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 8.1 (pp.77-78): THREE-DIMENSIONAL FIGURES, NETS & RECTANGULAR PRISMS

Methods: W1: Unfold into a 6-Face 2D Net | W2: Opposite Face Pairs SA = 2(lw + lh + wh) | W3: Lateral Strip Ph + 2B

PRACTICE A (p.77): KEY IDEAS (#1-#10)

ExA: 158 cm²

A1: 96 cm²
Setup: $4^2 \times 6$
SC-6: 4 [x²] 6 [×]

A2: 486 in.²
Setup: $9^2 \times 6$
SC-6: 9 [x²] 6 [×]

A3: 108 cm²
Setup: 2(24)+2(18)+2(12)
SC-6: Mental / Table Check

A4: 160 in.²
Setup: 2(50)+2(20)+2(10)
SC-6: Mental / Table Check

A5: 392 ft²
Setup: 2(96)+2(60)+2(40)
SC-6: Mental / Table Check

A6: 85 m²
Setup: 2(20)+2(12.5)+2(10)
SC-6: Mental / Table Check

A7: One face = 25 cm²; s = 5 cm
Setup: $\sqrt{(150 \div 6)}$
SC-6: 150 [ENTER] 6 [÷] [f][√x]

A8: 94 cm² (He only summed 3 faces instead of doubling for all 6..
Setup: 47 × 2
SC-6: 47 [ENTER] 2 [×]

A9: 2,100 cm²
Setup: 600 + 2(450) + 2(300)
SC-6: Mental / Table Check

A10: 500 in.² needed; Yes, with 50 in.² to spare
Setup: 2(150)+2(60)+2(40)
SC-6: Mental / Table Check

PRACTICE B (p.78): REAL LIFE (#1-#10)

ExB: Cube = 294 in²; Open-Top Bin = 260 cm²

B1: 2(160 + 600 + 240) = 2,000 cm²
Setup: (((20 × 8) + (20 × 30)) + (8 × 30)) × 2
SC-6: 20 [ENTER] 8 [×] 20 [ENTER] 30 [×] [+] 8 [ENTER] 30 [×] [+] 2..

B2: 1 face = 81 in²; Total SA = 486 in²
Setup: $9^2 \times 6$
SC-6: 9 [x²] 6 [×]

B3: 288 + 720 + 360 = 1,368 in²
Setup: ((24 × 12) + ((24 × 15) × 2)) + ((12 × 15) × 2)
SC-6: 24 [ENTER] 12 [×] 24 [ENTER] 15 [×] 2 [×] [+] 12 [ENTER] 15 [×..

B4: SA = 500 in²; yes (100 in² to spare)
Setup: (((15 × 10) + (15 × 4)) + (10 × 4)) × 2
SC-6: 15 [ENTER] 10 [×] 15 [ENTER] 4 [×] [+] 10 [ENTER] 4 [×] [+] 2..

B5: 120 + 192 + 160 = 472 ft²
Setup: ((12 × 10) + ((12 × 8) × 2)) + ((10 × 8) × 2)
SC-6: 12 [ENTER] 10 [×] 12 [ENTER] 8 [×] 2 [×] [+] 10 [ENTER] 8 [×]..

B6: 1 face = 25 cm²; edge = 5 cm
Setup: $\sqrt{(150 \div 6)}$
SC-6: 150 [ENTER] 6 [÷] [f][√x]

B7: 32 + 8 = 40 ft²
Setup: ((4 × 2) × 4) + ((2 × 2) × 2)
SC-6: 4 [ENTER] 2 [×] 4 [×] 2 [ENTER] 2 [×] 2 [×] [+]

B8: 2(24 + 60 + 40) = 248 cm²
Setup: (((6 × 4) + (6 × 10)) + (4 × 10)) × 2
SC-6: 6 [ENTER] 4 [×] 6 [ENTER] 10 [×] [+] 4 [ENTER] 10 [×] [+] 2 [×]

B9: 54 cm² vs. 216 cm² (4 times as great)
Setup: (6^2 × 6) ÷ 54
SC-6: 6 [x²] 6 [×] 54 [÷]

B10: 160 cm²
Setup: $4^2 \times 10$
SC-6: 4 [x²] 10 [×]

SEC 8.2 (pp.79-80): SURFACE AREAS OF TRIANGULAR PRISMS AND PYRAMIDS

Methods: W1: Add All Individual Faces on the 2D Net | W2: Base Area + Combined Lateral Area

PRACTICE A (p.79): KEY IDEAS (#1-#10)

ExA: 340 m²

A1: 96 cm²
Setup: $6^2 + (\frac{1}{2}(6\times5) \times 4)$
SC-6: 6 [x²] 6 [ENTER] 5 [g][TRI] 4 [×] [+]

A2: 224 in.²
Setup: $8^2 + (\frac{1}{2}(8\times10) \times 4)$
SC-6: 8 [x²] 8 [ENTER] 10 [g][TRI] 4 [×] [+]

A3: 360 ft²
Setup: $12^2 + (\frac{1}{2}(12\times9) \times 4)$
SC-6: 12 [x²] 12 [ENTER] 9 [g][TRI] 4 [×] [+]

A4: 62.4 cm²
Setup: $\frac{1}{2}(6\times5.2) \times 4$
SC-6: 6 [ENTER] 5.2 [g][TRI] 4 [×]

A5: 132 cm²
Setup: $\frac{1}{2}(4\times3) \times 2$
SC-6: 4 [ENTER] 3 [g][TRI] 2 [×] (12) + 12×10 (120)

A6: 150 ft²
Setup: $\frac{1}{2}(8\times3) \times 2$
SC-6: 8 [ENTER] 3 [g][TRI] 2 [×] (24) + 18×7 (126)

A7: 110.88 cm²
Setup: $\frac{1}{2}(8\times6.93) \times 4$
SC-6: 8 [ENTER] 6.93 [g][TRI] 4 [×]

A8: Triangular prism: 2 triangles + 3 rectangles; Square pyramid..
Setup: Count net faces
SC-6: Mental / Table Check

A9: 152 ft²
Setup: 24 + 48 + 80
SC-6: Mental / Table Check

A10: 108 ft²
Setup: $\frac{1}{2}(9\times6) \times 4$
SC-6: 9 [ENTER] 6 [g][TRI] 4 [×]

PRACTICE B (p.80): REAL LIFE (#1-#10)

ExB: 288 square feet

B1: 24 + 80 + 48 = 152 ft²
Setup: $(\frac{1}{2}(6\times4) \times 2) + ((5\times8) \times 2)) + (6\times8)$
SC-6: 6 [ENTER] 4 [g][TRI] 2 [×] 5 [ENTER] 8 [f][RECT] 2 [×] [+] 6 [..

B2: 144 + 240 = 384 cm²
Setup: $12^2 + (\frac{1}{2}(12\times10) \times 4)$
SC-6: 12 [x²] 12 [ENTER] 10 [g][TRI] 4 [×] [+]

B3: 96 ft²
Setup: $\frac{1}{2}(8\times6) \times 4$
SC-6: 8 [ENTER] 6 [g][TRI] 4 [×]

B4: 13.8 + 240 = 253.8 cm²
Setup: (6.9 × 2) + ((4×20) × 3)
SC-6: 6.9 [ENTER] 2 [×] 4 [ENTER] 20 [f][RECT] 3 [×] [+]

B5: 48 + 120 = 168 in²
Setup: $(\frac{1}{2}(8\times6) \times 2) + (((8 + 6) + 10) \times 5)$
SC-6: 8 [ENTER] 6 [g][TRI] 2 [×] 8 [ENTER] 6 [+] 10 [+] 5 [×] [+]

B6: 15.6 cm²
Setup: $\frac{1}{2}(3\times2.6) \times 4$
SC-6: 3 [ENTER] 2.6 [g][TRI] 4 [×]

B7: 96 + 600 = 696 ft²
Setup: $(\frac{1}{2}(16\times6) \times 2) + ((10\times30) \times 2)$
SC-6: 16 [ENTER] 6 [g][TRI] 2 [×] 10 [ENTER] 30 [f][RECT] 2 [×] [+]

B8: 15 in² per face; slant height = 5 in
Setup: (96 - 36) ÷ 4
SC-6: 96 [ENTER] 36 [-] 4 [÷]

B9: 96 + 480 = 576 cm²
Setup: $(\frac{1}{2}(12\times8) \times 2) + (((10 + 10) + 12) \times 15)$
SC-6: 12 [ENTER] 8 [g][TRI] 2 [×] 10 [ENTER] 10 [+] 12 [+] 15 [×] [+]

B10: Pyramid A (260 cm² > 224 cm²)
Setup: $8^2 + (\frac{1}{2}(8\times10) \times 4)$
SC-6: 10 [x²] 10 [ENTER] 8 [g][TRI] 4 [×] [+] 8 [x²] 8 [ENTER] 10 [g..

SEC 8.3 (pp.81-82): VOLUMES OF RECTANGULAR PRISMS

Methods: W1: Base Area × Height (V = Bh) | W2: Multiply Length × Width × Height (Fractions or Decimals)

PRACTICE A (p.81): KEY IDEAS (#1-#10)

ExA: Crate = 480 in.³; Aquarium = 10 1/2 ft³

A1: 3,024 cm³
Setup: Prism(18×6×28)
SC-6: 18 [ENTER] 6 [ENTER] 28 [f][PRISM]

A2: 125 in.³
Setup: 5^3
SC-6: 5 [x³]

A3: 500 m³
Setup: Prism(25×10×2)
SC-6: 25 [ENTER] 10 [ENTER] 2 [f][PRISM]

A4: 224 m³
Setup: 32 × 7
SC-6: 32 [ENTER] 7 [×]

A5: 10 1/2 ft³ (10.5)
Setup: Prism(3.5×1.5×2)
SC-6: 3.5 [ENTER] 1.5 [ENTER] 2 [f][PRISM]

A6: 27 in.³
Setup: Prism(4.5×3×2)
SC-6: 4.5 [ENTER] 3 [ENTER] 2 [f][PRISM]

A7: 6 in.
Setup: 270 ÷ (9 × 5)
SC-6: 270 [ENTER] 9 [ENTER] 5 [×] [÷]

A8: 8 half-inch cubes
Setup: 2^3
SC-6: 2 [x³]

A9: 8 inches
Setup: 360 ÷ 45
SC-6: 360 [ENTER] 45 [÷]

A10: 216 half-inch cubes
Setup: 27 × 2^3
SC-6: 27 [ENTER] 2 [x³] [×]

PRACTICE B (p.82): REAL LIFE (#1-#10)

ExB: 8 inches deep

B1: 500 m³
Setup: Prism(25×10×2)
SC-6: 25 [ENTER] 10 [ENTER] 2 [f][PRISM]

B2: 48 ft³
Setup: Prism(8×4×1.5)
SC-6: 8 [ENTER] 4 [ENTER] 1.5 [f][PRISM]

B3: 15,625 cm³
Setup: 25^3
SC-6: 25 [x³]

B4: 65 in³
Setup: Prism(6.5×4×2.5)
SC-6: 6.5 [ENTER] 4 [ENTER] 2.5 [f][PRISM]

B5: 8 per in³; 96 half-inch cubes
Setup: 12 × 8
SC-6: 12 [ENTER] 8 [×]

B6: 3 cm rise
Setup: 120 ÷ 40
SC-6: 120 [ENTER] 40 [÷]

B7: 77 ft³
Setup: Prism(5.5×4×3.5)
SC-6: 5.5 [ENTER] 4 [ENTER] 3.5 [f][PRISM]

B8: 36 ft³
Setup: (12 × 9) ÷ 3
SC-6: 12 [ENTER] 9 [×] 3 [÷]

B9: 8 inches
Setup: 360 ÷ 45
SC-6: 360 [ENTER] 45 [÷]

B10: 4,608 in³ ~ 19.95 (about 20) gallons
Setup: Prism(24×12×16) ÷ 231
SC-6: 24 [ENTER] 12 [ENTER] 16 [f][PRISM] 231 [÷]

SEC 8.4 (pp.83-84): CIRCLES, CONES, AND THE PYTHAGOREAN THEOREM

Methods: W1: Exact Form in Terms of π & $a^2 + b^2 = c^2$ | W2: Decimal Approximation (π ~ 3.14) & Scaled Triples

PRACTICE A (p.83): KEY IDEAS (#1-#10)

ExA: Pizza = 113.10 in.²; Ramp diagonal = 13 ft

A1: 78.54 cm²
Setup: $\pi \times 5^2$
SC-6: 5 [f][CIRC]

A2: 706.86 cm²
Setup: $\pi \times 15^2$
SC-6: 15 [f][CIRC]

A3: 5 cm
Setup: $\sqrt{(3^2 + 4^2)}$
SC-6: 3 [ENTER] 4 [g][HYP]

A4: 10 meters
Setup: $\sqrt{(6^2 + 8^2)}$
SC-6: 6 [ENTER] 8 [g][HYP]

A5: 13 ft
Setup: $\sqrt{(12^2 + 5^2)}$
SC-6: 12 [ENTER] 5 [g][HYP]

A6: r = 8 m; Area = 201.06 m²
Setup: $\pi \times (16 \div 2)^2$
SC-6: 16 [ENTER] 2 [÷] [f][CIRC]

A7: 18.85 m³
Setup: $1/3\pi(3^2)(2)$
SC-6: 3 [ENTER] 2 [g][CONE]

A8: 188.50 in.³
Setup: $1/3\pi(6^2)(5)$
SC-6: 6 [ENTER] 5 [g][CONE]

A9: Cylinder = 282.74 cm³; Cone = 94.25 cm³
Setup: $1/3\pi(3^2)(10)$
SC-6: 3 [ENTER] 10 [g][CONE]

A10: 40 inches
Setup: $\sqrt{(32^2 + 24^2)}$
SC-6: 32 [ENTER] 24 [g][HYP]

PRACTICE B (p.84): REAL LIFE (#1-#10)

ExB: 94.25 cm² (exact: 30π cm²)

B1: 153.94 ft²
Setup: $\pi \times 7^2$
SC-6: 7 [f][CIRC]

B2: r = 20 m; Area ~ 1,256.64 m²
Setup: $\pi \times 20^2$
SC-6: 20 [f][CIRC]

B3: 87.96 inches
Setup: (2 × π) × 14
SC-6: 2 [ENTER] [π] [×] 14 [×]

B4: 25 meters
Setup: $\sqrt{(15^2 + 20^2)}$
SC-6: 15 [ENTER] 20 [g][HYP]

B5: 20 inches
Setup: $\sqrt{(16^2 + 12^2)}$
SC-6: 16 [ENTER] 12 [g][HYP]

B6: 150.80 cm² (48π)
Setup: $1/3\pi(4^2)(9)$
SC-6: 4 [ENTER] 9 [g][CONE]

B7: 565.49 in² (180π)
Setup: $1/3\pi(6^2)(15)$
SC-6: 6 [ENTER] 15 [g][CONE]

B8: One 16-in pizza (201.06 in² > 157.08 in²)
Setup: $\pi \times 5^2 \times 2$
SC-6: 8 [f][CIRC] 5 [f][CIRC] 2 [×]

B9: 56.55 ft³
Setup: $\pi \times 6^2 \div 2$
SC-6: 6 [f][CIRC] 2 [÷]

B10: 17 feet
Setup: $\sqrt{(8^2 + 15^2)}$
SC-6: 8 [ENTER] 15 [g][HYP]

Unit 9: Statistical Measures & Data Displays — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapters 9 & 10 (Statistical Measures and Data Displays) • Hands-On Models: Counting & Array Tiles (Counting & Array Tiles) + Probability Dice Studio (Probability Dice & Spinner)

Target Misconception: Students forget to sort data before finding Median/Quartiles, lose their place adding long lists for Mean, and dread Mean Absolute Deviation (MAD) because doing it by hand takes 20 minutes of subtraction.

Hands-On Block Remediation: Show that Mean is physically leveling unequal cube towers to the same height. Then show that entering a list ONCE with [Σ+] unlocks Mean ([g] [x-bar]), Median ([f] [MED]), Mode ([g] [MODE]), Range ([RNG]), Quartiles ([f] [1Q], [g] [3Q], [IQR]), and MAD ([g] [MAD]) in 1 click each!



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 9.1 (pp.86-87): STATISTICAL QUESTIONS, DOT PLOTS, AND THE MEAN

Methods: W1: Mean Formula (Sum ÷ Count) | W2: Fair-Share Block Leveling | W3: Balance Point Deviations

PRACTICE A (p.86): KEY IDEAS (#1-#10)

ExA: Sum = 105, Count = 5, Mean = 21

A1: Sum = 20, Mean = 5
Setup: Mean(2, 4, 6, 8)
SC-6: [g][CLΣ] 2 [Σ+] 4 [Σ+] 6 [Σ+] 8 [Σ+] [g][x-bar]

A2: Sum = 70, Mean = 14
Setup: Sort/Load 5 values -> Stats
SC-6: Load 5 nums into [Σ+] -> [g][x-bar]

A3: 156 cm
Setup: Mean(150, 162, 156)
SC-6: 150 [Σ+] 162 [Σ+] 156 [Σ+] [g][x-bar]

A4: Mean = 4.0 g
Setup: Load data -> Stats
SC-6: Load into [Σ+] -> [g][x-bar]

A5: Sum = 55, Mean = 18.33
Setup: Mean(35, 25)
SC-6: ... 35 [f][Σ-] 25 [Σ+] [g][x-bar]

A6: 13.33 (80 ÷ 6)
Setup: Mean(10, 4, 20, 2)
SC-6: 10 [ENTER] 4 [FRQ] 20 [ENTER] 2 [FRQ] [g][x-bar]

A7: 70°F
Setup: Sort/Load 6 values -> Stats
SC-6: Load 6 temps into [Σ+] -> [g][x-bar]

A8: 60 total blocks
Setup: 5 × 12
SC-6: 5 [ENTER] 12 [×]

A9: 95
Setup: 360 - 265
SC-6: 360 [ENTER] 265 [-]

A10: 17 points
Setup: (90 + 29) ÷ 7
SC-6: 90 [ENTER] 29 [+] 7 [÷]

PRACTICE B (p.87): REAL LIFE (#1-#10)

ExB: 94

B1: Sum = 85; Mean = 17 points
Setup: Mean(14, 18, 12, 22, 19)
SC-6: [g][CLΣ] 14 [Σ+] 18 [Σ+] 12 [Σ+] 22 [Σ+] 19 [Σ+] [g][x-bar]

B2: 42 grams
Setup: Mean(41, 43, 42, 40, 44...)
SC-6: [g][CLΣ] 41 [Σ+] 43 [Σ+] 42 [Σ+] 40 [Σ+] 44 [Σ+] 42 [Σ+] [g][x-bar]

B3: 89 ÷ 10 = 8.9
Setup: (((3 × 8) + (5 × 9)) + (2 × 10)) ÷ 10
SC-6: 3 [ENTER] 8 [×] 5 [ENTER] 9 [×] [+] 2 [ENTER] 10 [×] [+] 10 [÷]

B4: 91
Setup: (5 × 85) - (((80 + 88) + 82) + 84)
SC-6: 5 [ENTER] 85 [×] 80 [ENTER] 88 [+] 82 [+] 84 [+] [-]

B5: Total = 36,000; Mean = 9,000 steps
Setup: (((8200 + 9500) + 7800) + 10500) ÷ 4
SC-6: 8200 [ENTER] 9500 [+] 7800 [+] 10500 [+] 4 [÷]

B6: 82.5 -> 90.0 (+7.5 points)
Setup: ((88 + 92) + 90) ÷ 3
SC-6: 88 [ENTER] 92 [+] 90 [+] 3 [÷]

B7: 12 books per stack
Setup: (((6 + 10) + 14) + 18) ÷ 4
SC-6: 6 [ENTER] 10 [+] 14 [+] 18 [+] 4 [÷]

B8: 4.8 g
Setup: ((((4.2 + 5.1) + 4.8) + 5.5) + 4.4) ÷ 5
SC-6: 4.2 [ENTER] 5.1 [+] 4.8 [+] 5.5 [+] 4.4 [+] 5 [÷]

B9: Reading time varies from student to student
Setup: 60 × 1
SC-6: 60 [ENTER] 1 [×]

B10: 260 ÷ 10 = 26 points
Setup: (80 + 180) ÷ 10
SC-6: 80 [ENTER] 180 [+] 10 [÷]

SEC 9.2 (pp.88-89): MEASURES OF CENTER: MEDIAN, MODE, AND RANGE

Methods: W1: Ordered List (Least to Greatest) | W2: Dot Plot / Frequency Table Analysis

PRACTICE A (p.88): KEY IDEAS (#1-#10)

ExA: Mean = 7.875, Median = 7.5, Mode = 7, Range = 4

A1: Mode = 7
Setup: Sort/Load 8 values -> Mode
SC-6: Load 8 nums -> [g][MODE]

A2: Median = 14, Mode = 12, Range = 8
Setup: Sort/Load 6 values -> Med/Mode/Range
SC-6: Load 6 nums -> [f][MED], [g][MODE], [RNG]

A3: Mean = 9, Median = 8, Mode = 8, Range = 9
Setup: Sort/Load 5 values -> Stats
SC-6: Load 5 nums into [Σ+]

A4: Min = 62°F, Max = 85°F, Range = 23°F
Setup: Sort/Load 7 values -> Range
SC-6: Load 7 nums -> [f][MIN], [g][MAX], [RNG]

A5: 75°F
Setup: Sort/Load 6 values -> Med
SC-6: Load 6 nums -> [f][MED]

A6: Mean = $29.17, Median = $17.50 (Median is best)
Setup: Sort/Load 6 values -> Mean/Med
SC-6: Load 6 nums -> [g][x-bar], [f][MED]

A7: Mean increases from 14 to 30.8; Median stays 14
Setup: Compare both sets
SC-6: Mental / Table Check

A8: 95
Setup: 64 + 31
SC-6: 64 [ENTER] 31 [+]

A9: Median = 14, Range = 13
Setup: Sorted: 8, 11, 14, 16, 21
SC-6: Mental / Table Check

A10: Mean = $370k, Median = $220k (Use Median due to the $980k ou..
Setup: Sort/Load 5 values -> Mean/Med
SC-6: Load 5 nums -> [g][x-bar], [f][MED]

PRACTICE B (p.89): REAL LIFE (#1-#10)

ExB: Mean = $29.17, Median = $17.50, Mode = $15 (Median is best)

B1: Median = 62, Mode = 62, Range = 22
Setup: Med/Mode/Range(55, 62, 48, 70, 62...)
SC-6: [g][CLΣ] 55 [Σ+] 62 [Σ+] 48 [Σ+] 70 [Σ+] 62 [Σ+] 50 [Σ+] 66 [Σ+] [f][MED] [g][MODE] [..

B2: Median = 12.5, Mode = 12, Range = 5
Setup: Med(11, 13, 12, 14, 12...)
SC-6: [g][CLΣ] 11 [Σ+] 13 [Σ+] 12 [Σ+] 14 [Σ+] 12 [Σ+] 16 [Σ+] [f][MED]

B3: Mean=$40, Med=$25, Mode=$20; Median (outlier $105 skews mean)
Setup: Mean/Med(20, 25, 20, 30, 105)
SC-6: [g][CLΣ] 20 [Σ+] 25 [Σ+] 20 [Σ+] 30 [Σ+] 105 [Σ+] [g][x-bar] [f][MED]

B4: Modes = 6 and 8 (bimodal); Median = 7
Setup: 8
SC-6: 6 [ENTER] 8

B5: Median = 11.8 s, Range = 1.2 s, No mode
Setup: 12.4 - 11.2
SC-6: 12.4 [ENTER] 11.2 [-]

B6: Min = 64, Max = 81, Range = 17°F, Median = 72°F
Setup: Med/Range(72, 68, 75, 81, 64...)
SC-6: [g][CLΣ] 72 [Σ+] 68 [Σ+] 75 [Σ+] 81 [Σ+] 64 [Σ+] 70 [Σ+] 74 [Σ+] [f][MED] [RNG]

B7: Mean = 11, Median = 11, Mode = 12, Range = 7
Setup: Mean/Med/Mode/Range(8, 12, 9, 15, 12...)
SC-6: [g][CLΣ] 8 [Σ+] 12 [Σ+] 9 [Σ+] 15 [Σ+] 12 [Σ+] 10 [Σ+] [g][x-bar] [f][MED] [g][MODE] [R..

B8: Example: 8, 8, 10, 12, 15
Setup: 15 - 8
SC-6: 15 [ENTER] 8 [-]

B9: Median increases by 5 (to 85); Range stays 20
Setup: 95 - 75
SC-6: 95 [ENTER] 75 [-]

B10: Mean = 11.75, Median = 11.5, Mode = 10, Range = 6
Setup: Mean/Med(9, 12, 10, 14, 11...)
SC-6: [g][CLΣ] 9 [Σ+] 12 [Σ+] 10 [Σ+] 14 [Σ+] 11 [Σ+] 15 [Σ+] 10 [Σ+] 13 [Σ+] [g][x-bar] [f][M..

SEC 9.3 (pp.90-91): QUARTILES, INTERQUARTILE RANGE (IQR) & BOX PLOTS

Methods: W1: Split Ordered Data into Lower & Upper Halves | W2: Box-and-Whisker Plot & 25% Quartile Sections

PRACTICE A (p.90): KEY IDEAS (#1-#10)

ExA: Min = 68, Q1 = 73.5, Med = 82, Q3 = 90, Max = 96; IQR = 16.5

A1: Min = 10, Q1 = 15, Med = 30, Q3 = 45, Max = 50, IQR = 30
Setup: Sort/Load 5 values -> Q1/Q3/IQR
SC-6: Load 5 nums -> [f][1Q], [g][3Q], [IQR]

A2: Q1 = 6, Med = 10, Q3 = 14, IQR = 8
Setup: Sort/Load 7 values -> Q1/Q3/IQR
SC-6: Load 7 nums -> [f][1Q], [g][3Q], [IQR]

A3: 34
Setup: 58 - 24
SC-6: 58 [ENTER] 24 [-]

A4: 75%
Setup: 3 quarters = 3 × 25%
SC-6: Mental / Table Check

A5: Q1 = 73.5, Med = 82, Q3 = 90, IQR = 16.5
Setup: Sort/Load 8 values -> Stats
SC-6: Load 8 nums into [Σ+]

A6: Q1 = 46.5, Med = 51, Q3 = 56.5, IQR = 10
Setup: Sort/Load 8 values -> Stats
SC-6: Load 8 nums into [Σ+]

A7: Player A (smaller IQR means the middle 50% of scores are clo..
Setup: Compare box widths
SC-6: Mental / Table Check

A8: 90 (Any score > 90 is an outlier)
Setup: 60 + (1.5 × 20)
SC-6: 60 [ENTER] 1.5 [ENTER] 20 [×] [+]

A9: Q1 = 15, Med = 22, Q3 = 28, IQR = 13
Setup: Sort/Load 7 values -> Q1/Q3/IQR
SC-6: Load 7 nums -> [IQR]

A10: About 100 students (50% of 200)
Setup: 50% of 200
SC-6: 200 [ENTER] 50 [%]

PRACTICE B (p.91): REAL LIFE (#1-#10)

ExB: IQR = 16; 20 students (50%)

B1: Min=40, Q1=45, Med=55, Q3=65, Max=70; IQR = 20
Setup: Q1/Q3/IQR(40, 45, 50, 55, 60...)
SC-6: [g][CLΣ] 40 [Σ+] 45 [Σ+] 50 [Σ+] 55 [Σ+] 60 [Σ+] 65 [Σ+] 70 [Σ+] [IQR]

B2: Q1 = 7.5, Med = 10, Q3 = 13, IQR = 5.5
Setup: Q1/Q3/IQR(6, 7, 8, 9, 11...)
SC-6: [g][CLΣ] 6 [Σ+] 7 [Σ+] 8 [Σ+] 9 [Σ+] 11 [Σ+] 12 [Σ+] 14 [Σ+] 15 [Σ+] [IQR]

B3: Q1 = 22.5, Med = 30, Q3 = 42.5, IQR = 20
Setup: Q1/Q3/IQR(30, 15, 45, 25, 35...)
SC-6: [g][CLΣ] 30 [Σ+] 15 [Σ+] 45 [Σ+] 25 [Σ+] 35 [Σ+] 20 [Σ+] 50 [Σ+] 40 [Σ+] 30 [Σ+] [IQR]

B4: Range = 28 ft; IQR = 13 ft
Setup: 31 - 18
SC-6: 40 [ENTER] 12 [-] 31 [ENTER] 18 [-]

B5: 75% = 45 trees
Setup: 75% of 60
SC-6: 60 [ENTER] 75 [%]

B6: 25% of 60 = 15 trees
Setup: 25% of 60
SC-6: 60 [ENTER] 25 [%]

B7: Class A (smaller IQR means less spread in the middle 50%)
Setup: 19 - 8
SC-6: 19 [ENTER] 8 [-]

B8: 30 + 15 = 45
Setup: 30 + (1.5 × 10)
SC-6: 30 [ENTER] 1.5 [ENTER] 10 [×] [+]

B9: Min=18, Q1=22, Med=27, Q3=34, Max=40; IQR = 12
Setup: Q1/Q3/IQR(18, 22, 25, 29, 34...)
SC-6: [g][CLΣ] 18 [Σ+] 22 [Σ+] 25 [Σ+] 29 [Σ+] 34 [Σ+] 40 [Σ+] [IQR]

B10: Range = 90, IQR = 8 (IQR is not distorted by the outlier 100)
Setup: Q1/Q3/IQR(10, 12, 14, 16, 18...)
SC-6: [g][CLΣ] 10 [Σ+] 12 [Σ+] 14 [Σ+] 16 [Σ+] 18 [Σ+] 20 [Σ+] 100 [Σ+] [IQR]

SEC 9.4 (pp.92-93): MEAN ABSOLUTE DEVIATION (MAD)

Methods: W1: 3-Step MAD Distance Table | W2: Number-Line Deviation Balance

PRACTICE A (p.92): KEY IDEAS (#1-#10)

ExA: Mean = 14, MAD = 2.4

A1: Mean = 156 cm, MAD = 4.0 cm
Setup: MAD(150, 162, 156)
SC-6: 150 [Σ+] 162 [Σ+] 156 [Σ+] [g][MAD]

A2: Mean = 14, MAD = 2.4
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

A3: Mean = 20, MAD = 6.0
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

A4: Mean = 7, MAD = 0 (Zero spread)
Setup: All distances from 7 are 0
SC-6: Mental / Table Check

A5: Mean = 28, MAD = 4.8
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

A6: Mean = 13.0 s, MAD = 0.40 s
Setup: Sort/Load 6 values -> Mean/MAD
SC-6: Load 6 decimals -> [g][x-bar], [g][MAD]

A7: Mean = 88, MAD = 3.6
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

A8: Team B (a larger MAD indicates greater variation from the me..
Setup: Compare MADs
SC-6: Mental / Table Check

A9: Mean = 10; deviations: 4, 0, 1, 3; MAD = 2.0
Setup: Mean=10; dists 4+0+1+3=8 ÷ 4
SC-6: Mental / Table Check

A10: Mean = 9, Med = 8, Mode = 8, Range = 9, IQR = 5.5, MAD = 2.4
Setup: 1 entry -> 6 stats!
SC-6: Mental / Table Check

PRACTICE B (p.93): REAL LIFE (#1-#10)

ExB: Mean = 13.0 s; MAD = 0.40 s

B1: Mean = 210°C; MAD = 3.0°C
Setup: MAD(206, 208, 212, 214)
SC-6: [g][CLΣ] 206 [Σ+] 208 [Σ+] 212 [Σ+] 214 [Σ+] [g][MAD]

B2: Mean = 10; MAD = 1.6 shots
Setup: Mean/MAD(7, 9, 10, 11, 13)
SC-6: [g][CLΣ] 7 [Σ+] 9 [Σ+] 10 [Σ+] 11 [Σ+] 13 [Σ+] [g][x-bar] [g][MAD]

B3: Mean = 50 g, MAD = 0 g (no deviation from the mean)
Setup: 0
SC-6: 0 [ENTER]

B4: Mean = 9 s, MAD = 2.0 s; Raven A is more consistent
Setup: MAD(6, 8, 10, 12)
SC-6: [g][CLΣ] 6 [Σ+] 8 [Σ+] 10 [Σ+] 12 [Σ+] [g][MAD]

B5: Mean = 85; MAD = 4.8
Setup: Mean/MAD(76, 82, 85, 88, 94)
SC-6: [g][CLΣ] 76 [Σ+] 82 [Σ+] 85 [Σ+] 88 [Σ+] 94 [Σ+] [g][x-bar] [g][MAD]

B6: Mean = 15 m, MAD = 1.0 m
Setup: Mean/MAD(15, 16, 14, 15, 17...)
SC-6: [g][CLΣ] 15 [Σ+] 16 [Σ+] 14 [Σ+] 15 [Σ+] 17 [Σ+] 13 [Σ+] [g][x-bar] [g][MAD]

B7: Mean = 85°F; deviations 5, 1, 1, 5; MAD = 3.0°F
Setup: MAD(80, 84, 86, 90)
SC-6: [g][CLΣ] 80 [Σ+] 84 [Σ+] 86 [Σ+] 90 [Σ+] [g][MAD]

B8: The signed deviations (-5 + 0 + 5) always add to 0
Setup: -5 + 5
SC-6: 5 [+/-] [ENTER] 5 [+]

B9: Mean = 5 min; MAD = 2.4 min
Setup: Mean/MAD(2, 3, 4, 5, 11)
SC-6: [g][CLΣ] 2 [Σ+] 3 [Σ+] 4 [Σ+] 5 [Σ+] 11 [Σ+] [g][x-bar] [g][MAD]

B10: Mean=16, Med=15, Mode=15, Rng=8, IQR=5.5, MAD=2.4
Setup: Mean/Med/Mode/Range/Q1/Q3/IQR/MAD(12, 15, 15, 18, 20)
SC-6: [g][CLΣ] 12 [Σ+] 15 [Σ+] 15 [Σ+] 18 [Σ+] 20 [Σ+] [g][x-bar] [f][MED] [g][MODE] [RNG] [..

Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability — Practice A & B Complete Key

Big Ideas Math Grade 7 Bridge — Chapters 1–3 & Chapter 7 • Hands-On Models: Ten-Frame Zero-Pair Counters (Ten-Frames & Two-Tone Counters) + Algebra Stage (Algebra & Area Tiles) + Probability Studio (Probability Dice & Spinner)

Target Misconception: Students guess signs when adding/subtracting negatives, get -25 instead of +25 for $(-5)^2$ on algebraic calculators, undo ×3 before +7 in $3x + 7 = 28$, and confuse Permutations (order matters) with Combinations (order doesn't).

Hands-On Block Remediation: Use Yellow/Red counters to show removing 4 red debt tokens (10 - (-4)) equals +14. Notice how RPN 5 [+/-] [x²] squares the entire register $(-5)^2 = +25$ without parentheses! Unwrap 2-step equations in reverse (untie ribbon ±b first, open box ×a second), and finish with [DICE], [f] [nCr], [g] [nPr], and [f] [?].



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 10.1 (pp.95-96): ADDING AND SUBTRACTING RATIONAL NUMBERS

Methods: W1: Zero-Pair Counters | W2: Number-Line Jumps | W3: Absolute Value Sign Rules & Add-the-Opposite

PRACTICE A (p.95): KEY IDEAS (#1-#10)

ExA: -8 + 5 = -3; -6 + (-4) = -10

A1: (a) -6; (b) -24
Setup: -15 + 9; -9
SC-6: 15 [+/-] [ENTER] 9 [+] / 9 [+/-] [+]

A2: -9
Setup: 14 + -23
SC-6: 14 [ENTER] 23 [+/-] [+]

A3: (a) 30; (b) -4
Setup: 12 - -18; -12
SC-6: 12 [ENTER] 18 [+/-] [-] / 12 [+/-]...

A4: -25
Setup: -9 - 16
SC-6: 9 [+/-] [ENTER] 16 [-]

A5: -14.2°C
Setup: (0 - 7.8) - 6.4
SC-6: 0 [ENTER] 7.8 [-] 6.4 [-]

A6: -4.8
Setup: 6.4 - 11.2
SC-6: 6.4 [ENTER] 11.2 [-]

A7: +2.7
Setup: -7.5 - -10.2
SC-6: 7.5 [+/-] [ENTER] 10.2 [+/-] [-]

A8: -1/8
Setup: -3/4 - -5/8
SC-6: [.] 3 [.] 4 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [-]

A9: 36°F
Setup: 14 - -22
SC-6: 14 [ENTER] 22 [+/-] [-]

A10: 2 7/8 (2.875)
Setup: (-2.5 + 4.25) + 1.125
SC-6: 2.5 [+/-] [ENTER] 4.25 [+] 1.125 [+]

PRACTICE B (p.96): REAL LIFE (#1-#10)

ExB: 14°C; and -1/8

B1: +8°F
Setup: -14 + 22
SC-6: 14 [+/-] [ENTER] 22 [+]

B2: -83 meters
Setup: -45 + -38
SC-6: 45 [+/-] [ENTER] 38 [+/-] [+]

B3: 4,800 - (-200) = 5,000 meters
Setup: 4800 - -200
SC-6: 4800 [ENTER] 200 [+/-] [-]

B4: -15 - (-25) = +10 points
Setup: -15 - -25
SC-6: 15 [+/-] [ENTER] 25 [+/-] [-]

B5: +$8.75
Setup: (-12.50 + 30) - 8.75
SC-6: 12.50 [+/-] [ENTER] 30 [+] 8.75 [-]

B6: 1/8
Setup: -5/8 + 3/4
SC-6: [.] 5 [.] 8 [+/-] [ENTER] [.] 3 [.] 4 [+]

B7: -9/6 = -1 1/2
Setup: -2/3 - 5/6
SC-6: [.] 2 [.] 3 [+/-] [ENTER] [.] 5 [.] 6 [-]

B8: -16.0 meters
Setup: (-18.4 + 7.9) - 5.5
SC-6: 18.4 [+/-] [ENTER] 7.9 [+] 5.5 [-]

B9: -15.0°C
Setup: -8.5 - 6.5
SC-6: 8.5 [+/-] [ENTER] 6.5 [-]

B10: -8
Setup: ((-14 - -20) + -9) - 5
SC-6: 14 [+/-] [ENTER] 20 [+/-] [-] 9 [+/-] [+] 5 [-]

SEC 10.2 (pp.97-98): MULTIPLYING AND DIVIDING RATIONAL NUMBERS

Methods: W1: Same/Different Sign Rules | W2: Repeated Addition / Equal Number-Line Jumps & Counter Flips

PRACTICE A (p.97): KEY IDEAS (#1-#10)

ExA: 6×(-4) = -24, (-6)×(-4) = +24, $(-5)^3$ = +25

A1: (a) -56; (b) +56
Setup: 7 × -8; -7
SC-6: 7 [ENTER] 8 [+/-] [×] / 7 [+/-]...

A2: (a) -5; (b) +5
Setup: -45 ÷ 9; -9
SC-6: 45 [+/-] [ENTER] 9 [÷] / 9 [+/-] [÷]

A3: +25 (2 negative factors = positive)
Setup: -5^2
SC-6: 5 [+/-] [x²]

A4: -8 (3 negative factors = odd count = negative)
Setup: -2^3
SC-6: 2 [+/-] [x³]

A5: +8.4
Setup: -3.5 × -2.4
SC-6: 3.5 [+/-] [ENTER] 2.4 [+/-] [×]

A6: -12
Setup: -14.4 ÷ 1.2
SC-6: 14.4 [+/-] [ENTER] 1.2 [÷]

A7: -3/5 (-0.6)
Setup: -2/3 × 9/10
SC-6: [.] 2 [.] 3 [+/-] [ENTER] [.] 9 [.] 10 [×]

A8: +2
Setup: -5/6 ÷ -5/12
SC-6: [.] 5 [.] 6 [+/-] [ENTER] [.] 5 [.] 12 [+/-] [÷]

A9: -120
Setup: (-4 × -5) × -6
SC-6: 4 [+/-] [ENTER] 5 [+/-] [×] 6 [+/-] [×]

A10: -76
Setup: (-18 × 4) - (-36 ÷ -9)
SC-6: 18 [+/-] [ENTER] 4 [×] 36 [+/-] [ENTER] 9 [+/-] [÷] [-]

PRACTICE B (p.98): REAL LIFE (#1-#10)

ExB: 2 minutes

B1: -120 feet
Setup: 8 × -15
SC-6: 8 [ENTER] 15 [+/-] [×]

B2: +40 points (same signs give a positive product)
Setup: -4 × -10
SC-6: 4 [+/-] [ENTER] 10 [+/-] [×]

B3: -4.5°F per hour
Setup: -27 ÷ 6
SC-6: 27 [+/-] [ENTER] 6 [÷]

B4: $(-4)^3$ = +16; $(-4)^3$ = -64
Setup: -4^6
SC-6: 4 [+/-] [x³] 4 [+/-] [x³]

B5: +2/3
Setup: -3/4 × -8/9
SC-6: [.] 3 [.] 4 [+/-] [ENTER] [.] 8 [.] 9 [+/-] [×]

B6: -2
Setup: -7/8 ÷ 7/16
SC-6: [.] 7 [.] 8 [+/-] [ENTER] [.] 7 [.] 16 [÷]

B7: -12
Setup: (-2.5 × -1.6) × -3
SC-6: 2.5 [+/-] [ENTER] 1.6 [+/-] [×] 3 [+/-] [×]

B8: -3.2 meters per year
Setup: -14.4 ÷ 4.5
SC-6: 14.4 [+/-] [ENTER] 4.5 [÷]

B9: +120
Setup: ((-2 × -3) × -4) × -5
SC-6: 2 [+/-] [ENTER] 3 [+/-] [×] 4 [+/-] [×] 5 [+/-] [×]

B10: 36 - (-20) + 6 = 62
Setup: $(-6^2 - (4 \times -5)) + (-18 \div -3)$
SC-6: 6 [+/-] [x²] 4 [ENTER] 5 [+/-] [×] [-] 18 [+/-] [ENTER] 3 [+/-..

SEC 10.3 (pp.99-100): SOLVING TWO-STEP EQUATIONS (AX + B = C)

Methods: W1: Two-Step Inverse Operations | W2: Balance Scale / Bar Model Unwrapping

PRACTICE A (p.99): KEY IDEAS (#1-#10)

ExA: x = 7 (Step 1: 3x = 21)

A1: 4x = 36; x = 9
Setup: (45 - 9) ÷ 4
SC-6: 45 [ENTER] 9 [-] 4 [÷]

A2: 5x = 50; x = 10
Setup: (38 + 12) ÷ 5
SC-6: 38 [ENTER] 12 [+] 5 [÷]

A3: 6y = 48; y = 8
Setup: (59 - 11) ÷ 6
SC-6: 59 [ENTER] 11 [-] 6 [÷]

A4: 8n = 64; n = 8
Setup: (49 + 15) ÷ 8
SC-6: 49 [ENTER] 15 [+] 8 [÷]

A5: m / 4 = 9; m = 36
Setup: (15 - 6) × 4
SC-6: 15 [ENTER] 6 [-] 4 [×]

A6: k / 3 = 12; k = 36
Setup: (7 + 5) × 3
SC-6: 7 [ENTER] 5 [+] 3 [×]

A7: -3x = -12; x = 4
Setup: (2 - 14) ÷ -3
SC-6: 2 [ENTER] 14 [-] 3 [÷/-] [÷]

A8: 2.5w = 15; w = 6
Setup: (23 - 8) ÷ 2.5
SC-6: 23 [ENTER] 8 [-] 2.5 [÷]

A9: 25h + 35 = 160; h = 5 hours
Setup: (160 - 35) ÷ 25
SC-6: 160 [ENTER] 35 [-] 25 [÷]

A10: 15w + 45 = 120; w = 5 weeks
Setup: (120 - 45) ÷ 15
SC-6: 120 [ENTER] 45 [-] 15 [÷]

PRACTICE B (p.100): REAL LIFE (#1-#10)

ExB: 4 hours

B1: 6r + 8 = 38 -> r = 5 laps
Setup: (38 - 8) ÷ 6
SC-6: 38 [ENTER] 8 [-] 6 [÷]

B2: 4h + 12 = 48 -> h = 9 hours
Setup: (48 - 12) ÷ 4
SC-6: 48 [ENTER] 12 [-] 4 [÷]

B3: 15w + 35 = 140 -> w = 7 weeks
Setup: (140 - 35) ÷ 15
SC-6: 140 [ENTER] 35 [-] 15 [÷]

B4: 4b - 10 = 46 -> b = $14 per book
Setup: (46 + 10) ÷ 4
SC-6: 46 [ENTER] 10 [+] 4 [÷]

B5: x = 6 (-4·6 + 18 = -6)
Setup: (-6 - 18) ÷ -4
SC-6: 6 [+/-] [ENTER] 18 [-] 4 [+/-] [÷]

B6: n = 35
Setup: (16 - 9) × 5
SC-6: 16 [ENTER] 9 [-] 5 [×]

B7: 3000 - 40m = 600 -> m = 60 minutes
Setup: (600 - 3000) ÷ -40
SC-6: 600 [ENTER] 3000 [-] 40 [+/-] [÷]

B8: x = 24
Setup: (11 + 5) ÷ 2/3
SC-6: 11 [ENTER] 5 [÷] [.] 2 [.] 3 [÷]

B9: 2w + 36 = 56 -> w = 10 cm
Setup: (56 - 36) ÷ 2
SC-6: 56 [ENTER] 36 [-] 2 [÷]

B10: 2.25m + 3.50 = 21.50 -> m = 8 miles
Setup: (21.50 - 3.50) ÷ 2.25
SC-6: 21.50 [ENTER] 3.50 [-] 2.25 [÷]

SEC 10.4 (pp.101-102): THEORETICAL PROBABILITY, COMBINATIONS & PERMUTATIONS

Methods: W1: Probability Ratio & Complement Rule | W2: Tree Diagram / Outcome List | W3: Fundamental Counting Principle

PRACTICE A (p.101): KEY IDEAS (#1-#10)

ExA: P(Blue) = 2/5 = 0.4 = 40%; 5! = 120 ways

A1: P(5) = 1/6; P(not 6) = 5/6
Setup: 1/6; and 1 - 1/6 = 5/6
SC-6: Mental / Table Check

A2: 0.65 (65% or 13/20)
Setup: 1 - 0.35
SC-6: 1 [ENTER] 0.35 [-]

A3: P(double 6s) = 1/36; P(sum 11 or 12) = 3/36 = 1/12
Setup: 1/36; and 3/36 = 1/12
SC-6: Mental / Table Check

A4: 24 ways
Setup: 4!
SC-6: 4 [x!]

A5: 720 ways
Setup: 6!
SC-6: 6 [x!]

A6: 56 teams
Setup: C(8,3)
SC-6: 8 [ENTER] 3 [f] [nCr]

A7: 336 podium arrangements
Setup: P(8,3)
SC-6: 8 [ENTER] 3 [g] [nPr]

A8: 1/8 (12.5%)
Setup: 1/2 × 1/2 × 1/2 = 1/8
SC-6: Mental / Table Check

A9: 1/3 (6/10 × 5/9 = 30/90)
Setup: 6/10 × 5/9 = 30/90
SC-6: Mental / Table Check

A10: 120 combinations vs. 720 permutations (differ by 3! = 6)
Setup: C(10,3); gnPr
SC-6: 10 [ENTER] 3 [f] [nCr] / [g] [nPr]

PRACTICE B (p.102): REAL LIFE (#1-#10)

ExB: 720 ordered podiums; 120 unordered 3-runner crews

B1: 4/20 = 1/5 = 0.20 = 20%
Setup: Simp(4/20)
SC-6: [.] 4 [.] 20 [g] [SIMP] [FDISP]

B2: 65% (13/20 = 0.65)
Setup: 100 - 35
SC-6: 100 [ENTER] 35 [-]

B3: P(black) = 2/5; P(not white) = 11/15
Setup: Simp(6/15)
SC-6: [.] 6 [.] 15 [g] [SIMP]

B4: 120 custom skateboards
Setup: (5 × 4) × 6
SC-6: 5 [ENTER] 4 [×] 6 [×]

B5: 5! = 120 orders
Setup: 5!
SC-6: 5 [x!]

B6: 210 batting orders (permutations)
Setup: P(7,3)
SC-6: 7 [ENTER] 3 [g] [nPr]

B7: 35 lab groups (combinations)
Setup: C(7,3)
SC-6: 7 [ENTER] 3 [f] [nCr]

B8: 28 fruit combinations
Setup: C(8,2)
SC-6: 8 [ENTER] 2 [f] [nCr]

B9: 60 times
Setup: 160 × 3/8
SC-6: 160 [ENTER] [.] 3 [.] 8 [×]

B10: 12 outcomes; P(H and 6) = 1/12
Setup: 2 × 6
SC-6: 2 [ENTER] 6 [×]

Middle School 10-Unit Misconception & Hands-On Remediation Matrix

UNIT & HANDS-ON MODEL	TARGET CLASSROOM MISCONCEPTION	HANDS-ON BLOCK & SC-6 REMEDIATION
Unit 1: Numerical Expressi Counting & Array Studio	Students interpret remainders as meaningless decimal tails (e.g., .875 of a person), read 2^3 as 2×3=6, lose track of PEMDAS parentheses, and confuse GCF (overlap) with LCM (union).	Have students build physical tile rows for division remainders, stack 2D/3D arrays for exponents, use the 4-line elevator (Y holds Room 1 while X solves Room 2) instead of rewriting PEMDAS lines, and stand logarithmic Prime Factor Towers side-by-side on the 12-socket lid.
Unit 2: Fractions & Decima Fraction Studio (Fractio	Students treat fractions, mixed numbers, and decimals as unrelated rules, expect multiplication to always make numbers bigger, forget which fraction to flip in division, and misalign decimal columns.	Use the 84.0 mm Fraction Studio towers so equivalence and common denominators are physical millimeter height matches. Use [f] [1/x] on Floor 1 (X) so students see only the divisor flip, and use [f] [IP] / [g] [FP] to split whole dollars from decimal cents.
Unit 3: Ratios, Rates & Pe Counting & Array Tiles (	Students confuse Part-to-Part (3:2) with Part-to-Whole (3/5), add instead of multiply in ratio tables, divide unit rates upside-down, and memorize 6 disconnected rules for moving decimal points in percents.	Use Blue and Gold Counting Tiles as physical Tape Diagram boxes, step through ratio tables rhythmically with [g] [REP], use [x swap y] when unit-rate division is entered upside-down, and use [f] [%] + [FDISP] to show Fraction/Decimal/Percent are three costumes of one number.
Unit 4: Algebraic Expressi Algebra Boxes & Factorin	Students panic when letters appear, erroneously combine $3x + 2$ into $5x$ or $x^2 + x$ into x^3, and don't see why $3(x + 4) = 3x + 12$.	Show that a variable x is literally a physical green x-rod (42×14mm) whereas 1 is a small yellow unit cube (14×14mm) and x^2 is a large blue flat (42×42mm)—different physical shapes cannot be counted as one pile! Build rectangles on the L-frame stage for the Distributive Property.
Unit 5: Areas of Polygons Counting & Array Tiles (	Students multiply the slanted side of a parallelogram instead of the vertical height, forget to divide by 2 for triangles, lose track of multi-piece composite areas, and subtract clock times in base-10 instead of base-60.	Slide the corner triangle of a parallelogram to form a straight [f] [RECT] rectangle, use the dedicated [g] [TRI] key ([g][7]) which divides by 2 automatically, accumulate composite chunks with [Σ+], and use the Planetary Clock + [CLK] key for base-60 time.
Unit 6: Integers, Rational Ten-Frame Two-Tone Count	Students confuse the subtraction operator [-] with the negative sign [+/-], think $-18 > -5$ because $18 > 5$, confuse Opposite ([+/-]) with Absolute Value ([g] [|x|]), and reverse (x, y) coordinates.	Use two-tone (+1 / -1) Zero-Pair counters and Zero-Pair tiles, plot signed jumps on the SC-6 Number Line ([f] [LINE]), test inequalities with [f] [>] and [g] [<], contrast [+/-] (flip sign) with [g] [|x|] (always positive distance from 0), and use the Elevator Rule (walk across x first, ride up/down y second).
Unit 7: Equations & Inequa Algebra Boxes & Stage (A	Students treat '=' as 'write the answer', translate '8 less than n' backwards as 8 - n, resent writing -14 under both sides of $x + 14 = 39$, and confuse open vs. solid inequality circles.	Model '=' as a Balance Scale using Algebra Boxes & Stage. Solve 1-step equations with inverse operations on both sides, verify your solution with the SC-6 Equality Test [!=] (which flashes NO when Left = Right!), and test inequalities on [f] [LINE] with [f] [>=] and [g] [<=] (which flash YES or NO).
Unit 8: Surface Area & Vol Counting & Array Tiles (	Students drop one of the 6 faces when finding rectangular prism surface area, tangle prism and pyramid formulas, and confuse 2D Surface Area (cardboard outside) with 3D Volume (cubes inside).	Have students touch the 3 pairs of faces on their physical box while logging each pair into [Σ+] (n=1, 2, 3), use [f] [PRISM] ([f][9]) to multiply 3 stack floors (l, w, h) at once for volume, and unlock [f] [CIRC], [g] [CONE], and [g] [HYP].
Unit 9: Statistical Measur Counting & Array Tiles (	Students forget to sort data before finding Median/Quartiles, lose their place adding long lists for Mean, and dread Mean Absolute Deviation (MAD) because doing it by hand takes 20 minutes of subtraction.	Show that Mean is physically leveling unequal cube towers to the same height. Then show that entering a list ONCE with [Σ+] unlocks Mean ([g] [x-bar]), Median ([f] [MED]), Mode ([g] [MODE]), Range ([RNG]), Quartiles ([f] [1Q], [g] [3Q], [IQR]), and MAD ([g] [MAD]) in 1 click each!
Unit 10: Grade 7 Bridge — S Ten-Frame Zero-Pair Coun	Students guess signs when adding/subtracting negatives, get -25 instead of +25 for $(-5)^2$ on algebraic calculators, undo ×3 before +7 in $3x + 7 = 28$, and confuse Permutations (order matters) with Combinations (order doesn't).	Use Yellow/Red counters to show removing 4 red debt tokens (10 - (-4)) equals +14. Notice how RPN 5 [+/-] [x^2] squares the entire register $(-5)^2 = +25$ without parentheses! Unwrap 2-step equations in reverse (untie ribbon ±b first, open box ×a second), and finish with [DICE], [f] [nCr], [g] [nPr], and [f] [?].

Photocopiable 10-Unit Middle School Student Mastery Tracker

STUDENT NAME / PERIOD	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10
01. _____										
02. _____										
03. _____										
04. _____										
05. _____										
06. _____										
07. _____										
08. _____										
09. _____										
10. _____										
11. _____										
12. _____										
13. _____										
14. _____										
15. _____										
16. _____										
17. _____										
18. _____										
19. _____										
20. _____										

TEACHER OBSERVATION NOTES & INTERVENTION LOG

- **Spiral-Bound Student Workbooks & Teacher Guides (Lulu): https://www.lulu.com/spotlight/stackcalc**
- Companion Book — From Blocks to the Stack (Amazon): https://link.amazon/B0f5cP0Ur | Manipulatives: https://stackcalc.io

General Student Lab Workbook Key — Part 1

Complete worked solutions and canonical SC-6 / SC-32 keystrokes for Core, Extension, and Lab Problems 1–4.

Domain / Unit 01: Counting, Cardinality & The One-More Step Counter (Pre-K – Kindergarten • SC-6)

Core Investigation: 5 pegs. Keystrokes: [4] [ENTER] [1] [+]

Core Keys: `4` `ENTER` `1` `+`

Extension Challenge: 4 more pegs. Keystrokes: [6] [ENTER] [2] [-]

Ext. Keys: `6` `ENTER` `2` `-`

Student Lab Sheet Answers: P1: 5 | P2: 3 | P3: 0 (Zero) | P4: 4

Domain / Unit 02: Part-Part-Whole & Doubles: The Math Domino Studio (Grades K–1 • SC-6)

Core Investigation: 9 pips. Keystrokes: [5] [ENTER] [4] [+]

Core Keys: `5` `ENTER` `4` `+`

Extension Challenge: Missing Part is 3. Keystrokes: [8] [ENTER] [5] [-]

Ext. Keys: `8` `ENTER` `5` `-`

Student Lab Sheet Answers: P1: 4 | P2: 9 | P3: 9 | P4: 3

Domain / Unit 03: Ten-Frames: Complements to 10 & 5-Structure (Grade 1 • SC-6)

Core Investigation: 4 counters. Keystrokes: [1][0] [ENTER] [6] [-]

Core Keys: `1` `0` `ENTER` `6` `-`

Extension Challenge: 15. Keystrokes: [9] [ENTER] [1] [+] [5] [+]

Ext. Keys: `9` `ENTER` `1` `+` `5` `+`

Student Lab Sheet Answers: P1: 6 | P2: 12 | P3: 1 | P4: 14

Domain / Unit 04: Number Racks (Rekenrek): 5-Structure & Doubles (Grades 1–2 • SC-6)

Core Investigation: 14 beads. Keystrokes: [7] [ENTER] [+]

Core Keys: `7` `ENTER` `+`

Extension Challenge: 13. Keystrokes: [6] [ENTER] [+] [1] [+]

Ext. Keys: `6` `ENTER` `+` `1` `+`

Student Lab Sheet Answers: P1: 16 | P2: 17 | P3: 18 | P4: 11

Domain / Unit 05: The 4-Register Stack: Four Operations & Automatic Lift (Grades 3–4 • SC-6)

Core Investigation: 3 cookies per bag. Keystrokes: [1][5] [ENTER] [6] [-] [3] [÷]

Core Keys: `1` `5` `ENTER` `6` `-` `3` `÷`

Extension Challenge: 19. Keystrokes: [2][4] [ENTER] [6] [÷] [1][5] [+]

Ext. Keys: `2` `4` `ENTER` `6` `÷` `1` `5` `+`

Student Lab Sheet Answers: P1: 27 | P2: 6 | P3: 4 | P4: 3 eggs

Domain / Unit 06: Fraction Studio: Height Equivalence & Two-Dot Notation (Grades 4–5 • SC-6)

Core Investigation: 3/4 (height 63 mm). Keystrokes: [.][1][.][2] [ENTER] [.][1][.][4] [+]

Core Keys: `•` `1` `•` `2` `ENTER` `•` `1` `•` `4` `+`

Extension Challenge: 1/4 (height 21 mm). Keystrokes: [.][2][.][3] [ENTER] [.][3][.][8] [×]

Ext. Keys: `•` `2` `•` `3` `ENTER` `•` `3` `•` `8` `×`

Student Lab Sheet Answers: P1: 1/2 (42 mm) | P2: 3/4 (63 mm) | P3: 0.375 | P4: 1/2 (42 mm)

Diagnostic Grading Rubric

Award credit for: (1) Concrete manipulative model sketch, (2) Written mathematical strategy & prediction before pressing keys, and (3) Stack verification on the SC-6 or SC-32.

General Student Lab Workbook Key — Part 2

Complete worked solutions and canonical SC-6 / SC-32 keystrokes for Core, Extension, and Lab Problems 1–4.

Domain / Unit 07: Postfix Evaluation & Chained Operations: The 4-Register Stack (Grades 5–6 • SC-6)

Core Investigation: 50. Keystrokes: [8] [ENTER] [2] [+] [1][4] [ENTER] [9] [-] [×]

Core Keys: `8` `ENTER` `2` `+` `1` `4` `ENTER` `9` `-` `×`

Extension Challenge: 2. Keystrokes: [1][5] [ENTER] [5] [-] [3] [ENTER] [2] [+] [÷]

Ext. Keys: `1` `5` `ENTER` `5` `-` `3` `ENTER` `2` `+` `÷`

Student Lab Sheet Answers: P1: 28 | P2: 2 | P3: 4 | P4: 17

Domain / Unit 08: Tactile Algebra: Area Models, Polynomials & Quadratic Factoring (Grades 7–8 • SC-32)

Core Investigation: (x + 3)(x + 4), roots -3 and -4. Keystrokes: [1] [ENTER] [7] [ENTER] [1][2] [g:][POLY]

Core Keys: `1` `ENTER` `7` `ENTER` `1` `2` `↷` `POLY`

Extension Challenge: (x - 2)(x - 3), roots 2 and 3. Keystrokes: [1] [ENTER] [5] [+/-] [ENTER] [6] [g:][POLY]

Ext. Keys: `1` `ENTER` `5` `+/-` `ENTER` `6` `↷` `POLY`

Student Lab Sheet Answers: P1: (x + 2)(x + 4); roots -2, -4 | P2: (x - 2)(x + 2); roots 2, -2 | P3: (x + 1)(x + 4); roots -1, -4 | P4: (x - 2)(x - 3); roots 2, 3

Domain / Unit 09: Trigonometry, Vectors & Polar/Rectangular Coordinates (Grades 9–11 • SC-32)

Core Investigation: Displacement: 100 m, Heading: 53.13°. Keystrokes: [6][0] [ENTER] [8][0] [f:][->ϑ,r] [x swap y]

Core Keys: `6` `0` `ENTER` `8` `0` `↶` `▸θ,r` `x⇄y`

Extension Challenge: Magnitude: 6.71. Resultant: <3, 6>.

Ext. Keys: `5` `ENTER` `2` `+/-` `+` `2` `ENTER` `4` `+` `↶` `▸θ,r` `x⇄y`

Student Lab Sheet Answers: P1: ϑ = 22.62°, r = 13 | P2: 25 | P3: 0.5 | P4: x = 7.07, y = 7.07

Domain / Unit 10: Exponential Growth, Compound Interest & Logarithms (Grades 10–12 • SC-32)

Core Investigation: 40,500 bacteria. Keystrokes: [3] [ENTER] [4] [y^x] [5][0][0] [×]

Core Keys: `3` `ENTER` `4` `y^x` `5` `0` `0` `×`

Extension Challenge: 19.80 years. Keystrokes: [2] [LN] [0][.][0][3][5] [÷]

Ext. Keys: `2` `LN` `0` `•` `0` `3` `5` `÷`

Student Lab Sheet Answers: P1: 7.3891 | P2: 8.66 years | P3: 3162.28 | P4: 2.6990

Domain / Unit 11: Combinatorics, Probability & Series Accumulation (Grades 11–12 / AP • SC-32)

Core Investigation: 210 combinations. Keystrokes: [1][0] [f:][x!] [4] [f:][x!] [÷] [6] [f:][x!] [÷]

Core Keys: `1` `0` `↶` `x!` `4` `↶` `x!` `÷` `6` `↶` `x!` `÷`

Extension Challenge: Count n = 4, Mean x-bar = 16.5.

Ext. Keys: `1` `2` `Σ+` `1` `5` `Σ+` `1` `8` `Σ+` `2` `1` `Σ+`

Student Lab Sheet Answers: P1: 720 | P2: 60 | P3: 21 | P4: Mean = 10

Domain / Unit 12: Digital Logic, Base Conversions & Computer Architecture (Grades 9–12 / AP CS • SC-32)

Core Investigation: HEX: FF, BIN: 11111111 (8 ones). Keystrokes: [2][5][5] [f:][BASE]

Core Keys: `2` `5` `5` `↶` `BASE`

Extension Challenge: Binary: 1111, Decimal: 15. Keystrokes: [1][2] [ENTER] [3] [OR]

Ext. Keys: `1` `2` `ENTER` `3` `OR`

Student Lab Sheet Answers: P1: 101010 (32 + 8 + 2) | P2: 42 | P3: 7 (0111) | P4: 6 (0110)

Diagnostic Grading Rubric

Award credit for: (1) Concrete manipulative model sketch, (2) Written mathematical strategy & prediction before pressing keys, and (3) Stack verification on the SC-6 or SC-32.

Proficiency Trials Master Key — Part 1

Canonical answers and RPN keystrokes for Part 1 (Trials 1–6 & Cumulative C1).

Unit 1 Time Trial: Step Counter & Accumulator Sprint (Max Stack 1)

1. 3 ([1] [ENTER] [1] [+] [1] [+])
2. 7 ([5] [ENTER] [1] [+] [1] [+])
3. 4 ([1] [ENTER] [+] [+] [+])
4. 8 ([1] [ENTER] [+] [+] [+] [+] [+])
5. 4 ([6] [ENTER] [1] [-] [1] [-])
6. 5 ([C] [1] [ENTER] [+] [+] [+] [+])
7. 6 ([2] [ENTER] [2] [+] [2] [+])
8. 10 ([7] [ENTER] [1] [+] [1] [+] [1])
9. 9 ([0] [ENTER] [9] [+])
10. 7 ([10] [ENTER] [1] [-] [1] [-] [1])

Unit 2 Time Trial: Part-Part-Whole & Stack Addition (Max Stack 2)

1. 7 ([4] [ENTER] [3] [+])
2. 10 ([5] [ENTER] [5] [+])
3. 3 ([9] [ENTER] [6] [-])
4. 6 ([10] [ENTER] [4] [-])
5. 15 ([8] [ENTER] [7] [+])
6. 6 ([14] [ENTER] [8] [-])
7. 13 ([6] [ENTER] [7] [+])
8. 9 ([18] [ENTER] [9] [-])
9. 20 ([12] [ENTER] [8] [+])
10. 14 ([25] [ENTER] [11] [-])

Unit 3 Time Trial: Ten-Frames & Complements to 10 (Max Stack 2)

1. 3 ([10] [ENTER] [7] [-])
2. 6 ([10] [ENTER] [4] [-])
3. 13 ([8] [ENTER] [2] [+] [3] [+])
4. 15 ([9] [ENTER] [1] [+] [5] [+])
5. 13 ([7] [ENTER] [3] [+] [3] [+])
6. 9 ([10] [ENTER] [1] [-])
7. 13 ([20] [ENTER] [7] [-])
8. 15 ([8] [ENTER] [2] [+] [5] [+])
9. 16 ([10] [ENTER] [6] [+])
10. 8 ([10] [ENTER] [2] [-])

Unit 4 Time Trial: Number Racks, Doubles & Groups (Max Stack 2)

1. 12 ([6] [ENTER] [+])
2. 16 ([8] [ENTER] [+])
3. 15 ([7] [ENTER] [+] [1] [+])
4. 13 ([6] [ENTER] [+] [1] [+])
5. 18 ([9] [ENTER] [+])
6. 17 ([8] [ENTER] [+] [1] [+])
7. 12 ([4] [ENTER] [3] [×])
8. 20 ([5] [ENTER] [4] [×])
9. 24 ([12] [ENTER] [+])
10. 19 ([9] [ENTER] [+] [1] [+])

Cumulative Trial C1: Primary Stack Fundamentals Championship (Max Stack 2)

1. 7 ([1] [ENTER] [+] [+] [+] [+] [+])
2. 16 ([9] [ENTER] [7] [+])
3. 9 ([17] [ENTER] [8] [-])
4. 7 ([10] [ENTER] [3] [-])
5. 28 ([14] [ENTER] [+])
6. 23 ([11] [ENTER] [+] [1] [+])
7. 17 ([9] [ENTER] [1] [+] [7] [+])
8. 30 ([5] [ENTER] [6] [×])
9. 18 ([30] [ENTER] [12] [-])
10. 21 ([8] [ENTER] [+] [5] [+])

Unit 5 Time Trial: 4-Register Stack & Four Operations (Max Stack 4)

1. 36 ([18] [ENTER] [6] [-] [3] [×])
2. 5 ([24] [ENTER] [16] [+] [8] [÷])
3. 5 ([45] [ENTER] [12] [ENTER] [3] [-] [÷])
4. 69 ([15] [ENTER] [4] [×] [18] [ENTER] [2])
5. 4 ([50] [ENTER] [14] [-] [3] [ENTER] [6])
6. 28 ([12] [ENTER] [5] [×] [8] [ENTER] [4])
7. 8 ([20] [ENTER] [8] [-] [4] [×] [6])
8. 24 ([72] [ENTER] [4] [ENTER] [5] [+] [÷])
9. 32 ([100] [ENTER] [16] [ENTER] [5] [×] [12])
10. 6 ([8] [ENTER] [4] [+] [9] [ENTER] [3])

Unit 6 Time Trial: Fraction Studio & Mixed Numbers (Max Stack 4)

1. 0.75 ([.1.2] [ENTER] [.1.4] [+])
2. 0.25 ([.3.4] [ENTER] [.1.2] [-])
3. 3.75 ([2.1.2] [ENTER] [1.1.4] [+])
4. 1.25 ([.4.5] [1/x])
5. 0.625 ([.3.8] [ENTER] [.1.4] [+])
6. 0.5 ([.2.3] [ENTER] [.3.4] [×])
7. 1.5 ([.3.4] [ENTER] [.1.2] [÷])
8. 6 ([1.1.2] [ENTER] [4] [×])
9. 2.25 ([5] [ENTER] [2.3.4] [-])
10. 6 ([.1.3] [ENTER] [.1.6] [+] [12] [×])

Speed Trial Scoring & Stack-Depth Discipline

Score (+5 × Correct) − (2 × Incorrect), plus +1 bonus point per full minute remaining when all 10 problems are solved within the allowed Max Stack Depth.

Proficiency Trials Master Key — Part 2

Canonical answers and RPN keystrokes for Part 2 (Trials 7–12 & Cumulative C2–C3).

Unit 7 Time Trial: Postfix Evaluation & Chained Operations (Max Stack 4)

1. 50 ([8] [ENTER] [2] [+] [9] [ENTER] [4])
2. 30 ([14] [ENTER] [6] [-] [3] [×] [24])
3. 20 ([30] [ENTER] [4] [ENTER] [5] [×] [-])
4. 12 ([5] [ENTER] [3] [+] [8] [ENTER] [2])
5. 40 ([18] [ENTER] [3] [÷] [4] [+] [15])
6. 5 ([10] [ENTER] [5] [ENTER] [2] [×] [+])
7. 70 ([25] [ENTER] [5] [ENTER] [3] [×] [-])
8. 8 ([7] [ENTER] [3] [+] [4] [×] [8])
9. 36 ([48] [ENTER] [2] [ENTER] [6] [+] [÷])
10. 30 ([16] [ENTER] [4] [ENTER] [2] [+] [-])

Unit 8 Time Trial: Tactile Algebra & Quadratic Factoring (Max Stack 4)

1. -2 ([1] [ENTER] [5] [ENTER] [6] [POLY])
2. -3 ([1] [ENTER] [7] [ENTER] [12] [POLY])
3. -3 ([1] [ENTER] [8] [ENTER] [15] [POLY])
4. 3 ([1] [ENTER] [1] [+/-] [ENTER] [6] [+/-])
5. 2 ([1] [ENTER] [5] [+/-] [ENTER] [6] [POLY])
6. 3 ([1] [ENTER] [0] [ENTER] [9] [+/-] [POLY])
7. 4 ([1] [ENTER] [0] [ENTER] [16] [+/-] [POLY])
8. 25 ([3] [ENTER] [x²] [2] [×] [3] [ENTER])
9. -0.5 ([2] [ENTER] [5] [ENTER] [2] [POLY])
10. -5 ([1] [ENTER] [10] [ENTER] [25] [POLY])

Cumulative Trial C2: Intermediate Stack Mastery Championship (Max Stack 4)

1. 15 ([32] [ENTER] [14] [-] [5] [×] [6])
2. 0.875 ([.1.2] [ENTER] [.1.4] [×] [.3.4] [+])
3. 10 ([9] [ENTER] [3] [-] [x²] [12] [ENTER])
4. 1.75 ([3.1.2] [ENTER] [1.3.4] [-])
5. 10 ([15] [ENTER] [5] [+] [8] [ENTER] [3])
6. 27 ([18] [ENTER] [+] [9] [-])
7. 3 ([10] [ENTER] [2] [+/-] [-] [3] [ENTER])
8. 32 ([80] [ENTER] [15] [ENTER] [4] [×] [-])
9. 2.4 ([.1.4] [ENTER] [.1.6] [+] [1/x])
10. 20 ([12] [x²] [16] [x²] [+] [√x])

Unit 9 Time Trial: Trigonometry, Vectors & Polar Pipeline (Max Stack 4)

1. 0.5 ([30] [SIN])
2. 0.5 ([60] [COS])
3. 1 ([45] [TAN])
4. 5 ([10] [ENTER] [30] [SIN] [×])
5. 6 ([12] [ENTER] [60] [COS] [×])
6. 10 ([10] [ENTER] [0] [yellow] [->y,x])
7. 5 ([3] [ENTER] [4] [yellow] [->θ,r])
8. 13 ([5] [ENTER] [12] [yellow] [->θ,r])
9. 10 ([8] [ENTER] [6] [yellow] [->θ,r])
10. 30 ([.5] [yellow] [ASIN])

Unit 10 Time Trial: Exponential Growth & Logarithms (Max Stack 4)

1. 32 ([2] [ENTER] [5] [y^x])
2. 81 ([3] [ENTER] [4] [y^x])
3. 1 ([1] [e^x] [LN])
4. 1 ([0] [e^x])
5. 3 ([1000] [yellow] [LOG])
6. 1102.5 ([1000] [ENTER] [1.05] [ENTER] [2] [y^x] [×])
7. 9.9021 ([2] [LN] [.07] [÷])
8. 12.5 ([100] [ENTER] [.5] [ENTER] [3] [y^x] [×])
9. 610.7014 ([.04] [ENTER] [5] [×] [e^x] [500] [×])
10. 1024 ([2] [ENTER] [10] [y^x])

Unit 11 Time Trial: Combinatorics, Stats & Probability (Max Stack 4)

1. 120 ([5] [yellow] [x!])
2. 720 ([6] [yellow] [x!])
3. 20 ([5] [yellow] [x!] [3] [yellow] [x!] [÷])
4. 120 ([6] [yellow] [x!] [3] [yellow] [x!] [÷])
5. 10 ([5] [yellow] [x!] [2] [yellow] [x!] [3])
6. 15 ([6] [yellow] [x!] [2] [yellow] [x!] [4])
7. 35 ([7] [yellow] [x!] [3] [yellow] [x!] [4])
8. 15 ([1] [Σ+] [2] [Σ+] [3] [Σ+] [4])
9. 24 ([4] [yellow] [x!])
10. 0.0667 ([6] [yellow] [x!] [2] [yellow] [x!] [4])

Unit 12 Time Trial: Digital Logic, Base Conversions & CS (Max Stack 4)

1. FF ([255] [yellow] [BASE] [HEX])
2. 16 ([HEX] [10] [yellow] [BASE] [DEC])
3. 1111 ([15] [yellow] [BASE] [BIN])
4. 10 ([BIN] [1010] [yellow] [BASE] [DEC])
5. 8 ([12] [ENTER] [10] [AND])
6. 14 ([12] [ENTER] [10] [OR])
7. 6 ([12] [ENTER] [10] [XOR])
8. 32 ([HEX] [20] [yellow] [BASE] [DEC])
9. 100 ([64] [yellow] [BASE] [OCT])
10. 15 ([255] [ENTER] [15] [AND])

Cumulative Trial C3: Grand RPN Invitational Championship (Max Stack 4)

1. 281.79 ([.06] [ENTER] [12] [÷] [1] [+] [24])
2. 17 ([5] [ENTER] [3] [+/-] [-] [x²] [10])
3. 12.0208 ([12] [ENTER] [45] [COS] [×] [5] [ENTER])
4. 3.0667 ([3.1.3] [ENTER] [1.1.4] [÷] [.2.5] [+])
5. 56 ([8] [yellow] [x!] [3] [yellow] [x!] [5])
6. 2473.08 ([.05] [ENTER] [10] [×] [e^x] [1500] [×])
7. 7 ([35] [ENTER] [7] [-] [4] [×] [14])
8. 192 ([192] [ENTER] [240] [AND])
9. 29 ([6] [ENTER] [+] [8] [ENTER] [+] [1])
10. FFF ([4095] [yellow] [BASE] [HEX])

Speed Trial Scoring & Stack-Depth Discipline

Score (+5 × Correct) − (2 × Incorrect), plus +1 bonus point per full minute remaining when all 10 problems are solved within the allowed Max Stack Depth.

Manipulative Kits & Keycap Quick-Reference

How each physical manipulative models the 4-register RPN stack, plus essential SC-6 and SC-32 keycap sequences.

Classroom Manipulative Kit	Target Domains	Core Mathematical & RPN Stack Connection
Ten-Frame Tray & Two-Tone Counters	Domains 01, 03, 12	5+5 subitizing, complements to 10, zero pairs & 8-bit binary registers
Math Dominoes & 12-Frame Tray	Domain 02	Left & right domino halves map directly to Register Y & Register X
Two-Row Number Rack (Rekenrek)	Domain 04	Doubles ([ENTER][+]), near-doubles & equal groups on 5+5 bead rows
Base-10 Place Value Blocks	Domains 05, 12	10:1 decimal trading across Hundreds, Tens, Ones & multi-step stack lift
84 mm Fraction Towers & Comparator	Domain 06	1 Whole = 84 mm; equal tower heights prove equivalence; two-dot entry
Counting & Rectangular Array Tiles	Domain 07	m × n area arrays, factor pairs & parenthesis-free postfix trees
Algebra Area Tiles & Factoring Stage	Domain 08	x^2, x & 1 area tiles model binomial products, zero pairs & factoring
Unit Circle Trigonometry Volvelle	Domain 09	Rotary angle disc reveals exact (cos ϑ, sin ϑ) & polar vector conversion
Logarithmic Prime Factor Towers	Domain 10	Tower height proportional to ln(p); stacking towers adds logs to multiply
Probability Dice & Modular Spinner	Domain 11	Fair dice & 10-sector spinner drive empirical [Σ+] data accumulation

CANONICAL SC-6 & SC-32 KEYCAP SEQUENCES (EXACT HARDWARE KEYCAPS)

Chained Expression (3×4)+(5×6) = 42	SC-6	3 ENTER 4 × 5 ENTER 6 × +
Two-Dot Fraction 2/3 + 1/4 = 11/12	SC-6	• 2 • 3 ENTER • 1 • 4 +
Mixed Number 2 3/8 -> Improper 19/8	SC-6	2 • 3 • 8 ENTER f d/c
Quotient & Remainder 23 ÷ 4 (Y=5, X=3)	SC-6	2 3 ENTER 4 g ÷R
Prime Factorization & GCF(24, 36) = 12	SC-6	2 4 g 2 4 ENTER 3 6 GCF
Quadratic Roots of $x^2 + 5x + 6 = 0$	SC-32	1 ENTER 5 ENTER 6 POLY
2D Polar Vector (x=3, y=4) -> r=5, ϑ=53.13°	SC-32	4 ENTER 3 ▸θ,r
Continuous Growth 500 × e^(0.04×5)	SC-32	• 0 4 ENTER 5 × e^x 5 0 0 ×

Concept Pacing & Classroom Observation Log

Photocopiable yearly planning tracker for Concept Domains 01–12 and Middle School Units 01–10.

Concept Domain & Topic	Dates Taught	Components Completed	Student Mastery Notes
Domain 01: Cardinality, Counting & Step Accumulation	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 02: Part-Part-Whole Number Bonds & Missing Addends	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 03: The Ten-Frame, Subitizing & Make-10 Bridging	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 04: Two-Row Number Rack: Doubles & Equal Groups	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 05: Place Value & The 4-Level Stack Elevator	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 06: Fraction Studio: 84 mm Towers & Two-Dot Entry	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 07: Postfix Trees, Order of Operations & Factors	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 08: Tactile Algebra: Area Tiles, Zero Pairs & Roots	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 09: Coordinate Geometry, Trigonometry & Vectors	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 10: Exponents, Logarithms & Continuous Growth	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 11: Combinatorics, Probability & Series Statistics	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	
Domain 12: Digital Logic, Base-N & Computer Architecture	____/____/20__	[ ] Core [ ] Ext [ ] Lab [ ] Trial	

CLASSROOM MISCONCEPTION NOTES & SMALL-GROUP INTERVENTION LOG

PART III • REPRODUCIBLE FAMILY HANDOUT (PHOTOCOPY & SEND HOME ON DAY 1)

Welcome to Hands-On & Stack Mathematics!

A 2-Minute Parent & Family Guide: Why Our Classroom Uses Physical Blocks and a Calculator with No [=] Sign.

DEAR PARENTS & FAMILIES: WHY YOUR CHILD'S CALCULATOR HAS AN [ENTER] KEY INSTEAD OF [=]

This year, our math classroom pairs hands-on blocks (Ten-Frames, 84 mm Fraction Towers, and Algebra Tiles) with a 4-line StackCalc RPN (Reverse Polish Notation) calculator. If you grew up pressing 8 + 5 = on a standard calculator, you might wonder why our calculator has an [ENTER] button and no [=] key—and no parentheses keys!

- Standard 1-line calculators hide the math: Students often type long expressions blindly from left to right (or get lost in parentheses) without knowing which step happens first.
- The 4-Line Stack works just like paper-and-pencil math: On paper, you write the first number on the top line (Line Y), write the second number below it (Line X), and then add (+). That is exactly how our calculator works! Every step stays visible on the 4-line screen so your child always sees the 'why' behind the answer.

THE 3-EXAMPLE KITCHEN-TABLE CHEAT SHEET (TRY THIS WITH YOUR CHILD TONIGHT!)

Example 1: Everyday Addition — How to Calculate 8 + 5 = 13

1. Type [8] and press [ENTER] to place 8 on Waiting Shelf Y. 2. Type [5] on Workbench X. 3. Press [+] to combine Y + X = 13!

Keys to Press: `8` `ENTER` `5` `+`

Example 2: Order of Operations Without Parentheses — (8 + 4) × (9 − 3) = 72

1. Solve the first group: [8] [ENTER] [4] [+] -> 12 waits visibly on Line Y! 2. Solve the second group: [9] [ENTER] [3] [-] -> 6 sits on Line X. 3. Press [×] to multiply 12 × 6 = 72!

Keys to Press: `8` `ENTER` `4` `+` `9` `ENTER` `3` `-` `×`

Example 3: Real Fractions Using Two Dots — 1/3 + 1/4 = 7/12

Pressing [.] twice types a real fraction ([.] [1] [.] [3] = 1/3). Press [ENTER], type 1/4 ([.] [1] [.] [4]), and press [+] to see 7/12!

Keys to Press: `•` `1` `•` `3` `ENTER` `•` `1` `•` `4` `+`

3 GREAT QUESTIONS TO ASK YOUR CHILD DURING HOMEWORK

1. "Before you touch the calculator keys, can you show me or sketch what this problem looks like with blocks?"

2. "Which inner part of this problem do we need to solve first so it can wait on Shelf Y?"

3. "What number is sitting on Waiting Shelf Y right now, and what number is on Workbench X?"

Explore free printable family math activities & interactive guides at: https://stackcalc.io/curriculum/

OPTIONAL FAMILY RETURN SLIP (TRY THE 3 CHEAT-SHEET EXAMPLES TOGETHER!)

Student Name: _____ Parent/Guardian Signature: _____

[ ] We tried the 3 Stack Calculator examples together at home! Teacher / Class: _____